Deploy Dell Technologies PowerFlex v3.6.x

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Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

This guide contains deployment procedures for PowerFlex.

I

Linux

Deployment procedures for Linux-based systems

2

Overview—deployment steps and options

This section summarizes deployment steps and options.

In physical environments and in non-VMware virtual environments, use the PowerFlex Installer to install and configure the PowerFlex components on multiple nodes from one central server, using a web client. The PowerFlex Installer is configurable.

As part of the installation process, the PowerFlex Lightweight Installation Agent (LIA) is installed on each node. The LIA is a required component for many maintenance operations. During initial installation, the PowerFlex Installer establishes trust with the LIA, which facilitates its operation. You can limit the operations that LIA can perform by making changes to its configuration file.

In Windows environments, LIA will only install and upgrade packages that are signed by PowerFlex and are associated with the PowerFlex system.

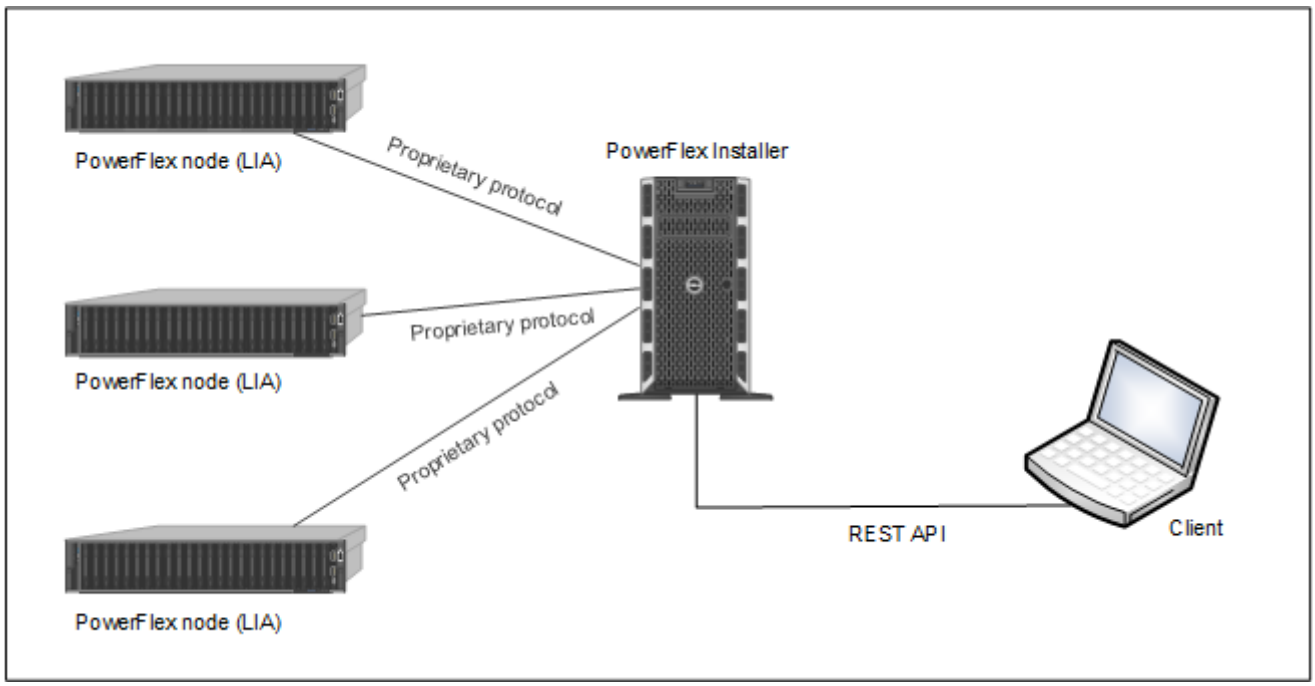


Figure 1. Node management architecture

NOTE: The PowerFlex Installer also has a REST API that provides install, extend, and uninstall functionality. Deployment procedures are included in this guide. All the REST API commands are described in the *PowerFlex REST API Reference Guide*.

Deployment workflow—Linux

The PowerFlex deployment workflow on physical servers, and in non-VMware hyper-converged infrastructure (HCI) environments such as Citrix Hypervisor (XenServer) and KVM, is described in this summary.

For VxFlex Ready Node or VxRack Node 100 Series systems, refer to your product's *Hardware Configuration and Operating System Installation Guide* and *Deployment Guide* for step-by-step installation instructions on how to prepare the server and operating system prior to deployment of PowerFlex.

The main steps in the PowerFlex deployment workflow are:

1. Prepare the environment.
2. Start the deployment.

3. Monitor the deployment.
4. For some two-layer system configurations, install SDCs on external servers.
5. Install the PowerFlex presentation server.
6. Install the license.
7. Perform mandatory, recommended, and optional post-deployment tasks.
8. Move the system to production.

Deployment options—Linux

Deploy PowerFlex on physical servers using the PowerFlex Installer wizard, or fully customized option.

The following deployment options are available:

- Fully customizable installation

  This mode, which enables the highest level of customization, uses a user-prepared CSV topology file and the PowerFlex Installer to install PowerFlex and configure nodes. To add additional servers in the future, use the combination of CSV topology file and PowerFlex Installer again. For this reason, it is recommended to save the file in a secure encrypted folder on your operating system, after deployment.

- Using the PowerFlex Installer wizard

  This abbreviated mode gets a PowerFlex system up and running in the simplest manner, with preset node configuration, where all management and data communications are on the same network. This mode is perfect for a single-Protection Domain, fully converged system. No CSV file is required.

Deployment checklist—Linux

Checklist for deployment requirements in Linux-based environments.

The following table provides a checklist of items required for PowerFlex deployment.

Table 1. Pre-deployment checklist

Item	Checked
General	
All servers intended for hosting PowerFlex components and the PowerFlex Gateway meet the system requirements described in the *Getting to Know Dell PowerFlex Guide*. **NOTE:** The Dell Technologies-supplied hardware configurations satisfy all the hardware requirements.	-
Secure Boot on servers is not supported. Ensure that Secure Boot is disabled in each server's BIOS.	-
All nodes must have unique hostnames.	-
Servers that contained PowerFlex components from other systems must be cleaned before deployment of a new system. Use operating system tools to remove all those PowerFlex components.	-
Configure disks so they are visible to the operating system according to operating system and vendor hardware guidelines.	-
To use secure authentication mode, ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system.	-
Prepare passwords for the PowerFlex Gateway, MDM, and LIA. These passwords will be required for future access to system components for upgrade and certain configuration procedures. Store them in a secure location.	-
On nodes configured with NVDIMMs, running on Red Hat Enterprise Linux version 7.6, ensure that the following package is installed: `kmod-redhat-nfit-3.10.0_957-1.el7_6.x86_64.rpm` For more information, see *Dell Technologies EMC NVDIMM-N Persistent Memory User Guide*.	-
On nodes configured with NVDIMMs, ensure that the ndctl and daxio packages are installed:	-

Table 1. Pre-deployment checklist (continued)

Item	Checked

ndctl package—for example: `ndctl-64.1-2.el7.x86_64.rpm`
 	
daxio package—for example: `daxio-1.5.1-2.1.el7.x86_64.rpm`
 	
Verify that there are no bad blocks on the NVDIMMs. If there are bad blocks, replace the corresponding NVDIMMs before deploying the PowerFlex system. Perform this check on every SDS that uses NVDIMMs:
 	
1. Using shell, run the following command on each Linux-based node that uses NVDIMMs:
 	
`ndctl list --media-errors`
 	
2. If the output for any of the hosts displays a result for `"badblock_count":`, the NVDIMM (devdax device) has bad blocks. For example:
 	
`{ "dev":"namespace1.0", "mode":"devdax", "map":"dev", "size":62914560, "uuid":"47072fee-1447-4e04-b4ca-f28e2c03af80", "chardev":"dax1.0", "align":4096, "badblock_count":13, "badblocks": }`
 	
3. Refer to the Linux NVDIMM replacement procedure for the specific server model.	

For SELinux, follow the guidelines included in "Preparation for Deployment" before deploying the system.	-
For SELinux activation and SDC deployment, ensure that the relevant packages are installed. For more information, refer to the operating system requirements described in the *Getting to Know Dell PowerFlex Guide*.	-
Gateway	
The server that hosts the PowerFlex Gateway must have adequate memory to run the PowerFlex Installer and any other applications. Ensure that this server meets all the requirements listed in the *Getting to Know Dell PowerFlex Guide*.	-
The PowerFlex Gateway must not be installed on a server that hosts either the SDC component, or on an SDS component that uses RFcache acceleration feature.	-
The PowerFlex Gateway cannot be installed on a Fedora\RH CoreOS-based server. For a Fedora\RH CoreOS environment, prepare a supported Linux or Windows-based server on which the PowerFlex Gateway can be installed.	-
Oracle JRE 1.8 build 291 or OpenJDK JDK 8 build 292.b10 is installed. On Linux-based servers, OpenJDK 11.0.11.09 is also supported.	-
If LDAP authentication is used for LIA authentication, before deploying the PowerFlex Gateway, perform configurations on the server that will host the PowerFlex Gateway, as described in *Dell PowerFlex User Roles and LDAP Usage Technical Notes*.	-
PowerFlex presentation server	
Oracle JRE 1.8 build 291, OpenJDK JDK 8 build 292.b10, or OpenJDK 11.0.11.09 is installed.	-
ESXi	
For two-layer system SDCs running on VMware, the host from which you run the PowerShell (.ps1) script meets the following prerequisites: ● Runs on supported Windows operating systems. For more information, refer to the VMware compatiblity matrix: VMware PowerCLI compatibility matrix ● PowerCLI is installed. For vCenter v6.7, PowerCLI v6.5 must be used.	-

Table 1. Pre-deployment checklist (continued)

Item	Checked
• Oracle JRE 1.8 build 291 or OpenJDK JDK 8 build 292.b10 is installed. • Has incoming and outgoing communication access to the vCenter.	
The required network interfaces and IP addresses are configured.	-
Networking	
For a fully converged Linux-based system, follow the networking recommendations in the *Getting to Know Dell PowerFlex Guide*.	-
For two-layer system SDCs running on VMware, the vSphere web client (Virgo) server has network connectivity to the host on which the PowerShell script will be used.	-
The server which will host the PowerFlex Gateway has connectivity to both data and management PowerFlex networks.	-
For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM, disable the IPv6 DAD setting on the server's relevant interfaces, using the command: `sysctl net.ipv6.conf.<interface_name>.dad_transmits=0`	-
Replication	
Use the information provided in the *Getting to Know Dell PowerFlex Guide* to calculate the percentage of replication journal capacity that is required for each Storage Pool.	-

3

Preparation for deployment

Prepare the software packages, CSV file (when required), PowerFlex Gateway, and other items for PowerFlex deployment.

Download installation packages—Linux

Download and extract the complete software package for this version.

About this task

The software package contains:

- Management software:
  - PowerFlex Gateway packages for Linux and Windows. The PowerFlex Gateway includes the PowerFlex Installer, which is used to deploy PowerFlex.
  - PowerFlex presentation server packages for the PowerFlex GUI
- PowerFlex core component files. You need these files for all operating systems of nodes in the system, for LIA, MDM, SDC and SDS components.
- RFcache `xcache` package (optional for systems where RFcache acceleration is enabled)

Deploy the PowerFlex Gateway on a node whose operating system matches your environment needs:

Table 2. PowerFlex nodes for PowerFlex Gateway deployment

To deploy PowerFlex nodes on this operating system	Create a PowerFlex Gateway on this operating system
Linux	Windows or Linux
Linux and Windows	Windows

Steps

1. From the Online Support site (PowerFlex Software Drivers & Downloads page), download the complete software for this version.
2. Extract the packages from the downloaded file.
3. Save the extracted files in a location with network connectivity with the server on which PowerFlex Gateway will be installed.

   The PowerFlex Installer on the gateway deploys the packages on all the servers in the system.

Linux package names

PowerFlex Linux installation packages have several name formats.

Package formats

Throughout this section, Linux packages are displayed in the following format:

`EMC-ScaleIO-<component>-3.6-X.<build>.<flavor>.x86_64.rpm`

where:

- *<component>* is the PowerFlex component: MDM, LIA, SDS, SDR, and so on.
- *<flavor>* is the operating system for your environment, according to the following table:

Table 3. Linux package formats

Linux flavor	Package format	Example
RHEL/OL/CentOS	el<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.el<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-<build>.X.el6.x86_64.rpm`
SUSE	sles<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.sles<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-<build>.X.sles12.2.x86_64.rpm`
Ubuntu	Ubuntu.<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.Ubuntu.<version>.x86_64.tar` Example: `EMC-ScaleIO-mdm-3.6-X.<build>.Ubuntu.16.04.x86_64.tar` Ubuntu packages may need to be extracted before use. This is described where relevant.
Citrix Hypervisor (XenServer)	xs<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.xs<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-X.<build>.xs7.0.0.x86_64.rpm`

NOTE: RedHat CoreOS and Fedora CoreOS are supported on CSI containers. Refer to CSI documentation for more information.

Use the packages and the installation commands that match your Linux operating system environment.

On Citrix Hypervisor (XenServer)-based nodes, the syntax for PowerFlex CLI commands is `siocli`, instead of `scli`.

NOTE: Before deploying or upgrading Oracle Linux (OL) or Ubuntu-based systems, ensure that the PowerFlex environment is prepared, as described in the preparation tasks for Oracle Linux, SLES and Ubuntu in the *Deploy Dell PowerFlex* .

Deploying PowerFlex with SELinux

PowerFlex does not support native deployment on an operating system configured with Security-Enhanced Linux (SELinux) in enforcing mode. Therefore, prerequisites must be adhered to, and preparation guidelines must be followed prior to system deployment. These guidelines also apply to an existing system that is being moved to SELinux enforcing mode.

CAUTION: If you are not sure whether all other processes have been configured with SELinux policies, do not enable SELinux in enforcing mode. Contact your system administrator for guidance.

For more information about SELinux, see What is SELinux.

SELinux Prerequisites

The following packages are prerequisites for Security-Enhanced Linux (SELinux).

Table 4. SELinux prerequisites

Package	Description
policycoreutils	Provides utilities for managing SELinux
selinux-policy	Provides SELinux reference policy
selinux-policy-targeted	Provides SELinux targeted policy
libselinux	Provides SELinux configuration commands
libselinux-utils	Provides SELinux configuration commands
policycoreutils-python	Required to enable the `semanage` command
policycoreutils-devel	Provides troubleshooting capabilities in case an issue is encountered

Download and copy the SELinux module ZIP file

Initially, an administrator is required to download the Security-Enhanced Linux (SELinux) Policy for PowerFlex to the relevant operating systems.

Steps

1. Download the latest version of the file `PowerFlex_SELinux_DDMMYYYY.zip` from the PowerFlex Drivers & Downloads page.
2. On all relevant nodes, create a folder called `/var/PowerFlex_SELinux/`.

   Use the command:

   ```
   mkdir /var/PowerFlex_SELinux/
   ```

   The relevant nodes are:
   - MDM
   - SDS
   - SDR (if used; this includes the relevant rules for LIA)
   - SDC
   - LIA
   - PowerFlex Gateway
   - PowerFlex presentation server
3. Copy or transfer the `PowerFlex_SELinux_DDMMYYYY.zip` file to the nodes that will be used for PowerFlex.
4. On each node, perform the following:
   a. Change the directory to the folder that you created:

      ```
      cd /var/PowerFlex_SELinux/
      ```

   b. Unzip the file:

      ```
      tar -xvf PowerFlex_SELinux_DDMMYYYY.zip
      ```

Verify that SELinux is enabled

Verify that Security-Enhanced Linux (SELinux) is enabled on all relevant nodes and operating systems.

Steps

1. Run the following command on all the nodes to check SELinux status:

   ```
   sestatus
   ```

   Expected output if SELinux is enabled and set to enforcing mode:

   ```
   SELinux status:                 enabled
   SELinuxfs mount:                /sys/fs/selinux
   SELinux root directory:         /etc/selinux
   Loaded policy name:             targeted
   Current mode:                   enforcing
   Mode from config file:          enforcing
   Policy MLS status:              enabled
   Policy deny_unknown status:     allowed
   Max kernel policy version:      31
   ```

   Expected output if SELinux is disabled:

   ```
   SELinux status:                 disabled
   ```
2. If SELinux is disabled, follow the corresponding operating system's guidelines in order to enable it.

For example, for CentOS, the steps are:

a. Run the command:

```
b.    vi /etc/selinux/config
```

b. Change the SELINUX line to enforcing or permissive, as needed:

```
SELINUX=enforcing
...
SELINUXTYPE=targeted
```

c. Schedule a reboot of the operating system.

CAUTION: If this process is running on a pre-deployed PowerFlex system, follow graceful shutdown guidelines to reboot a node with MDM\SDS\SDR\SDC. Refer to "Shutdown or restart a node gracefully" the *Upgrade Dell PowerFlex* matching your system version.

Load the PowerFlex SELinux module

Load the PowerFlex Security-Enhanced Linux (SELinux) module and prepare to deploy it or run it on the system.

Steps

1. Ensure that you are in the correct folder for SELinux, using the command:

```
pwd
```

Expected location: `/var/PowerFlex_SELinux/`

Go to the folder, if necessary.

2. To load the policy and additional settings without impacting a deployed system, run the following command to set the current state of SELinux to Permissive state:

```
setenforce 0
```

3. Run the following command to load the SELinux module:

```
semodule -i dell-vxflex.pp -vv
```

Expected result:

```
Attempting to install module 'dell-vxflex.pp':
Ok: return value of 0.
Committing changes:
Ok: transaction number 0
```

4. Run the following command:

```
restorecon -RF /opt/emc
```

This command corrects any missing labels in all directories and files in `/opt/emc`.

5. Configure the PowerFlex ports in SELinux:

NOTE: Only run commands on the relevant node. For example, if the PowerFlex presentation server or PowerFlex Gateway is on a stand-alone node, do not enable the MDM, SDS, SDR, or LIA ports.

For relevant port information, see "Port usage and changing default ports" in the *PowerFlex Security Guide* corresponding to the PowerFlex version you are using.

If the default ports settings were changed for any process, update the command below accordingly (for example, for the PowerFlex presentation server https port).

```
/usr/sbin/semanage port -N -a -t vxflex_mdm_port_t -p tcp 25620;
/usr/sbin/semanage port -N -a -t vxflex_mdm_port_t -p tcp 9011;
/usr/sbin/semanage port -N -a -t vxflex_mdm_port_t -p tcp 6611;
/usr/sbin/semanage port -N -a -t vxflex_mdm_port_t -p tcp 7611;
/usr/sbin/semanage port -N -a -t vxflex_mdm_port_t -p tcp 25600;
/usr/sbin/semanage port -N -a -t vxflex_lia_port_t -p tcp 9099 ;
/usr/sbin/semanage port -N -a -t vxflex_sdr_port_t -p tcp 11088;
/usr/sbin/semanage port -N -a -t vxflex_sds_port_t -p tcp 7072;
/usr/sbin/semanage port -N -a -t vxflex_sds_port_t -p tcp 25640;
/usr/sbin/semanage port -N -a -t vxflex_gateway_port_t -p tcp 443;
/usr/sbin/semanage port -N -a -t vxflex_gateway_port_t -p tcp 8080;
```

NOTE: Ports 443 and 8080 are usually enabled by default in SELinux and might return the following error:

```
ValueError: Port tcp/443 already defined
ValueError: Port tcp/8080 already defined
```

In this case, proceed. This is a valid error.

```
/usr/sbin/semanage port -N -a -t vxflex_mgmt_port_t -p tcp 8443;
 /usr/sbin/semanage port -N -a -t vxflex_mgmt_port_t -p tcp 8080;
```

6. Run the following command:

```
setenforce 1
```

This command changes the SELinux state to the targeted state for the current run time.

7. Repeat the steps above on all nodes that will run with SELinux.

Results

Preparation of the operating system on all nodes for PowerFlex running in SELinux is now complete. Proceed with deployment, or continue to use your system if it is already deployed.

Troubleshoot SELinux with PowerFlex issues

When troubleshooting Security-Enhanced Linux (SELinux) issues with PowerFlex, confirm that the issues are related to PowerFlex.

About this task

This procedure requires moving the relevant nodes from SELinux enforcing mode to permissive mode. To return to enforcing mode later on, use the command `setenforce 1`.

Steps

1. Change the mode of the relevant nodes from SELinux enforcing mode to permissive mode:

```
setenforce 0
```

2. If the issue disappears in permissive mode, stay in this mode and run the following commands:

```
semodule -l |grep vxflex
audit2allow -a
semanage port -l |grep vxflex
```

3. Save the output to a file, for sharing when opening a customer support ticket.
4. Zip the `/var/log/audit/` folder and share the file in the support ticket.

In rare cases, the support person might ask you to change SELinux logging to a higher level using this command:

```
semodule
-DB
```

This might require repeating the failed operation with this setting enabled. After the information is collected, run the following command to revert to the standard logging level:

```
  semodule
-B
```

See the following for troubleshooting information: Troubleshooting problems related to SELinux.

Prepare server operating systems for support of large-scale systems (up to 2048 SDCs)

For large-scale systems with up to 2048 SDCs, prepare the operating systems of servers that will host MDM, SDS, SDR, and PowerFlex Gateway components.

Prerequisites

Superuser permissions are required for this procedure.

About this task

Configuration of servers hosting MDM, SDS, SDR, and PowerFlex Gateway components is required, in order to support large-scale systems with up to 2048 SDCs.

- A system setting must be validated and configured on servers hosting MDM, SDS, SDR, and the PowerFlex Gateway.
- For all servers hosting MDM, SDS, and SDR components, certain threshold parameters must be configured. This change enables a larger number of records in the operating system's ARP Cache table.

Steps

1. On servers hosting MDM, SDS, SDR, or PowerFlex Gateway components, do the following:
   a. Run this command:

      ```
      cat /proc/sys/fs/file-max
      ```

      The output for the command is the number of files that can be opened by the operating system.
   b. If the number is 24576 or higher, go to step 2. If the number is less than 24576, go to the next sub-step.
   c. Open the file `/etc/sysctl.conf` using vi.
   d. If the following line exists, edit it with vi. If it does not exist, add the full line with the following setting:

      ```
      fs.file-max = 24576
      ```

2. On servers hosting MDM, SDS, or SDR components, if the following lines exist in the file `/etc/sysctl.conf`, edit them with vi. If they do not exist, add the full lines with the following settings:

   ```
   net.ipv4.neigh.default.gc_thresh1 = 8192
   net.ipv4.neigh.default.gc_thresh2 = 16384
   net.ipv4.neigh.default.gc_thresh3 = 32768
   ```

3. Save the file.
4. Apply the changes using the command:

   ```
   sysctl -p
   ```

5. Repeat this procedure on the next server, until all servers are configured.

Prepare the DAX devices

If the deployment will be using NVDIMM devices for acceleration, perform the following tasks to prepare the NVDIMMs as DAX acceleration devices.

NVDIMM information table

Before you prepare the NVDIMM as a DAX device, create or print the following NVDIMM information table. Use it to record the information that you discover while performing the following task. You will need the information to add the acceleration device to PowerFlex.

Table 5. NVDIMM information table

Item	Replacement NVDIMM
Serial Number	-
Device name (NMEM)	-
Namespace	-
DAX device name (chardev)	-
Acceleration device path	-

Prepare an NVDIMM as a DAX (acceleration) device

Prepare a new or replacement NVDIMM as a DAX/acceleration device before adding it to the SDS in a Linux system. This step is optional when replacing an NVDIMM battery.

About this task

NOTE: If you replaced the system board, perform the following steps for each NVDIMM mounted on the server.

Steps

1. Using SSH, log in to the Linux server.
2. View information for the new/replacement NVDIMM:

```
dmidecode --type memory | grep "Non-" -B 3 -A 3 | grep -E 'Locator|Serial' | grep -v
Bank
```

   Output similar to the following appears:

```
Locator: A7
Serial Number: 17496594
Locator: B7
Serial Number: 174965AC
```

3. Find the serial number in the output and record it in the NVDIMM information table.
4. Display the correlation between the ID and NMEM device name of each NVDIMM mounted on the server:

```
ndctl list -Dvvv | jq '.[].dimms[]'
```

   Output similar to the following appears:

```
{
  "dev": "nmem1",
  "id": "802c-0f-1722-174965ac",
  "handle": 4097,
  "phys_id": 4370,
  "health": {
```

```
    "health_state": "ok",
    "temperature_celsius": 255,
    "life_used_percentage": 30
  }
}
{
  "dev": "nmem0",
  "id": "802c-0f-1711-16492521",
  "handle": 1,
  "phys_id": 4358,
  "health": {
    "health_state": "ok",
    "temperature_celsius": 255,
    "life_used_percentage": 30
  }
}
```

5. In the output from the previous step, find the device (dev) with the `id` that partially correlates with the serial number you discovered previously for the failed device.

   For example:

   - The NVDIMM output displays serial number 16492521 for the NVDIMM device.
   - In the previous step, the output displays the ID of device nmem0 as 802c-0f-1711-16492521.

Table 6. NVDIMM device details

Serial number	Device ID

`Locator: A7`
`Serial Number: 16492521`
`Locator: B7`
`Serial Number: 1649251B`
	`"dev": "nmem0",`
	`"id": "802c-0f-1711-16492521",`

   In the above example, the NMEM name is nmem0.

6. Record the NMEM name in the `Device name` row of the NVDIMM information table.
7. Correlate between the NMEM DIMM and the namespace/DAX device (acceleration device):

```
ndctl list -Dvvv | jq '.[].dimms[]'
```

   Output similar to the following appears:

```
{
  "dev": "nmem1",
  "id": "802c-0f-1722-174965ac",
  "handle": 4097,
  "phys_id": 4370,
  "health": {
    "health_state": "ok",
    "temperature_celsius": 255,
    "life_used_percentage": 30
  }
}
{
  "dev": "nmem0",
  "id": "802c-0f-1722-17496594",
  "handle": 1,
  "phys_id": 4358,
  "health": {
    "health_state": "ok",
    "temperature_celsius": 255,
    "life_used_percentage": 30
  }
}

ndctl list -Dvvv | jq '.[].regions[].namespaces[]'

{
```

```
  "dev": "namespace1.0",
  "mode": "raw",
  "size": 17179869184,
  "sector_size": 512,
  "blockdev": "pmem1",
  "numa_node": 1
}
{
  "dev": "namespace0.0",
  "mode": "raw",
  "size": 17179869184,
  "sector_size": 512,
  "blockdev": "pmem0",
  "numa_node": 0
}
```

8. In the output displayed in the previous step, locate the namespace that correlates with the NMEM name and DIMM serial number, and record it in the NVDIMM information table.

   In the above example, nmem0's namespace is namespace0.0.

9. Destroy the default namespace that was created for the replacement NVDIMM, using the namespace discovered in the previous step:

   ```
   ndctl destroy-namespace namespaceX.0
   ```

   For example, if the replacement NVDIMM maps to namespace0.0, the command is:

   ```
   ndctl destroy-namespace namespace0.0
   ```

10. Create a new, raw nmem device using the region associated with the namespace of the failed device, as recorded in the NVDIMM information table:

    ```
    ndctl create-namespace -r regionX -m raw -f
    ```

    For example, if the NVDIMM you replaced mapped to region 0, the command is:

    ```
    ndctl create-namespace -r region0 -m raw -f
    ```

11. Convert the namespace device to the acceleration device name of type /dev/daxX.X:

    ```
    ndctl create-namespace -f -e namespaceX.X --mode=devdax -align=4K
    ```

    For ndctl build 61.2 or later, run:

    ```
    ndctl create-namespace -f -e namespaceX.X --mode=dax -align=4K --no-autolabel
    ```

    Where namespace*X.X* is the namespace of the NVDIMM.

    For example, to convert namespace0.0 to /dev/dax0.0, where /dev/daxX.X is the acceleration device name, depending on the ndctl build run:

    ```
    ndctl create-namespace -f -e namespace0.0 --mode=dax -align=4K
    ```

    Or

    ```
    ndctl create-namespace -f -e namespace0.0 --mode=dax -align=4K --no-autolabel
    ```

    NOTE: `--no-autolabel` is only required for RHEL7.6.

12. Record the acceleration device name in the NVDIMM information table.

13. Run the `namespace-to-dax-device` correlation command to find the DAX/acceleration device name of the replacement NVDIMM:

```
ndctl list -Dvvv | jq '.[].regions[].namespaces[]'
```

Output similar to the following appears:

```
{
  "dev": "namespace1.0",
  "mode": "devdax",
  "map": "dev",
  "size": 16909336576,
  "uuid": "c59d6a2d-7eeb-4f32-b27a-9960a327e734",
  "daxregion": {
    "id": 1,
    "size": 16909336576,
    "align": 4096,
    "devices": [
      {
        "chardev": "dax1.0",
        "size": 16909336576
      }
    ]
  },
  "numa_node": 1
}
{
  "dev": "namespace0.0",
  "mode": "devdax",
  "map": "dev",
  "size": 16909336576,
  "uuid": "eff6429c-706f-469e-bab4-a0d34321c186",
  "daxregion": {
    "id": 0,
    "size": 16909336576,
    "align": 4096,
    "devices": [
      {
        "chardev": "dax0.0",
        "size": 16909336576
      }
    ]
  },
  "numa_node": 0
}
```

The DAX/acceleration device name appears in the output as the chardev value.

In the example output above, the DAX/acceleration device name is dax0.0.

14. Record the DAX/acceleration device name in the NVDIMM information table.
15. Find the full acceleration device path:

```
    ls -ltr /dev/ | grep dax
```

The following output should appear:

```
/dev/daxX.X
```

For example:

```
/dev/dax0.0
```

16. Record the acceleration device path in the NVDIMM information table.

Results

You are now ready to add the DAX/acceleration device to the NVDIMM Acceleration Pool.

Install and configure the PowerFlex Gateway

Install the PowerFlex Gateway. The gateway serves as the PowerFlex Installer, the PowerFlex REST API gateway, and the SNMP trap sender. The PowerFlex Gateway can be installed on Linux or on Windows. A PowerFlex Gateway installed on Windows is capable of deploying PowerFlex system components on both Linux and Windows based servers.

Install PowerFlex Gateway on Linux

Install the PowerFlex Gateway on a Linux-based server. The gateway includes the PowerFlex Installer.

Prerequisites

Before deploying any PowerFlex components on a server that was part of another PowerFlex system, use the operating system tools to remove all of those PowerFlex components.

Ensure that the version of the PowerFlex Gateway that you are about to install is the same software version as the other PowerFlex system components that you intend to install.

If LDAP authentication will be used for LIA, the server on which the PowerFlex Gateway will be installed must be prepared before PowerFlex Gateway installation. For more information, refer to *Dell PowerFlex User Roles and LDAP Usage Technical Notes*.

About this task

- The PowerFlex Gateway can be installed on a node that will also be used for PowerFlex SDS components. However, do not install the PowerFlex Gateway on an SDS upon which RFcache acceleration will be enabled (in other words, do not install the PowerFlex Gateway on SDS nodes on which the RFcache `xcache` package will be installed).
- Do not install PowerFlex Gateway on a server that will be used for a PowerFlex MDM or SDC component.
- You can use a Linux-based PowerFlex Installer to deploy PowerFlex only on Linux-based servers.
- PowerFlex Gateway cannot be installed on the Fedora\RH CoreOS operating system. In a Fedora\RH CoreOS environment, install PowerFlex Gateway on a supported Windows or Linux-based server instead.

NOTE: When deploying the PowerFlex Gateway, if one of the following ports is not free, the deployment will fail: 80, 8080, 443, 8443. If this occurs, free the above-mentioned ports and then repeat the deployment procedure.

NOTE: To install with a customized Java configuration, see "Advanced and optional PowerFlex Gateway tasks".

Steps

1. Copy the PowerFlex Gateway package from the extracted download file to the server on which you will install the package. The recommended location on the server is the `/tmp/` directory.
2. Install the PowerFlex Gateway on the Linux server, by running the following command (all on one line):
   - RHEL/Centos/Oracle Linux/SLES

     ```
     GATEWAY_ADMIN_PASSWORD=<new_GW_admin_password> rpm -i /tmp/EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
     ```

   - Ubuntu

     ```
     GATEWAY_ADMIN_PASSWORD=<new_GW_admin_password> dpkg -i /tmp/EMC-ScaleIO-gateway-3.6-X.<build>.amd64.deb
     ```

     where `/tmp/` is the local location where you saved the package, and *<new_GW_admin_password>* is a password that you define to access the PowerFlex Installer.

     The password must meet the following criteria:

     - At least 8 characters long
     - Include at least 3 of the following groups: [a-z], [A-Z], [0-9], special chars (!@#$ …)

     To install without assigning a PowerFlex Gateway password, see "Install the PowerFlex Gateway without assigning an admin password".

NOTE: To include special characters in the password, such as ! @ $, when assigning the password, add the "\" character before the special character. For example, to assign the password Scaleio$$, enter it as Scaleio\$\$. When logging in, the "\" character is not used.

3. Proceed with the remaining preparation tasks, as relevant for your system and installation method. For installation on Ubuntu, Oracle Linux, or SLES servers, pay special attention to the preparation tasks for these operating systems.

   NOTE: The PowerFlex Gateway and the PowerFlex Installer can now be configured or disabled, as explained in "Configure PowerFlex Gateway properties".

Install PowerFlex Gateway on Windows

Install the PowerFlex Gateway on a Windows-based server. The PowerFlex Gateway includes the PowerFlex Installer.

Prerequisites

Before deploying any PowerFlex components on a server that was part of another PowerFlex system, use the operating system tools to remove all of those PowerFlex components.

Ensure that the version of the PowerFlex Gateway that you are about to install is the same software version as the other PowerFlex system components that you intend to install.

If LDAP authentication will be used for LIA, the server on which the PowerFlex Gateway will be installed must be prepared before PowerFlex Gateway installation. For more information, refer to *Dell PowerFlex User Roles and LDAP Usage Technical Notes*.

About this task

- Do not install PowerFlex Gateway on a server that will be used for a PowerFlex SDC component.
- A Windows-based PowerFlex Installer can be used to deploy PowerFlex on both Windows and Linux-based servers.

NOTE: When deploying the PowerFlex Gateway, if one of the following ports is not free, the deployment will fail: 80, 8080, 443, 8443. If this occurs, free the above-mentioned ports and then repeat the deployment procedure.

Steps

1. From the extracted download file, copy the PowerFlex Gateway MSI file to the server where it will be installed:
   `EMC-ScaleIO-gateway-3.6-X.<build>-x64.msi`
2. Run the file, and enter a new PowerFlex Gateway admin password that will be used to access the PowerFlex Installer.
   The password must meet the following criteria:
   - At least 8 characters long
   - Include at least 3 of the following groups: [a-z], [A-Z], [0-9], special chars (!@#$ …)
3. From the extracted download file, run the `Clean_XC_registry.bat` script on every Windows machine to be part of the system.
   The script is included in the Complete Windows download package.
4. Proceed with the remaining preparation tasks, as relevant for your system and installation method. For installation on Ubuntu, Oracle Linux, or SLES servers, pay special attention to the preparation tasks for these operating systems.

   NOTE: The PowerFlex Gateway and the PowerFlex Installer can now be configured or disabled, as explained in "Configure PowerFlex Gateway properties".

Configure PowerFlex Gateway properties

Configure PowerFlex Gateway properties and additional PowerFlex features, using the `gatewayUser.properties` file.

Steps

1. Using a text editor, open the `gatewayUser.properties` file, located in the following directory on the PowerFlex Gateway server:

PowerFlex Gateway operating system	Location of gatewayUser.properties file
Windows	C:\Program Files\EMC\ScaleIO\Gateway\webapps\ROOT\WEB-INF\classes\
Linux	/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes

2. Edit the file with the required changes, as follows:

Table 7. Use case details for the `gatewayUser.properties` file

Use case	Property	Action	Example
Enable or disable the reuse of previously used devices in the following use cases: • Add SDS devices that were used in a previous PowerFlex. system. (Adding them without the `true` flag causes deployment failure.) • Extend a system with an SDS whose devices were used in a previous system. • Extend a system by adding SDSs that are already installed on hosts	`add.sds.with.force.on.next.run`	Default: false To enable, set to `true`. To disable, set to `false`. **NOTE:** After finishing and marking the deployment as complete, the flag will revert to `false`, so you must set it again for future deployments, as necessary.	`add.sds.with.force.on.next.run=true`
Use a PowerFlex Gateway that did not initially deploy your PowerFlex system	`mdm.ip.addresses`	Add the IP addresses of the system's primary and secondary MDMs, separated by semicolons, to this property. Both management and data IP addresses must be listed.	`mdm.ip.addresses=10.76.60.232;10.76.60.233;10.76.60.234;10.76.30.10;10.76.30.11;10.76.30.12`
Enable or disable the PowerFlex Installer	`features.enable_IM`	Default: true To disable, set to `false`. You can completely disable the use of the PowerFlex Installer default port, 443, by setting both this property and the `features.enable_gateway` property to `false`.	`features.enable_IM=false`
Enable PowerFlex Gateway	`features.enable_gateway`	Default: true To disable, set to `false` You can disable the use of the default port, 443, by setting both this property and the `features.enable_IM` to `false`.	`features.enable_gateway=false`
Enable SNMP	`features.enable_snmp`	Default: false To enable, set to `true`	`features.enable_snmp=true`
Enable Secure Remote Services or email alert notifications	`features.notification_method`	Default: srs To disable Secure Remote Services, set to `none`	`features.notification_method=srs`

Table 7. Use case details for the `gatewayUser.properties` file (continued)

Use case	Property	Action	Example
		To enable Secure Remote Services, set to `srs` To enable email notifications, set to `email` Perform additional configurations for the other email notification properties in this configuration file.	
Set the email notification type	`notifications.emailSID.1.type`	Default (only option currently available): `callHome`	`notifications.emailSID.1.type=callHome`
Set email notification SID: email sender identification (identity shown in the email's "From" field)	`notifications.emailSID.1.SID`	-	`notifications.emailSID.1.SID=DDD@dell.com`
Set email notification user name, required only when email notification uses SMTP server authentication	`notifications.emailSID.1.username`	-	`notifications.emailSID.1.username=user123`
Set email notification SMTP server name	`notifications.emailSID.1.smtp`	-	`notifications.emailSID.1.smtp=mailhub.lss.emc.com`
Enable SMTP authenticated mode for email notification	`notifications.emailSID.1.authenticate`	Default: false To disable, set to `false` To enable, set to `true` In addition, configure the user name, using the property `notifications.emailSID.1.username`, and configure the password using REST API, as described in the *Configure and Customize Dell PowerFlex*	`notifications.emailSID.1.authenticate=false`
Set a custom port for email notification (optional; will override default port number 25)	`notifications.emailSID.1.port`	Default: 25	`notifications.emailSID.1.port=25`
To restrict login to LDAP users only (disable local user login)	`gateway-admin.disable.local.login`	Default: false To enable, set to `true` For more information about setting up the PowerFlex Gateway admin login credentials for using LDAP server\servers, see the *PowerFlex User Roles and LDAP Usage Technical Notes*.	`gateway-admin.disable.local.login=true`
Change the name of the VASA Provider replica set. When there are three VASA Providers, a replica set is created	`vasa.provider.replica.set.name`	Default: `vasa-rs`	`vasa.provider.replica.set.name=vasa-rs`

Table 7. Use case details for the `gatewayUser.properties` file (continued)

Use case	Property	Action	Example
during deployment to facilitate communication between VASA Providers.			
Enable or disable monitoring of VASA Providers	`features.enable_vasa_monitor`	Default: true	`features.enable_vasa_monitor=true`
Set the list of IP addresses for VASA Providers	`vasa.provider.ips`	A comma-separated list of IP addresses of all the VASA Providers. The installation process sets this property, but it can also be set manually.	`vasa.provider.ips=<ip>,<ip>,...`
Set the port of the VASA Providers' REST service, in order to monitor the page `http://<ip>:<port>/actuator/health`	`vasa.provider.monitor.port`	Default: 8080	`vasa.provider.monitor.port=8080`
Set the number of seconds to sleep between each sample	`vasa.provider.monitor.interval.seconds`	Default: 60	`vasa.provider.monitor.interval.seconds=60`
When installing VASA in a VMware environment, set this property to false in order to enable extending the system with the PowerFlex Installer.	`vmware`	In a Linux environment, default: false. In a VMware environment, default: true When installing VASA in a VMware environment, set the value to `false`.	`vmware=false`
When CloudIQ is used, set the IP address of the PowerFlex presentation server.	`presentationserver.ip`	IP address of the PowerFlex presentation server that hosts the PowerFlex GUI	`presentationserver.ip=10.76.60.200`
When LIA uses LDAP authentication on a Linux-based LDAP server, configure username DN format. For Windows-based LDAP servers, do not use this property, because internal defaults are used.	`lia.ldap.initial.usernameDnFormat`	Values for `CN`, `OU` and `DC`. `CN` is the user assigned to LIA `OU` is the name of the organizational unit `DC` fields are the domain parts of the Base DN	`lia.ldap.initial.usernameDnFormat=CN=LIA_user,OU=group_1,DC=ldap,DC=com`
When LIA uses LDAP authentication on a Linux-based LDAP server, configure a search filter format. For Windows-based LDAP servers, do not use this property, because internal defaults are used.	`lia.ldap.initial.searchFilterFormat`	Values for the search filter format `CN` is the user assigned to LIA `memberOf` is the organizational unit to which the user is assigned `(&(objectClass=userSecurityInformation)` should be used as is shown in the example	`lia.ldap.initial.searchFilterFormat=(&(objectClass=userSecurityInformation)(cn=LIA_user)(memberOf=group_1))`

Table 7. Use case details for the `gatewayUser.properties` file (continued)

Use case	Property	Action	Example
When LIA uses LDAP authentication on a Linux-based LDAP server, this property allows the editing of an existing configuration for an LDAP server, instead of removing the configuration from the system and adding the LDAP server from scratch.	`lia.ldap.replaceExisting`	`true` allows editing of existing LDAP parameters	`lia.ldap.replaceExisting=true`
When SDCs are deployed in containers, and therefore no LIA is present, this property is required in order to ignore them during a non-disruptive upgrade.	`sdc.allowNoLiaOnSDCOnlyNode`	Default: false. Set this property to `false` when SDCs are not deployed in containers. Set this property to `true` when the PowerFlex environment runs with containerized SDCs (in other words, there is no LIA on these nodes).	`sdc.allowNoLiaOnSDCOnlyNode=true`

3. Save and close the file.
4. Restart the PowerFlex Gateway service:
   - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux: Type the following command:

```
service scaleio-gateway restart
```

Results

Configuration is complete.

Advanced and optional PowerFlex Gateway tasks

This section contains advanced, optional tasks for the PowerFlex Gateway, depending on your specific environment's setup.

Install the PowerFlex Gateway without assigning an admin password

You can install the PowerFlex Gateway without adding the `GATEWAY_ADMIN_PASSWORD` variable.

About this task

To add this variable later, perform one of the following:

Steps

1. Use a REST API command, as described in the *PowerFlex REST API Reference Guide*.
2. Edit the user properties file.
3. Restart the PowerFlex Gateway service:
   - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux: Type the following command:

```
service scaleio-gateway restart
```

Reset PowerFlex Gateway password

Reset the PowerFlex Gateway admin password.

Prerequisites

Run the FOSGWTool from the following location:

- Linux: `/opt/emc/scaleio/gateway/bin/FOSGWTool.sh`
- Windows: `C:\Program Files\EMC\ScaleIO\Gateway\bin\FOSGWTool.bat`

Steps

1. Log in as root user to the system on which the PowerFlex Gateway is running.
2. Run FOSGWTool in command line to reset the PowerFlex Gateway admin password:

   ```
   FOSGWTool --reset_password --password <PASSWORD> --config_file <PATH>
   ```

   - The `--password` is the new password for login to the PowerFlex Gateway
   - The `--config_file` is the full location path of the configuration file which is `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes/gatewayUser.properties`.
3. Restart the PowerFlex Gateway service:
   - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux: Type the following command:

     ```
     service scaleio-gateway restart
     ```

Results

The PowerFlex Gateway admin password is reset

Store Tomcat's keystore password in the lockbox

After PowerFlex Gateway installation or upgrade, Tomcat's keystore password is written to file in plain text. Optionally, if your organization requires it, follow this procedure to store it in the lockbox.

About this task

The PowerFlex Gateway's keystore password is stored as plain text in a configuration file, and the keystore contains the PowerFlex Gateway's TLS private key. To prevent the private key from being extracted unencrypted, perform the following procedure.

NOTE: This procedure requires a PowerFlex Gateway restart.

Steps

1. On the PowerFlex Gateway, open the `catalina.properties` file in the following location:
   - Linux: `/opt/emc/scaleio/gateway/conf`
   - Windows: `C:\Program Files\EMC\Scaleio\Gateway\conf`
2. Copy the *keystore.password* value.
   For example, for `keystore.password = ubSF2TqZgQpmLZU2GMvyyAdGgIkt1w`, copy `ubSF2TqZgQpmLZU2GMvyyAdGgIkt1w`.
3. Log in as a root user to the system on which the PowerFlex Gateway is running.
4. Run the following FOSGWTool command in command line:

   NOTE: If a lockbox does not exist yet, add the *--create_default_lockbox* flag to the command in order to create one.

   NOTE: You can overwrite an existing value using the *--overwrite* flag.

- Linux:

```
/opt/emc/scaleio/gateway/bin/FOSGWTool.sh --store_value_in_lockbox --key
lb.keystore.password --value <keystore.password value>
```

- Windows:

```
C:\Program Files\EMC\ScaleIO\Gateway\bin\FOSGWTool.bat
            --store_value_in_lockbox --key lb.keystore.password --value
<keystore.password value>
```

5. Edit the `server.xml` file:
   a. Using a text editor, open the file:
      - Linux: `/opt/emc/scaleio/gateway/conf/server.xml`
      - Windows: `C:\Program Files\EMC\Scaleio\Gateway\conf\server.xml`
   b. Change the entry `keystorePass="${keystore.password}"` to `keystorePass="${lb.keystore.password}"`, and save the file.

   The *${keystore.password}* reference has been replaced with a reference to the key name that you used to store the password in the lockbox.
6. Remove the plain text password from the `catalina.properties` file.
   a. Using a text editor, open the `catalina.properties` file mentioned in Step 1.
   b. Delete the password value for *keystore.password*, and save the file.
7. Restart the PowerFlex Gateway service:
   - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux: Type the following command:

```
etc/init.d/scaleio-gateway restart
```

Results

The plain text password has been moved from the Tomcat keystore to the lockbox.

PowerFlex Gateway installation with a custom Java configuration

About this task

To use a custom Java configuration:

Steps

Use the following command to install the PowerFlex Gateway, in place of the normal command:

```
GATEWAY_ADMIN_PASSWORD=<new_GW_admin_password> SIO_GW_JAVA=<path_to_java> rpm -i /tmp/
EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
```

Where *<new_GW_admin_password>* is a password that you define to access the PowerFlex Installer administration commands and web interface, and *<path_to_java>* is the path to the desired Java files.

Install Java on SUSE 12 servers

Install Java on SUSE 12 servers.

About this task

Installation of Java is different in SLES-based distributions because SLES uses update-alternatives commands. For SUSE, we use a TGZ file in place of RPM.

To install Java on SUSE 12 servers:

Steps

1. Untar the TGZ (for example, `jre-XXXX-linux-x64.tar.gz`) to `/usr/java`.

   This creates a directory of `/usr/java/jre1.X.0_XX/`.

2. Apply the std update-alternatives procedure (the following example shows use of Java 8 (`jre1.8.0_XX`)):

```
/usr/sbin/update-alternatives --install "/usr/bin/java" "java" "/usr/java/
jre1.8.0_XX/bin/java" 40
/usr/sbin/update-alternatives --config java
/usr/sbin/update-alternatives --install "/usr/bin/keytool” "keytool" "/usr/java/
jre1.8.0_XX/bin/keytool" 40
/usr/sbin/update-alternatives --config keytool
```

Certificate management for PowerFlex Gateway

This section explains how to replace the PowerFlex Gateway's self-signed security certificate with your organization's "trusted" certificate, and how to create a new "trusted" certificate. The PowerFlex Gateway automatically creates its own self-signed security certificate when it is installed or upgraded. If your organization has no special security certificate requirements, you can keep working with the default certificate.

Replace the default self-signed security certificate with your own trusted certificate

Create your own trusted certificate, and then replace the default certificate with the one that you created.

Steps

1. Find the location of `keytool` on your server, and open it.

   It is a part of the Java (JRE or JDK) installation on your server, in the `bin` directory. For example:

   - `C:\Program Files\Java\jdk1.8.0_XX\bin\keytool.exe`
   - `/usr/bin/keytool`

2. Generate your RSA private key:

```
keytool -genkey -alias <YOUR_ALIAS> -keyalg RSA -keystore <PATH_TO_NEW_KEYSTORE_FILE>
```

   a. If you want to define a password, add the following parameters to the command. Use the same password for both parameters.

```
-storepass <KEYSTORE_PASSWORD> -keypass <KEYSTORE_PASSWORD>
```

   NOTE: Specify a directory outside the PowerFlex Gateway installation directory for the newly created keystore file. This will prevent it from being overwritten when the PowerFlex Gateway is upgraded or reinstalled.

3. If you already have a Certificate Signing Request (CSR), skip this step.

   If you need a CSR, generate one by typing the following command. (If you did not define a keystore password in the previous step, omit the password flags.)

```
keytool -certreq -keyalg RSA -alias <YOUR_ALIAS> -file certreq.txt
-keystore <PATH_TO_NEW_KEYSTORE_FILE> -storepass <KEYSTORE_PASSWORD> -keypass
<KEYSTORE_PASSWORD>
```

4. If you already have an SSL certificate, skip this step.

   If you need an SSL certificate, use your CSR to obtain a new certificate from a third-party trusted SSL certificate provider. Save the certificate file on your server, outside the PowerFlex Gateway installation directory.

5. Import the Trusted Root, by typing this command. (If you did not define a keystore password, omit the password flags.)

```
keytool -import -alias root -keystore <PATH_TO_NEW_KEYSTORE_FILE> -trustcacerts
-file <LOCATION OF_YOUR_root.cer_FILE> -storepass <KEYSTORE_PASSWORD> -keypass
<KEYSTORE_PASSWORD>
```

NOTE: The certificate must be in x.509 format.

If a message appears saying that the root is already in the system-wide store, import it anyway.

6. Import the intermediate certificates, by typing the command. (If you did not define a keystore password, omit the password flags.)

```
keytool -import -alias intermediateCA -keystore <PATH_TO_NEW_KEYSTORE_FILE>
-trustcacerts -file <LOCATION_OF_YOUR_intermediate.cer_FILE> -storepass <keystore
password> -keypass <keystore password>
```

You must provide a unique alias name for every intermediate certificate that you upload with this step.

7. Install the SSL Certificate under the same alias that the CSR was created from (<YOUR_ALIAS> in previous steps), by typing the command (if you did not define a keystore password, omit the password flags):

```
keytool -import -alias <YOUR_ALIAS> -keystore <PATH_TO_NEW_KEYSTORE_FILE>
-trustcacerts -file <LOCATION_OF_SSL_CERTIFICATE> -storepass <keystore password>
-keypass <keystore password>
```

8. Edit the following items in the file *<POWERFLEX_GATEWAY_INSTALLATION DIRECTORY>*`\conf\catalina.properties`:
   a. `keystore.file=`*`<PATH_TO_NEW_KEYSTORE_FILE>`*
   b. `keystore.password=`*`<PASSWORD_DEFINED_DURING_KEYSTORE_CREATION>`*

   If you did not define a password, the default password is `changeit`.
9. Restart the PowerFlex Gateway service:
   - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux: Type the following command:

```
service scaleio-gateway restart
```

Replacement of the security certificate is complete.

Replace the default self-signed security certificate with your own self-signed certificate

Replace the default self-signed security certificate with your own self-signed security certificate.

Steps

1. Find the location of `keytool` on your server, and open it.

   It is usually a part of the Java (JRE or JDK) installation on your server, in the `bin` directory. For example:
   - `C:\Program Files\Java\jdk1.8.0_XX\bin\keytool.exe`
   - `/usr/bin/keytool`
2. Generate your RSA private key:

```
keytool -genkey -alias <YOUR_ALIAS> -keyalg RSA -validity 360 -keysize 2048 -keystore
<PATH_TO_NEW_KEYSTORE_FILE>
```

   a. If you want to define a password, add the following parameters to the command. Use the same password for both parameters.

```
-storepass <KEYSTORE_PASSWORD> -keypass <KEYSTORE_PASSWORD>
```

   NOTE: Specify a directory outside the PowerFlex Gateway installation directory for the newly created keystore file. This will prevent it from being overwritten when the PowerFlex Gateway is upgraded or reinstalled.
3. Edit the following items in the file *<POWERFLEX_GATEWAY_INSTALLATION DIRECTORY>*`\conf\catalina.properties`:
   a. `keystore.file=`*`<PATH_TO_NEW_KEYSTORE_FILE>`*

b. `keystore.password=<PASSWORD_DEFINED_DURING_KEYSTORE_CREATION>`

   If you did not define a password, the default password is `changeit`.

4. Restart the PowerFlex Gateway service:
   - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux: Type the following command:

```
service scaleio-gateway restart
```

Results

Replacement of the security certificate is complete.

Configure OpenStack interoperation with the PowerFlex Gateway

Configure the PowerFlex Cinder driver to verify the PowerFlex Gateway SSL certificate.

About this task

The OpenStack PowerFlex Cinder driver communicates with the PowerFlex Gateway through HTTPS (in other words, over SSL). By default, the driver ignores the gateway SSL certificate verification. However, the PowerFlex Cinder driver can be configured to verify the certificate.

NOTE: You can generate a self-signed certificate (`.PEM` file), using the keytool utility.

To enable certificate verification:

Steps

Add the following parameters to the file `/etc/cinder/cinder_scaleio.config` on the Cinder node:

- `verify_server_certificate=true`
- `server_certificate_path=<PATH_TO_PEM_FILE>`

Generate a self-signed certificate using the keytool utility

Generate self-signed certificates using the keytool utility. The certificates can by used by the OpenStack PowerFlex driver to communicate with the PowerFlex Gateway.

About this task

To generate a self-signed certificate using the keytool utility, perform the following steps:

Steps

1. Create a keystore file (`.JKS`):

```
keytool -genkeypair -keysize 1024 -alias herong_key -keypass keypass -keystore
herong.jks -storepass jkspass
```

2. Export the keystore file to a `.PEM` file:

```
keytool -exportcert -alias herong_key -keypass keypass -keystore herong.jks
-storepass jkspass -rfc -file keytool_crt.pem
```

   The certificate is stored in the file `<keytool_crt.pem>`. During configuration of the Cinder driver, the path to this `.PEM` file is required.

Using SSH authentication on the PowerFlex Gateway

Create SSH keys for servers in a new or existing PowerFlex system.

Prerequisites

Lockbox must be configured on the PowerFlex Gateway.

About this task

A manually generated public-private key pair can be used to perform SSH key authentication, instead of passwords, between the PowerFlex Gateway and PowerFlex system servers. The public key is placed on all servers that will allow access to the PowerFlex Gateway. The PowerFlex Gateway is the owner of the matching private key.

FOSGWTool is used to upload the private key to the PowerFlex Gateway. FOSGWTool is located in:

- Linux: `/opt/emc/scaleio/gateway/bin/FOSGWTool.sh`
- Windows: `C:\Program Files\EMC\ScaleIO\Gateway\bin\FOSGWTool.bat`

NOTE: For systems using SSH key authentication, there is no need to add node user name and password credentials to the CSV file used for system deployment.

Steps

1. Create the private and public key pair on the node where the PowerFlex Gateway is installed:
   a. In command line, run the command:

   ```
   cd ~/.ssh/
   ```

   b. Run the command:

   ```
   ssh-keygen -t rsa
   ```

   This command generates a private and public key. When performing this command, you can add a passphrase, or generate the key without a passphrase. Two SSH keys are produced: `id_rsa` and `id_rsa.pub`.
2. On one of the PowerFlex servers, run this command to store the public key in the `authorized_keys` file:

   ```
   cat ~/.ssh/id_rsa.pub | cat >> ~/.ssh/authorized_keys
   ```

3. Copy the `.ssh` directory to the rest of the PowerFlex servers.
4. Copy `id_rsa.pub` to the file system on the rest of the PowerFlex servers.
5. Use FOSGWTool in command line to upload the private key to thePowerFlex Gateway:

   ```
   FOSGWTool --update_ssh_key --ssh_key <PATH_TO_SSH_KEY_FILES> [--key_passphrase
   <KEY_PASSPHRASE>]
   ```

   The `--key_passphrase` is optional. Add this if you generated it when you created the key pair. During deployment, the gateway will use the private key for authentication.

6. Restart the PowerFlex Gateway service:
   - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux: Type the following command:

   ```
   service scaleio-gateway restart
   ```

Preparation tasks Oracle Linux

Steps that must be taken when deploying SDC or RFcache on SLES, Oracle Linux (OL), or Ubuntu servers.

Before you begin:

Ensure that the PowerFlex SDC and RFcache (RF) drivers are compatible with the Linux kernel.

NOTE: All references to Ubuntu in this section apply equally to supported versions of SLES and OL.

To ensure that a client node always has an SDC/RF kernel driver that matches the client kernel, the SDC/RF startup script uses `driver_sync.sh`, a synchronization utility that fetches SDC/RF kernel drivers from a configured repository. This repository can be either the Dell Technologies repository, or a local repository in your private network.

For Linux nodes with PowerFlex SDC/RF drivers:

- The `driver_sync.sh` script is located under `<base_dir>`.
- In the repository, a driver package tarball is located at `<base_dir>/driver_cache/<distribution>/<PowerFlex_version>/<kernel_version>/<driver>.tar`.

Where:

- `<base_dir>` is `/bin/emc/scaleio/<driver>_sync` by default.
- `<distribution>` is one of theseCentosStream8, CentosStream9, CoreOS, RH, RHEL8, RHEL9, SLES, SLES15.3, Ubuntu, Ubuntu16.04, Ubuntu18.04, Ubuntu20.04, or Ubuntu22.04.
- `<kernel_version>` is the output of uname -r.
- `<driver>` is scini for SDC or xcache for RF.

This is relevant for all supported Linux distributions, and mandatory for SLES, OEL, and Ubuntu. It may be necessary to manually restart the driver/s and services on hosts.

In a normal work flow, an SDC/RF driver is loaded on boot. If there is a mismatch between the version of the running kernel and the kernel against which the driver was compiled, the `driver_sync.sh` script is run automatically to fetch the appropriate driver. If the script fails for any reason (for example, due to incorrect configuration) after fixing the issue with the `driver_sync.sh` script, the driver services must be restarted manually, as described below. Root permission is required to run the commands.

To manually restart driver/s and services, run these commands:

```
service scini restart
```

```
service xcache restart
```

Ensure that a driver is available for the running kernel

Before installing PowerFlex, ensure that the kernel version is correct by performing the following steps.

Steps

1. Ensure that your kernel version matches a version in the repository, by performing the following:

   NOTE: If all servers have the same kernel, you can run this command on a single server.

   a. Run this command on every Ubuntu server:

   ```
   uname -r
   ```

   Output similar to the following is displayed:

   ```
   3.16.0-62-generic
   ```

   b. Copy the following text into a browser, and compare the output to the Ubuntu kernel version in the PowerFlex repository:

   ```
   https://mft.dell.com/
   ```

   Username:`PowerFlexRepo`

   Password:`V3Die83$BK`

   NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. To use a mirror repository, you must ensure that the SSH public and SSH private keys exist in all system nodes in the same path.

   The private key should have directory-level permission only (chmod 600 <private_key_path>).
3. The GPG key must be located in all system nodes in the same path. This key is required for using a mirror or Dell Technologies repository.

Next steps

The configuration of `driver_sync.sh` is specified in `driver_sync.conf` file, which can be created using the following methods:

- Create the configuration file using the PowerFlex Gateway
- Create the configuration file manually

NOTE: The SDC package now supports automated installation on Linux operating systems with SELinux. For supported operating systems, see the support matrix. To ensure that SELinux is activated and that SDCs are deployed as expected, verify that the relevant pre-requisite packages are installed. For more information, see the operating system requirements in the *Getting to Know Dell PowerFlex Guide*.

Create the configuration file using the PowerFlex Gateway

The `driver_sync.conf` file can be created during the PowerFlex Gateway deployment process. This topic is relevant for deployments on Linux flavors other than RedHat. Before deploying, you set parameters that will be used by the PowerFlex Gateway, during deployment, to create the file for your environment.

About this task

To edit the properties, perform the following:

Steps

1. Use a text editor to open the `gatewayUser.properties` file, located in the following directory on the PowerFlex Gateway server:

   `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes`
2. Edit the file by adding the parameters described below.
3. Save and close the file.
4. Import the GPG key by running the following command, on every Ubuntu node that will contain SDC or RFcache:

   ```
   apt-key add <new_public_key>
   ```
5. Restart the PowerFlex Gateway service:

   ```
   service scaleio-gateway restart
   ```

   After the changes are made, use the PowerFlex Installer to deploy the system, using the installation files for Ubuntu.

Parameters to add to the driver sychronization utility

Add the following parameters to the driver synchronization utility.

Steps

1. Validate that the repository contains the kernel your operating system requires.

   ```
   sftp://mft.dell.com/
   ```

   Or

   ```
   https://mft.dell.com/
   ```

NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. Download `sftp_keys_pkb.tar` from `<location needed>` and extract the contents to your SDC/RF host, as user root.
3. Place scini_repo_key.pub at the location specified in repo_public_rsa_key, `/etc/emc/scaleio/<driver>_sync/scini_repo_key.pub` by default.
4. Place scini_key at the location specified in user_private_rsa_key, `/etc/emc/scaleio/<driver>_sync/scini_key` by default. Where `<driver>` is scini for SDC or xcache for RF.
5. Change the permissions of a private key file.

```
chmod 600 /etc/emc/scaleio/scini_sync/scini_key
```

6. Edit /etc/emc/scaleio/scini_sync/driver_sync.confsimilar to the example below:

```
repo_address = sftp://mft.dell.com
repo_user = PowerFlexRepo
user_private_rsa_key = /etc/emc/scaleio/scini_sync/scini_key
repo_public_rsa_key = /etc/emc/scaleio/scini_sync/scini_repo_key.pub
```

NOTE: The `repo_password` line is not used with SFTP and can be left with the default value. Use the real password supplied above to check driver availability.

7. Restart the Storage Data Client RFCache driver:
   - Option1: Run from the operating system command prompt:

     ```
     systemctl restart <driver>
     ```

     NOTE: You can restart the Storage Data Client from the operating system only if the operating system is not using any volumes.
   - Option2: If needed, plan for a reboot cycle of the operating system for this setting to take effect.
8. Check that the `<driver>` is running:

```
systemctl status <driver>
```

9. Once you are finished, perform an operating system update and reboot to load the new kernel. The Storage Data Client driver sync automatically downloads the kernel object (KO) file, if available.

Create the configuration file manually

The `driver_sync.conf` file can be created or edited manually using a text editor. This topic is relevant for deployments on Linux flavors other than Red Hat.

About this task

If PowerFlex was deployed with the PowerFlex Gateway, the file exists in the `etc` folder, with dummy values, and you must edit that file, and then copy it to the `etc` folder on every SDC/RFcache (RF) server in the system. The `etc` folder is located in:

- SDC

Table 8. SDC file location

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/scini_sync/driver_sync.conf`

- RFcache

Table 9. RF cache location

Host	Path
Ubuntu/OL	/etc/emc/scaleio/xcache_sync/driver_sync.conf

The configuration of the driver_sync script is read from driver_sync.conf (an example appears below), which contains the following parameters:

Table 10. driver_sync.conf parameters

Parameter	Description
repo_address	Encodes the chosen method and address.
repo_user	Identifies the username that is used when logging in to the drivers server. Required for SFTP and FTP methods.
repo_password	The FTP server login password (for user repo_user). Required for FTP method
user_private_rsa_key	Contains the path of the private RSA key file that is used to SFTP connection. This key file (provided by PowerFlex on installation) enables SFTP connection without a password.
local_dir	The local directory in which the downloaded SDC/RF drivers are stored. For SDC/RF clients, this value is set up during installation and does not require a change.
repo_public_rsa_key	The path of the public RSA key of the repository server machine (also known as the host public key).
module_sigcheck	Determines whether to verify (1) or not to verify (0) the downloaded kernel drivers.
emc_public_gpg_key	A key that is used to verify signatures of downloaded drivers. Enter the path to the public GPG signing key of PowerFlex (like the one that comes with PowerFlex LIA). This is needed only if module_sigcheck is set to 1.
The following parameters are used when driver_sync.sh is used to synchronize a local repository against a remote one.	
sync_pattern	A regular expression that can be used to fetch several drivers at once. This parameter is not needed, and becomes relevant only when your own private repository uses driver_sync.sh to fetch from a remote (For example: Dell EMC) repository.
is_local_repo	Set to 0 (default) or omit, for SDC/RF clients. Set to 1 when driver_sync.sh is used to synchronize local repository against a remote repository. Setting to 1 causes the fetched tar files not to be extracted.

Example:

```
################################################
#driver_sync Configuration file
#Everything after a '#' until the end of the line is ignored
################################################
#Repository address, prefixed by protocol
repo_address = sftp://localhost/path/to/repo/dir
#repo_address = ftp://localhost/path/to/repo/dir
#repo_address = file://local/path/to/repo/dir
# Repository user (valid for ftp/sftp protocol)
repo_user = scini
# Repository password (valid for ftp protocol)
repo_password = scini
# Local directory for modules
local_dir = /bin/emc/scaleio/scini_cache/
# User's RSA private key file (sftp protocol)
user_private_rsa_key = /bin/emc/scaleio/scini_key
# Repository host public key (sftp protocol)
repo_public_rsa_key = /bin/emc/scaleio/scini_repo_key.pub
# Should the fetched modules' signatures be checked [0, 1]
module_sigcheck = 1
# EMC public signature key (needed when module_sigcheck is 1)
emc_public_gpg_key = /bin/emc/scaleio/emc_key.pub
# Sync pattern (regular expression) for massive retrieve
sync_pattern = .*
```

Perform the following steps for updating the driver_sync file on the Storage Data Client:

Steps

1. Validate that the SFTP contains the kernel your operating system requires.

```
sftp://mft.dell.com/
```

Or

```
https://mft.dell.com/
```

NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. Download and extract the attached sftp_keys to your SDC host, as user root, to /etc/emc/scaleio/scini_sync. Find the file that is attached to the article: sftp_keys_pkb.tar.
3. Change the permissions of a private key file.

```
chmod 600 /etc/emc/scaleio/scini_sync/scini_key
```

4. Edit /etc/emc/scaleio/scini_sync/driver_sync.confsimilar to the example below:

```
repo_address = sftp://mft.dell.com
                                    repo_user = PowerFlexRepo
                                    user_private_rsa_key = /etc/emc/scaleio/
scini_sync/scini_key
                                    repo_public_rsa_key = /etc/emc/scaleio/scini_sync/
scini_repo_key.pub
```

NOTE: The repo_password line is not used with SFTP and can be left with the default value. Use the real password supplied above to check driver availability.

5. Restart the Storage Data Client driver:
   - Option1: Run from the operating system command prompt:

```
systemctl restart scini
```

NOTE: This is possible only if no volumes are in use by the operating system.

   - Option2: If needed, plan for a reboot cycle of the operating system for this setting to take effect.
6. Check that the `sciniservice` is running:

```
systemctl status scini
```

7. Once you are finished, perform an operating system update and reboot to load the new kernel. The Storage Data Client driver sync automatically downloads the kernel object (KO) file, if available.

Update the PowerFlex signature key

If the PowerFlex package-signing key expires, you must update the configuration for each SDC or SDS node that uses RFcache.

About this task

When a PowerFlex package-signing key expires, a new public key is issued. You must update `driver_sync.conf` to use the new public key.

Steps

1. Download the new public key from the Dell support site.

2. On each SDC or RFcache node, edit `driver_sync.conf` so that the value of `emc_public_gpg_key` is the path to the new key file.

   The expired public key will still be in the gpg keyring, and older packages that were signed with the old key can still be verified.

   Location of `driver_sync.conf`:

   - SDC

     Table 11. SDC location

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/scini_sync/driver_sync.conf`

   - RFcache

     Table 12. RF cache location

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/xcache_sync/driver_sync.conf`

Create a mirror repository

For Ubuntu or Oracle Linux (OL, also known as OEL), set up a local mirror repository (optional).

This section describes how to set up a local repository. This is necessary for scenarios when connecting all SDC/RFcache (RF) nodes to the PowerFlex repository is not possible, perhaps due to security considerations or other limitations.

A common workflow would be to configure your mirror repository to synchronize with the PowerFlex repository, while your SDC/RF nodes will fetch drivers (synchronize) from the mirror repository.

Driver repository directory hierarchy

The directory hierarchy in the repository consists of the following levels:

<Distro>/<PowerFlex *Version*>/*<Kernel Version>*/

Where:

- *<Distro>* is either `Ubuntu` or `OL` (for Oracle Linux)
- <PowerFlex *Version*> is encoded as `x.x.x.x`
- *<Kernel Version>* is the output of `uname -r`.

The files themselves are tar files named `scini.tar` and `xcache.tar`.

For example:

<Repo_base>/OL/3.6-<build>/X.X.X-OL/scini.tar is a tar file of the SDC driver for PowerFlex version 3.6-<build> for Oracle Linux with kernel version X.X.X-OL.

Synchronize the mirror repository

Synchronizing is performed by calling `driver_sync.sh sync`.

In normal workflow, you do not have to (and should not) modify the directory structure manually. Instead, synchronizing is done by simply calling `driver_sync.sh sync`.

In the `driver_sync.conf` file, you must set the `is_local_repo` parameter to **1**. This will instruct `driver_sync.sh` not to untar the downloaded files.

It is also recommended that you change the `sync_pattern` parameter, a regular expression, to something more specific than `".*"`. Otherwise the sync operation will mirror the entire Dell EMC driver repository every time.

SFTP-based repository setup script

To save time in setting up a new SFTP-based driver repository, use the `driver_repo_wizard.sh` script.

The `driver_repo_wizard.sh` script can be found in the following locations on Ubuntu or Oracle Linux-based servers:

- :
  - `/bin/emc/scaleio/scini_sync`
  - `/bin/emc/scaleio/xcache_sync`

The script does the following:

1. Creates a new user with a name you provide. The user will be created shell-less.
2. Creates a pair of RSA keys that will be used for passwordless SSH connection. You must provide the created private key to all your client nodes, while the public key will be added to the `authorized_keys` file of the created user.
3. Updates the SSH service configuration file (`/etc/ssh/ssh_config`) for the new user, instructing to `chroot` the user to its home directory, which should also be the repository base directory.

Prepare the CSV file

Create a CSV topology file to deploy PowerFlex with the PowerFlex Installer on Linux or Windows-based nodes. Follow the instructions that match your environment and the PowerFlex support matrix for your version. CSV file templates are provided as part of the software download, in the PowerFlex Gateway software packages. They are also accessible from the PowerFlex Installer user interface.

Prerequisites

Ensure that the settings you enter comply with PowerFlex product limits, as described in the "Product limits" table in the *Getting to Know PowerFlex guide*.

Ensure that you have a copy of the following CSV templates, which are provided in the PowerFlex Gateway software packages. Decide which template is most suitable for your needs. In this document, we refer to and illustrate the CSV file as a spreadsheet.

- Complete—this spreadsheet template contains all available fields, both required and optional. In particular, it contains an optional section for Storage Pool configuration, which is relevant for multiple nodes.
- Minimal—this spreadsheet template contains only the required fields. The optional fields (those that are in the complete spreadsheet, but not in the minimal one) are assigned default values.

About this task

NOTE: During CSV file preparation, note that there are important differences between hyperconverged and two-layer topologies.

- In hyperconverged topologies, each row in the upper section represents a server that can host MDM, SDS, and SDC components
- In two-layer topologies where frontend servers are Linux or Windows-based, SDCs must be represented by separate, additional rows. The values for SDC-specific rows are described in "CSV topology for two-layer deployment". Alternatively, omit the SDCs from the CSV file, and install them manually after deploying the backend.
- In two-layer topologies where frontend servers are ESXi-based, omit them from the CSV file. In this case, the SDCs are installed after deploying the backend, either manually, or using the vSphere PowerFlex plug-in.

NOTE: Save a copy of the CSV file for future use. It is recommended to save the file in a secure encrypted folder on your operating system, after deployment. When you add nodes to the system in the future, you need the CSV file that you used for initial deployment of the system.

You can edit a CSV file with Microsoft Excel or other file-editing software.

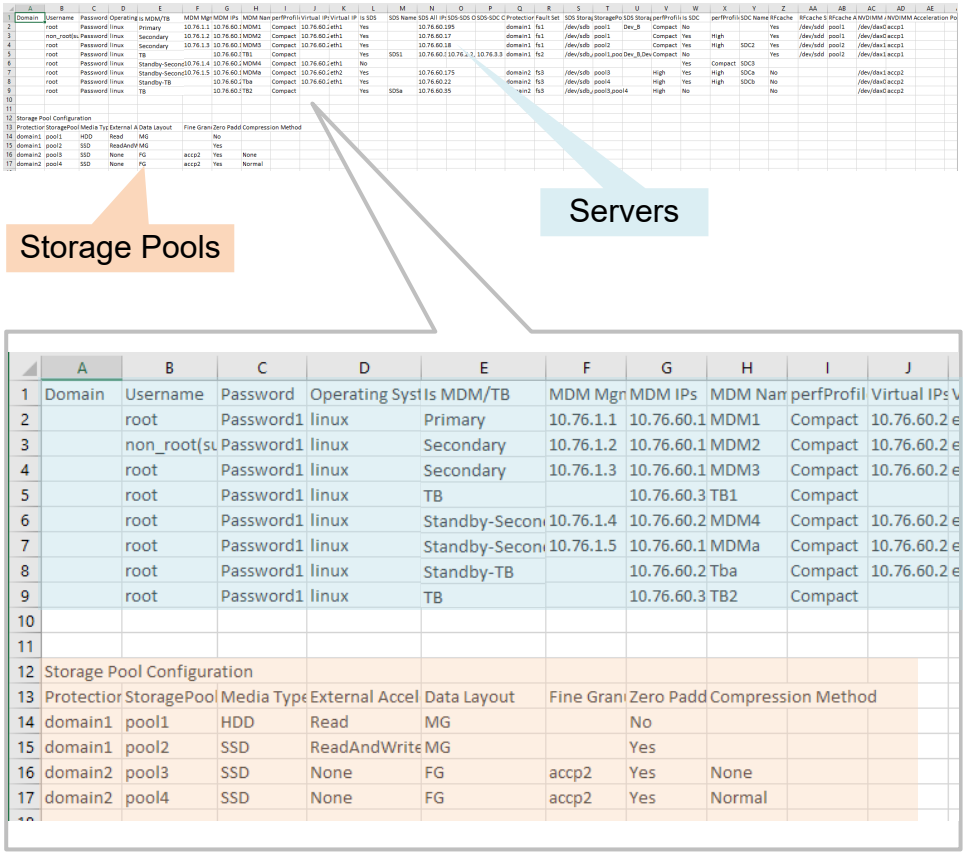
Steps

1. In the CSV file, enter your site-specific information in the appropriate places, overwriting the default information provided in the file.

   You only need to use one spreadsheet for the installation, as follows:

   - To manually enter all configuration details, use the Complete spreadsheet.
   - To use default values for the non-mandatory fields, use the Minimal spreadsheet.
   - To configure non-default values for columns that are not in the Minimal spreadsheet, either use the Complete spreadsheet, or copy the column heading from there into the Minimal spreadsheet, and enter your custom values into the Minimal spreadsheet.

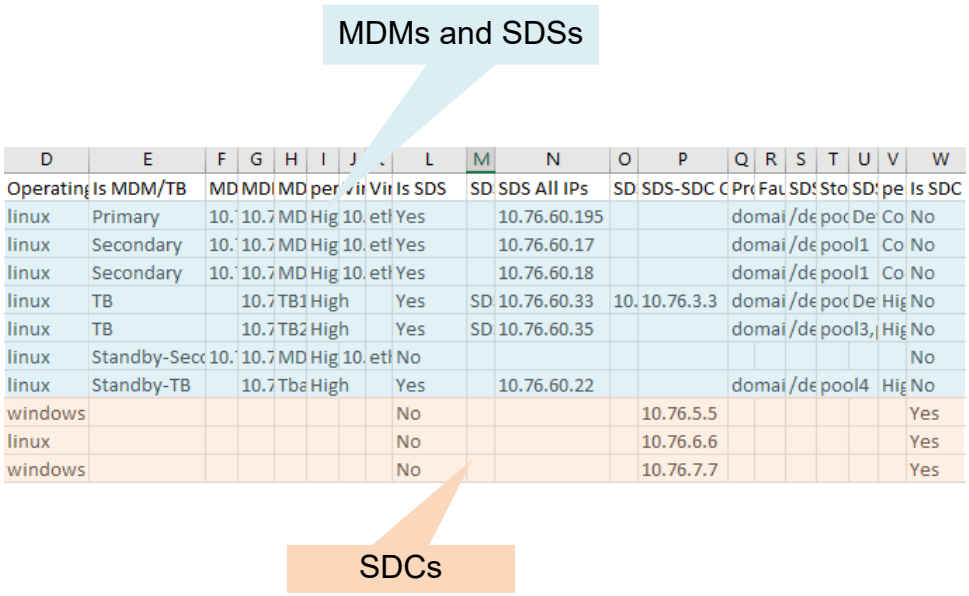
   The following figure shows part of a sample Complete CSV file used to deploy Linux nodes in a hyper-converged system:



	A	B	C	D	E	F	G	H	I	J
1	Domain	Username	Password	Operating Syst	Is MDM/TB	MDM Mgr	MDM IPs	MDM Nan	perfProfil	Virtual IPs
2		root	Password1	linux	Primary	10.76.1.1	10.76.60.1	MDM1	Compact	10.76.60.2 e
3		non_root(su	Password1	linux	Secondary	10.76.1.2	10.76.60.1	MDM2	Compact	10.76.60.2 e
4		root	Password1	linux	Secondary	10.76.1.3	10.76.60.1	MDM3	Compact	10.76.60.2 e
5		root	Password1	linux	TB		10.76.60.3	TB1	Compact	
6		root	Password1	linux	Standby-Secon	10.76.1.4	10.76.60.2	MDM4	Compact	10.76.60.2 e
7		root	Password1	linux	Standby-Secon	10.76.1.5	10.76.60.1	MDMa	Compact	10.76.60.2 e
8		root	Password1	linux	Standby-TB		10.76.60.2	Tba	Compact	10.76.60.2 e
9		root	Password1	linux	TB		10.76.60.3	TB2	Compact	
10										
11										
12	Storage Pool Configuration									
13	Protectior	StoragePool	Media Type	External Accel	Data Layout	Fine Grani	Zero Padd	Compression Method		
14	domain1	pool1	HDD	Read	MG		No			
15	domain1	pool2	SSD	ReadAndWrite	MG		Yes			
16	domain2	pool3	SSD	None	FG	accp2	Yes	None		
17	domain2	pool4	SSD	None	FG	accp2	Yes	Normal		

Figure 2. CSV—Complete example

The following figure shows a close up of a sample Complete CSV file, where SDCs are defined below the rows for MDMs and SDSs. In a two-layer configuration, the Linux servers form the MDM cluster, and each of those servers will also be SDSs. SDCs can be added here, in separate rows, or added to the system later.



D	E	F	G	H	I	J	L	M	N	O	P	Q	R	S	T	U	V	W
Operating	Is MDM/TB	MD	MDI	MD	per	Vir	Is SDS	SD	SDS All IPs	SD	SDS-SDC C	Pr	Fau	SDS	Sto	SD	pe	Is SDC
linux	Primary	10.	10.7	MD	Hig	10.	Yes		10.76.60.195			domai	/de	poc	De		Co	No
linux	Secondary	10.	10.7	MD	Hig	10.	Yes		10.76.60.17			domai	/de	pool1			Co	No
linux	Secondary	10.	10.7	MD	Hig	10.	Yes		10.76.60.18			domai	/de	pool1			Co	No
linux	TB		10.7	TB1	High		Yes	SD	10.76.60.33	10.	10.76.3.3	domai	/de	poc	De		Hig	No
linux	TB		10.7	TB2	High		Yes	SD	10.76.60.35			domai	/de	pool3,			Hig	No
linux	Standby-Secc	10.	10.7	MD	Hig	10.	No											No
linux	Standby-TB		10.7	Tba	High		Yes		10.76.60.22			domai	/de	pool4			Hig	No
windows							No				10.76.5.5							Yes
linux							No				10.76.6.6							Yes
windows							No				10.76.7.7							Yes

Figure 3. CSV—Two-layer example

2. For Linux nodes, the **Domain** column is not relevant. Leave the column blank or remove the column.

   The following table describes the fields in the spreadsheets. The required fields appear in both complete and minimal spreadsheets. Field names are case-sensitive; the order of the columns is not significant.

Table 13. CSV topology spreadsheets

Field	Description	Required
Domain	If using a domain user, the name of the domain (not relevant for Linux)	-
Username	The name of the user. • Linux nodes: Enter either `root` or a non-root sudo user, indicated by appending "(sudo)" to the end of the username (default: `root`). For more information about deploying with a non-root user, see "Deployment of a non-root user". • Windows nodes: Enter a user with administrator rights (default: `administrator`).	-
Password	Password used to log in to the node. This should be the password which the user entered in the **Username** column. To authenticate with SSH instead of node passwords, see "Using SSH authentication on the PowerFlex Gateway.	Yes
Operating System	The operating system of the server: `linux` or `windows`	Yes
IPs	IP address to be used for multiple purposes. Use this field to designate one IP address that will be assigned to all the following: MDM IP, MDM Mgmt IP and SDS All IP. This option is provided for use cases where separate networks for data and management are not required.	-
Is MDM/TB	The MDM role to deploy on this node: `Primary`, `Secondary`, `TB`, `Standby-Secondary`, `Standby-TB`, or blank (if not an MDM). For more information, see "The MDM cluster" in the Architecture section of *Getting to Know PowerFlex*.	Yes
MDM Mgmt IP	The IP address for the management-only network. The management IP address is not required for: Tiebreaker, Standby Tiebreaker, and any host that is not an MDM. In such cases, leave this field blank.	-

Table 13. CSV topology spreadsheets (continued)

Field	Description	Required
MDM IPs	MDM IP addresses used to communicate with other PowerFlex components in the storage network. This is required for all MDMs, Tiebreakers and Standbys. Leave this field blank for hosts that are not part of the MDM cluster. Maximum of eight addresses, comma-separated, no spaces.	-
MDM Name	The name to be assigned to an MDM node, up to 31 characters in length. MDM names must be unique and may not contain spaces. Ensure that each server on which MDM is installed has a unique hostname.	-
perfProfileForMDM	Optional performance profile to set for MDMs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the Fine Granularity data layout feature.	-
Virtual IPs	A virtual IP address (VIP) for each possible manager MDM. This virtual IP address can be used for communication between the MDM cluster and SDCs on the PowerFlex data networks (not applicable for the management network). Only one virtual IP address can be mapped to each NIC, with a maximum of four virtual IP addresses per system.	-
Virtual IP NICs	The NIC to which the virtual IP addresses are mapped. Maximum length of 11 characters.	-
Is SDS	Deploy the SDS component on this node: `Yes` or `No`	Yes
SDS Name	The name for the SDS node, up to 31 characters in length. The name must not contain spaces. Ensure that each server on which SDS is installed has a unique hostname.	-
SDS All IPs	SDS IP addresses to be used for communication among all nodes. Maximum of eight addresses, comma-separated, no spaces. **NOTE:** When using this option, all roles use the same IP addresses, and there is no need to fill the SDS\SDC only columns.	-
SDS-SDS Only IPs	SDS IP addresses to be used for communication among SDS nodes. When the replication feature is used, these addresses are also used for SDS-SDR communication. Maximum of eight addresses, comma-separated, no spaces.	-
SDS-SDC Only IPs	SDS IP addresses to be used for communication among SDS and SDC nodes only. Maximum of eight addresses, comma-separated, no spaces. For SDC-only nodes, enter the IP address in this column.	-
Protection Domain	The Protection Domain to which to assign this SDS or SDR. The name can be up to 31 characters in length.	-
Fault Set	The Fault Set to which to assign this SDS. **NOTE:** When using Fault Sets, you must assign a minimum of three, and follow the guidelines described in “Fault Sets” in the Architecture section of *Getting to Know PowerFlex*.	-
SDS Storage Device List	Storage devices to be added to an SDS. For more than one device, use a comma-separated list, with no spaces. Ensure that the device names and paths do not contain any special characters, such as: `?, ` ` `, !, @, #` Ensure that devices are prepared as described in "Configuring direct attached storage" in the Architecture section of *Getting to Know PowerFlex*. Device name format on Linux: `/dev/sdb,/dev/sdc`	-

Table 13. CSV topology spreadsheets (continued)

Field	Description	Required

	When specifying the SDS device path on a Linux node, use the path according to how it is listed in `cat/proc/partitions` (and not according to the output of `fdisk -l`). Use persistent device names.
	For example:
	`fdisk` output: `/dev/mapper/samah-lv1, /dev/sdb`
	`cat /proc/partitions` output: `dm-3, sdb`
	Use these values in the CSV file: `/dev/dm-3, /dev/sdb`
	To enable volume creation, you must add (at least) one device to (at least) three SDSs, where each SDS is in a separate Fault Set, and each device has a minimum of 130 GB free storage capacity (an SDS which has not been assigned to a Fault Set is treated as a Fault Set of its own). You can do that using the CSV file, or at a later stage. The maximum number of devices per SDS is listed in the "Product limits" table in the *Getting to Know PowerFlex* guide.
	Device data is erased when devices are added to an SDS. When adding a device to an SDS, PowerFlex checks that the device is clear before adding it. An error will be returned, per device, if it is found not to be clear.
	PowerFlex might not perform optimally if there are large differences between the sizes of the devices in the Storage Pool—for example, if one device is as big as the rest of the devices. After adding devices, you can define how much of the device capacity is available to PowerFlex by using the SCLI command `modify_sds_device_capacity`.
	For optimal performance, try to balance the number of devices of a Storage Pool, and the capacity of those devices among the relevant SDSs.
	When adding devices that were used in a previous PowerFlex system, follow the instructions in "Prepare previously used SDS devices".	

StoragePool List	Sets Storage Pool names, up to 31 characters in length. Use this option in one of the following ways to assign Storage Pools to the devices in the **SDS Storage Device List**: • A comma-separated list of StoragePool names; the length of the list must be the same length as the list of devices in the **SDS Storage Device Names** list. In this case, the Storage Pools will be mapped to the devices in that list, respectively. The list must be comma-separated, with no spaces. • If one Storage Pool is entered here, the same Storage Pool will be mapped to all the devices in the **SDS Storage Device List**. • If no Storage Pool is listed here, one will be automatically created during installation, named `default`.	-
SDS Storage Device Names	Sets names for devices. Assigns device names to the devices in the **SDS Storage Device List**, respectively. The amount of names must equal the amount of devices. Comma-separated, no spaces.	-

SDS Storage Device Path	The length of the SDS device path must not exceed 63 characters. For example, this path for the following NVMe disk under `/dev/disk/by-id` is not supported because it is too long:
	`/dev/disk/by-id/Dell_Express_Flash_NVMe_PM1725_1.6TB_SFF_____S2JPNA0J500141`
	Alternatively, this equivalent name is supported:
	`/dev/disk/by-id/nvme-eui.002538957100082e`
		-

Table 13. CSV topology spreadsheets (continued)

Field	Description	Required
perfProfileForSDS	Optional performance profile to set for SDSs: `High` (default) or `Compact` When this field is left empty, the default option is applied. High performance is required for the Fine Granularity data layout feature. For more information, see "Configure performance profile during deployment".	-
Is SDC	`Yes` or `No`.	Yes
perfProfileForSDC	Optional performance profile to set for SDCs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the Fine Granularity data layout feature. For more information, see "Configure performance profile during deployment".	-
SDC Name	Sets a name for the SDC, which can be up to 31 characters in length. Ensure that each server on which SDC is installed has a unique hostname.	-
RFcache	Install the RFcache `xcache` package on an SDS: `Yes` or `No` (default). **NOTE:** If you wish to configure RFcache devices as part of the deployment, set this column to `Yes`, **RFcache SSD Device List** column to a list of comma-separated devices, and the **RFcache Accelerated StoragePool List** to a corresponding comma-separated list of Storage Pools. **NOTE:** Ensure that the RFcache feature is supported by the Linux Kernel version used in your system. For more information, refer to the *Getting to Know Dell PowerFlex Guide*.	-
RFcache SSD Device List	List of SSD devices to provide RFcache acceleration for Medium Granularity data layout Storage Pools. Up to eight devices, comma-separated, with no spaces. Ensure that the device names and paths do not contain any special characters, such as: `?, `, !, @, #` If RFcache is `Yes`, and this field is left blank, you can add RFcache devices after installation. **NOTE:** The RFcache feature is only supported by 512 Byte devices for the cache use case. RFcache is not supported on 4K devices used for cache. However,RFcache acceleration of 4K storage devices is supported.	-
RFcache Accelerated StoragePool List	The names of the Storage Pools for which RFcache acceleration is to be provided. The amount of StoragePool names must equal the amount of devices in the **RFcache SSD Device List**. Comma-separated, no spaces. The specified Storage Pools must be defined in **StoragePool List**.	-
NVDIMM Acceleration Device List	List of NVDIMM devices used to provide acceleration for Storage Pools using Fine Granularity data layout. Up to six devices, comma-separated, with no spaces. Ensure that the NVDIMM devices are configured for DAX mode.	-
NVDIMM Acceleration Pool List	Sets NVDIMM Acceleration Pool names. Assigns NVDIMM Acceleration Pools to the devices in the **NVDIMM Acceleration Device List**, respectively. The amount of NVDIMM Acceleration Pools must equal the amount of devices. Comma-separated, no spaces.	-
Is SDR	Deploy the SDR component on this node: `Yes` or `No` (default). A minimum of two SDRs per peer system must be deployed for the replication feature. SDRs must be deployed on nodes that also contain the SDS component.	-
SDR Name	The name of the SDR node, up to 31 characters in length. The name must not contain spaces. Default: empty field.	-
SDR Port	The port used by the SDR. Default: 11088	-

Table 13. CSV topology spreadsheets (continued)

Field	Description	Required
SDR Application IPs	The IP addresses through which the SDC communicates with the SDR. When an application writes data, it will work with the SDR through these IP addresses. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for Application, Storage, External. IPv6 is supported.	-
SDR Storage IPs	The IP addresses through which the SDR communicates with the MDM for server side control communications. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for Application, Storage, External. IPv6 is supported.	-
SDR External IPs	The IP addresses through which the SDR communicates with peer systems' SDRs. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for Application, Storage, External. IPv6 is supported.	-
SDR All IPs	SDR IP addresses to be used for communication among all nodes (including all three roles) Maximum of eight addresses, comma-separated, no spaces. This is the default setting.	-
perfProfileForSDR	Optional performance profile to set for SDRs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the Fine Granularity data layout feature. For more information, see "Configure performance profile during deployment".	-
Is Vasa Provider	A VASA provider is used (for VMware vSphere Virtual Volumes (vVols) support): `Yes` means that the node contains a VASA provider. Any other value means that it does not contain one. A cluster may contain a single VASA provider, three VASA providers, or none. VASA providers must be installed on nodes where SDC is also installed. **NOTE:** You can only add VASA providers to a system that previously contained no VASA providers. You must begin from a state where no VASA providers are installed in the cluster.	-
Vasa Provider IP	The IP address of the VASA provider **NOTE:** If no IP address is defined in this column, the IP address will be taken from the "IPs" column.	-
Storage Pool Configuration section	*Optional section of the CSV file that lets you configure properties for each of the Storage Pools to which the SDS devices are assigned*	-
ProtectionDomain	The Protection Domain to which the Storage Pool belongs	-
StoragePool	The Storage Pool name	-
Media Type	The expected device media type in the Storage Pool: `HDD` or `SDD`	-
External Acceleration	External acceleration type, if used: • `Read`—All devices are accelerated by a non-PowerFlex read cache • `Write`—All devices are accelerated by a non-PowerFlex write cache • `ReadAndWrite`—All devices are accelerated by both non-PowerFlex read and write cache	-
Data Layout	Data layout type:	-

Table 13. CSV topology spreadsheets (continued)

Field	Description	Required
	• `FG`—Fine Granularity layout • `MG`—Medium Granularity layout	
Fine Granularity ACCP	The name of the Fine Granularity Acceleration Pool associated with the Storage Pool	-
Zero Padding	Zero padding enabled on the Storage Pool (mandatory for Fine Granularity): • `Yes`—zero padding enabled • `No`—zero padding disabled	-
Compression Method	The compression method used in the Fine Granularity Storage Pool. Compression method might affect performance. For more information about recommended use cases, refer to the *Getting to Know Dell PowerFlex Guide*. • `None`—no compression is used • `Normal`—compression is enabled	-
Replication journal capacity percentage	The percentage of capacity dedicated to the replication journal: a value between 1–100 The percentage is calculated from the net capacity, and not from the total capacity. Use the information provided in the *Getting to Know Dell PowerFlex Guide* to calculate the percentage of capacity that is required. This calculation also applies to expansion use cases where SDR is being added to an existing system. Each Protection Domain that contains SDRs should have at least one Storage Pool with replication journal capacity higher than 0%. Each SDR needs at least 108 GB of replication journal capacity in order to work. Take this into consideration when calculating capacity requirements.	-

3. Save the CSV file and retain for future use.

CSV topology file for two-layer deployment

When deploying SDC nodes on a Linux or Windows server in a two-layer environment, add rows to the CSV file specifically for SDCs, that only contain the values listed in this table.

Table 14. CSV topology spreadsheet—Two-layer

Field	Description	Required
Domain	If using a domain user, the name of the domain (not relevant for Linux)	-
Username	The name of the user. • Linux nodes: Enter either `root` or a non-root sudo user, indicated by appending "(sudo)" to the end of the username (default: `root`). For more information about deploying with a non-root user, see "Deployment of a non-root user". • Windows nodes: Enter a user with administrator rights (default: `administrator`).	-
Password	Password used to log in to the node. This should be the password of the user entered in the **Username** column. To authenticate with SSH instead of node passwords, see "Using SSH authentication on the PowerFlex Gateway.	Yes
Operating System	The server's operating system: `linux` or `windows`	Yes

Table 14. CSV topology spreadsheet—Two-layer (continued)

Field	Description	Required
SDS-SDC Only IPs	SDC IP addresses to be used for communication among SDS and SDC nodes only. Comma-separated, no spaces. **NOTE:** For SDC-only nodes, enter the IP address in this column.	-
Is SDC	`Yes`	Yes
perfProfileForSDC	Optional performance profile to set for SDCs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the Fine Granularity data layout feature. For more information, see "Configure performance profile during deployment". -	-
SDC Name	Name for the SDC.	-

CSV file examples for typical configurations

Refer to the examples provided in this section to see how the CSV file is used to configure various system topologies and features.

Four data networks: two frontend and two backend, no fault sets

This example shows a CSV topology file for four data networks, where two are frontend networks, and two are backend networks. No fault sets are configured.

Four data networks are configured as follows:

- Two frontend networks for SDCs
- Two backend networks for MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- No fault sets

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary.

Table 15. Critical parameters for this example

Parameter	Example
Username	root
Password	Scaleio123
Operating System	linux
Is MDM/TB	Primary Other options: Secondary, TB
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.153.10,192.168.154.10
Virtual IP NICs	ens153, ens154
Is SDS	Yes
SDS Name	SDS1
SDS-SDS Only IPs	192.168.151.11,192.168.152.11

Table 15. Critical parameters for this example (continued)

Parameter	Example
SDS-SDC Only IPs	192.168.153.11,192.168.154.11
Protection Domain	PD01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01, SP01, SP02, SP02
perfProfileForSDS	high
Is SDC	No
perfProfileForSDC	high
SDC Name	SDC1
RFcache	No
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Four data networks: two frontend and two backend, with fault sets

This example shows a CSV topology file for four data networks, where two are frontend networks, and two are backend networks. Fault sets are also configured.

Four data networks are configured as follows:

- Two frontend networks for SDCs
- Two backend networks for MDMs, SDSs, and SDRs (when replication is required)
- Two external networks for replication (when applicable)
- Fault sets

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary.

Table 16. Critical parameters for this example

Parameter	Example
Username	root
Password	Scaleio123
Operating System	linux
Is MDM/TB	Primary Other options: Secondary, TB
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.153.10,192.168.154.10
Virtual IP NICs	ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS-SDS Only IPs	192.168.151.11,192.168.152.11
SDS-SDC Only IPs	192.168.153.11,192.168.154.11
Protection Domain	PD01
Fault Set	FS01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	Yes
perfProfileForSDC	high
SDC Name	SDC1
RFcache	No
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD

Table 16. Critical parameters for this example (continued)

Parameter	Example
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Four data networks: frontend and backend on all four networks, no fault sets

This example shows a CSV topology file for four data networks, where all four are both frontend and backend networks. No fault sets are configured.

Four data networks are configured as follows:

- Four networks for both frontend and backend: SDCs, MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- No fault sets

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary.

Table 17. Critical parameters for this example

Parameter	Example
Username	root
Password	Scaleio123
Operating System	linux
Is MDM/TB	Primary Other options: Secondary, TB
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.151.10,192.168.152.10,192.168.153.10,192.168.154.10
Virtual IP NICs	ens151,ens152,ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS All IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
Protection Domain	PD01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	No

Table 17. Critical parameters for this example (continued)

Parameter	Example
perfProfileForSDC	high
SDC Name	SDC1
RFcache	No
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
SDR All IPs	-
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Four data networks: frontend and backend on all four networks, with fault sets

This example shows a CSV topology file for four data networks, where all four are both frontend and backend networks. Fault sets are also configured.

Four data networks are configured as follows:

- Four networks for both frontend and backend: SDCs, MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- Fault sets

You can download an example CSV file, saved in Excel format for convenience.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary.

Table 18. Critical parameters for this example

Parameter	Example
Username	root

Table 18. Critical parameters for this example (continued)

Parameter	Example
Password	Scaleio123
Operating System	linux
Is MDM/TB	Primary Other options: Secondary, TB
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.151.10,192.168.152.10,192.168.153.10,192.168.154.10
Virtual IP NICs	ens151,ens152,ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS All IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
Protection Domain	PD01
Fault Set	FS01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	No
perfProfileForSDC	high
SDC Name	SDC1
RFcache	No
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG

Table 18. Critical parameters for this example (continued)

Parameter	Example
Fine Granularity ACCP	accp01
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Configure performance profile during deployment

During deployment, system performance is enhanced by default implementation of the high performance profile for the MDM, SDC, SDS, and SDR components. Most servers support the requirements for this configuration and can benefit considerably from its implementation.

The high performance profile is mandatory for Fine Granularity data layout. However, for Medium Granularity data layout, you can choose to use the compact performance profile instead.

High performance profile should be enabled on servers equipped with processors based on Haswell-equivalent or newer architectures and 48 GB of RAM. Other configurations may benefit from this setting as well.

The performance profile is set to High by default, but if you want to change the profile to Compact in the CSV file, use the `perfProfileFor<COMPONENT>` columns.

For more information on the difference between the compact and the high performance profile, see the *Configure and Customize PowerFlex Guide*.

Adding new components or features by editing the existing CSV file

Use your PowerFlex system's existing CSV topology file to add components or new features to your system.

The CSV file that was used to deploy a system can be used later, in several scenarios, for adding features or components to an existing PowerFlex system. Such scenarios include:

- Add SDSs, with or without new storage and acceleration devices
- Add MDMs, or tiebreakers. The PowerFlex Installer adds them to the MDM cluster and configure them automatically.
- Add SDRs for the replication feature
- Add SDCs
- Configure Storage Pools
- Configure PowerFlex system features which can be configured using the CSV file, such as performance profiles, acceleration, VASA, and more.

To add and configure components and features, make a copy of your system's latest CSV topology file. Add rows or columns, or edit existing rows and columns. For new features, you can copy columns from the latest complete CSV template, and paste them into the copy of your system's CSV file. Get the latest complete template by downloading it from the PowerFlex Installer **Install** tab. Once you have finished editing your file, follow the procedures described in "Add components to a PowerFlex system—Linux."

The following examples show how the CSV topology file can be edited for various system modification scenarios.

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Username	Passwor	Operat	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Prote	Fault Set	SDS Storage Device L	StoragePool
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
3	root	Scaleio1	linux	Secondar	MDM_115	10.55.137.	10.55.137	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
4	root	Scaleio1	linux	Secondar	MDM_116	10.55.137.	10.55.137	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb	sp2
7														
8														

Edited CSV file—new rows for added SDSs

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Username	Passwor	Operati	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	Fault Set	SDS Storage Device List	StoragePool	Is SDC	RfCache	RFcache SSD Dev	RFcache Accelerated S
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1	Yes			
3	root	Scaleio1	linux	Secondar	MDM_115	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1	Yes			
4	root	Scaleio1	linux	Secondar	MDM_116	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1	Yes			
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137.	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1	Yes			
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137.	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb	sp2	Yes	Yes	/dev/sde	sp2
7	root	Scaleio1	linux					Yes	sds_idd_6	10.55.137.120	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2		Yes	/dev/sde	sp2
8	root	Scaleio1	linux					Yes	sds_idd_7	10.55.137.121	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2		Yes	/dev/sde	sp2
9																		

Figure 4. Add storage and acceleration

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Username	Password	Operating	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	SDS Storag	StoragePo	Is SDC	Is SDR
2	root	Scaleio12	linux	Primary	MDM_118	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
3	root	Scaleio12	linux	Secondary	MDM_120	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
4	root	Scaleio12	linux	TB	MDM_121	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
5															

Edited CSV file—new rows added for an additional MDM and TB

	A	B	C	D	E	F	G	H	I	J	
1	Username	Password	Operating	Is MDM/TI	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Pr
2	root	Scaleio12	linux	Primary	MDM_118	10.55.137.118	10.55.137.118	Yes	sds_idd_1	10.55.137.	pc
3	root	Scaleio12	linux	Secondary	MDM_120	10.55.137.120	10.55.137.120	Yes	sds_idd_2	10.55.137.	pc
4	root	Scaleio12	linux	TB	MDM_121	10.55.137.121	10.55.137.121	Yes	sds_idd_4	10.55.137.	pc
5	root	Scaleio12	linux	Secondary	MDM_122	10.55.137.122	10.55.137.122				
6	root	Scaleio12	linux	TB	MDM_123	10.55.137.123	10.55.137.123				
7											

Figure 5. Add an MDM and a tiebreaker

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Username	Password	Opera	Is MDM/TI	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	SDS Storag	StoragePc	Is SDC			
2	root	Scaleio12	linux	Primary	MDM_118	10.55.137.118	10.55.137.118	Yes	sds_idd_1	10.55.137.118	pd1	/dev/sdb,	sp1	Yes			
3	root	Scaleio12	linux	Secondary	MDM_120	10.55.137.120	10.55.137.120	Yes	sds_idd_2	10.55.137.120	pd1	/dev/sdb,	sp1	Yes			
4	root	Scaleio12	linux	TB	MDM_121	10.55.137.121	10.55.137.121	Yes	sds_idd_4	10.55.137.121	pd1	/dev/sdb,	sp1	Yes			
5																	

Edited CSV file—new columns added for SDRs

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Username	Password	Opera	Is MDM/TI	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	SDS Storag	StoragePc	Is SDC	Is SDR	SDR Name	SDR Port	SDR Application IPs	SDR Storage IPs	SDR External IPs	SDR All IPs	
2	root	Scaleio12	linux	Primary	MDM_118	10.55.137.118	10.55.137.118	Yes	sds_idd_1	10.55.137.118	pd1	/dev/sdb,	sp1	Yes	Yes	SDR1		10.55.137.118	10.55.137.118	10.55.137.118	10.55.137.118	
3	root	Scaleio12	linux	Secondary	MDM_120	10.55.137.120	10.55.137.120	Yes	sds_idd_2	10.55.137.120	pd1	/dev/sdb,	sp1	Yes	Yes	SDR2		10.55.137.120	10.55.137.120	10.55.137.120	10.55.137.120	
4	root	Scaleio12	linux	TB	MDM_121	10.55.137.121	10.55.137.121	Yes	sds_idd_4	10.55.137.121	pd1	/dev/sdb,	sp1	Yes	Yes	SDR3		10.55.137.121	10.55.137.121	10.55.137.121	10.55.137.121	
5																						

	N	O	P	Q	R	S	T	U
gePc	Is SDC	Is SDR	SDR Name	SDR Port	SDR Application IPs	SDR Storage IPs	SDR External IPs	SDR All IPs
	Yes	Yes	SDR1		10.55.137.118	10.55.137.118	10.55.137.118	10.55.137.118
	Yes	Yes	SDR2		10.55.137.120	10.55.137.120	10.55.137.120	10.55.137.120
	Yes	Yes	SDR3		10.55.137.121	10.55.137.121	10.55.137.121	10.55.137.121

Figure 6. Add SDRs for replication

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Username	Passwor	Operating	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	SDS-SDC Only IPs	Protectior	Fault Set	SDS Storag	StoragePc	Is SDC	RfCache	RFcache S	RFcache A	Is SDR
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.114		pd1	fs1_mg	/dev/sdb,	sp1	Yes				Yes
3	root	Scaleio1	linux	Secondary	MDM_115	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.115		pd1	fs1_mg	/dev/sdb,	sp1	Yes				Yes
4	root	Scaleio1	linux	Secondary	MDM_116	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.116		pd1		/dev/sdb,	sp1	Yes				Yes
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137.	Yes	sds_idd_5	10.55.137.117		pd1		/dev/sdb,	sp1	Yes				
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137.	Yes	sds_idd_3	10.55.137.119		pd2		/dev/sdb	sp2	Yes	Yes	/dev/sdc	sp2	
7	root	Scaleio1	linux					Yes	sds_idd_6	10.55.137.120		pd2		/dev/sdb,	sp2		Yes	/dev/sdc	sp2	
8	root	Scaleio1	linux					Yes	sds_idd_7	10.55.137.121		pd2		/dev/sdb,	sp2		Yes	/dev/sdc	sp2	
9																				

Edited CSV file—new rows added for SDCs

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Username	Passwor	Operating Sys	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	SDS-SDC Only IPs	Protectior	Fault Set	SDS Storag	StoragePc	Is SDC	RfCache
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.114		pd1	fs1_mg	/dev/sdb,	sp1	Yes	
3	root	Scaleio1	linux	Secondary	MDM_115	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.115		pd1	fs1_mg	/dev/sdb,	sp1	Yes	
4	root	Scaleio1	linux	Secondary	MDM_116	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.116		pd1		/dev/sdb,	sp1	Yes	
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137.	Yes	sds_idd_5	10.55.137.117		pd1		/dev/sdb,	sp1	Yes	
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137.	Yes	sds_idd_3	10.55.137.119		pd2		/dev/sdb	sp2	Yes	Yes
7	root	Scaleio1	linux					Yes	sds_idd_6	10.55.137.120		pd2		/dev/sdb,	sp2		Yes
8	root	Scaleio1	linux					Yes	sds_idd_7	10.55.137.121		pd2		/dev/sdb,	sp2		Yes
9	root	Scaleio1	windows					No			10.55.5.1					Yes	
10	root	Scaleio1	windows					No			10.55.5.2					Yes	
11	root	Scaleio1	windows					No			10.55.5.3					Yes	

Figure 7. Add SDCs

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Username	Password	Operating S	Is MDM/TB	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	Fault Set	SDS Storage Device List	StoragePool	Is SDC
2	root	Scaleio123	linux	Primary	MDM_114	10.55.137.114	10.55.137.	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1	Yes
3	root	Scaleio123	linux	Secondary	MDM_115	10.55.137.115	10.55.137.	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1	Yes
4	root	Scaleio123	linux	Secondary	MDM_116	10.55.137.116	10.55.137.	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1	Yes
5	root	Scaleio123	linux	TB	MDM_117	10.55.137.117	10.55.137.	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1	Yes
6	root	Scaleio123	linux	TB	MDM_119	10.55.137.119	10.55.137.	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2	Yes
7	root	Scaleio123	linux					Yes	sds_idd_6	10.55.137.120	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2	
8	root	Scaleio123	linux					Yes	sds_idd_7	10.55.137.121	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2	

Edited CSV file—new rows for Storage Pool configuration

	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	Username	Password	Operating S	Is MDM/TB	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	Fault Set	SDS Storage Device List	Stora
2	root	Scaleio123	linux	Primary	MDM_114	10.55.137.114	10.55.137.	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
3	root	Scaleio123	linux	Secondary	MDM_115	10.55.137.115	10.55.137.	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
4	root	Scaleio123	linux	Secondary	MDM_116	10.55.137.116	10.55.137.	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1
5	root	Scaleio123	linux	TB	MDM_117	10.55.137.117	10.55.137.	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1
6	root	Scaleio123	linux	TB	MDM_119	10.55.137.119	10.55.137.	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2
7	root	Scaleio123	linux					Yes	sds_idd_6	10.55.137.120	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2
8	root	Scaleio123	linux					Yes	sds_idd_7	10.55.137.121	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2
9														
10	Storage Pool Configuration													
11	Protectior	StoragePo	Media Type	External Accele	Data Layout	Fine Granularity ACCP	Zero Padd	Compression Method	Replication journal capacity percentage					
12	pd1	sp1	HDD	ReadAndWrite	MG		Yes		50					
13	pd2	sp2	SSD	Read	FG	accp2	Yes	Normal	20					
14														

Figure 8. Configure Storage Pools

Clean the PowerFlex VMware environment

For two-layer systems, or ESXi-based systems, clean the PowerFlex VMware environment in order to perform a clean installation. This procedure is recommended for cases when previously defined networks will be used again, or when PowerFlex components such as SDCs that ran on other networks exist in the environment.

Prerequisites

NOTE: This procedure is only required when PowerFlex plug-in installation failed.

Before you begin, unmap and delete any PowerFlex volumes in your system.

If necessary, unregister PowerFlex from within the vSphere PowerFlex plug-in, and delete all the PowerFlex SVMs.

Steps

1. Close the existing PowerCLI sessions, set PowerCLI to **Run as administrator**, and then open a new session.
2. Using the PS1 script, unregister the PowerFlex plug-in.
3. Stop the vSphere web client service:
   VC Linux: `service-control --stop vsphere-client`
4. Delete the contents of the PowerFlex plug-in folder.
   The vSphere web client (Virgo) plug-in folders are located at:

Table 19. Location of vSphere web client plug-in folders

vCenter	Operating system	Path to file
6.x	Windows	C:\ProgramData\VMware\vCenterServer\cfg\vsphere-client\vc-packages\vsphere-client-serenity
	Linux	/etc/vmware/vsphere-client/vc-packages/vsphere-client-serenity

5. Delete the `scaleio` folder or its contents.
   The `scaleio` folders are located at:

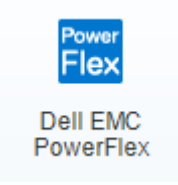
Table 20. Location of `scaleio` folder

vCenter	Operating system	Path to file
6.x	Windows	C:\Users\vspherewebclientsvc\AppData\Roaming\VMware\scaleio
	Linux	/etc/vmware/vsphere-client/vc-packages/scaleio

6. Start the vSphere web client service:
   VC Linux: `service-control --start vsphere-client`
7. Clear your web browser's cache and cookies, or open a different web browser.
8. Using the PS1 script in PowerCLI, register the PowerFlex plug-in.

   NOTE: Do not press ENTER at this point.

9. After you have logged in to the vSphere web client to complete the registration and you see the PowerFlex icon

   , press ENTER in the PowerCLI session.

   This stops the embedded Tomcat server.
10. If necessary, remove the SDC module parameters and VIB from the ESXi servers:
    a. Connect to each ESXi server via SSH.
    b. Run:

    ```
    ~ # esxcli system module parameters set -m scini -p ""
    ~ # esxcli software vib remove -n scaleio-sdc-esx6.x
    ```

    c. Reboot each ESXi server.
11. Proceed to the procedures for re-registering the PowerFlex plug-in, and then deploy PowerFlex.

Deploying PowerFlex using a non-root user

PowerFlex can be deployed or extended in Linux environments using a non-root sudo user in non-interactive mode.

Sudo is a program that allows a user to run or install a program as the root user. A sudo user can be created to deploy PowerFlex.

In order to successfully deploy or extend PowerFlex with a non-root user, the non-root user must meet the following conditions:

- The username included in the CSV file must already exist.
- The non-root user must be a sudo user.
- The non-root user must be in non-interactive mode.
- The requirement for TTY must be disabled.

In the CSV file used for deployment, you must indicate that you are intending to use a sudo non-root username by appending the string "(sudo)" to the user name in the Username field. For example, if you are using a non-root user with the username "non_root", enter the string "non_root(sudo)" in the username field of the CSV file.

Configure a non-root non-interactive sudo user

In Linux, you can deploy or extend PowerFlex with a non-root user. You must configure a non-root sudo user in non-interactive mode.

About this task

The following procedure details one method for configuring a non-root non-interactive sudo user. Perform the commands from the operating system console of where you want the PowerFlex Gateway to deploy PowerFlex.

Steps

1. Create a user group named "admin".

```
groupadd admin
```

2. Create a user named "non_root" and add it to the admin group.

```
useradd -G admin non_root
```

3. Change the password of the non_root user.

```
passwd non_root
```

When prompted, enter the new password and then confirm it by entering it again.

4. Open the sudoers `/etc/sudoers` file for editing.

```
vi /etc/sudoers
```

5. Search the sudoers file for "## Same thing without a password".

```
:s/## Same thing without a password
```

6. In the line below the search result, add the text `%admin ALL=(ALL) NOPASSWD: ALL` to the file.
7. Search the sudoers file for "Defaults requiretty", and replace it with `Defaults ! requiretty`.
8. Exit the vi editor by typing the following command to exit: `:wq!`
9. Create a hidden directory in the non_root user's home directory to store the SSH configuration.

```
mkdir /home/non_root/.ssh
```

10. Copy the SSH configuration from the root user to the non_root user's directory.

```
cp -rf /root/.ssh/* /home/non_root/.ssh/
```

Configure the OpenStack Nova Compute service

If your system will operate in an OpenStack environment, configure the openstack-nova-compute service to start after the scini service.

About this task

The openstack-nova-compute service must start after the scini service, to ensure that all volumes are recognized and configured by the SDC before the compute service starts, and the VM instances are started.

Steps

1. In command line on the Nova Compute server, run

```
systemctl status openstack-nova-compute
```

2. Identify the service file that is currently loaded.

For example, in the following output, the file is: `/usr/lib/systemd/system/openstack-nova-compute.service`

```
Loaded: loaded (/usr/lib/systemd/system/openstack-nova-compute.service; enabled;
vendor preset: disabled)
```

3. Open the file in a text editor, and insert the following line under the `[Unit]` section:

```
[Unit]
...
After=scini.service
```

4. Save the file.

Results

The openstack-nova-compute service will start after the scini service.

4

Deploy PowerFlex

Installation wizard

This section describes how to use the PowerFlex Installer wizard, the quickest way to get a PowerFlex system up and running. The wizard is most suitable for environments where all servers are converged and in one Protection Domain.

Prerequisites

Ensure that you have performed all the preparation tasks before using the wizard.

About this task

The wizard guides you through the following tasks:

Steps

1. Setting up the installation.
2. Running the installation.

   After installation is complete, a number of post-installation tasks must be performed, including GUI installation, configuration, and license activation.

Log in and upload installation packages

Log in to the PowerFlex Installer and upload the PowerFlex installation packages.

Prerequisites

Ensure that you have network access to the software packages that you downloaded and extracted as part of the preparation tasks.

Ensure that you know the password that was defined for the PowerFlex Gateway.

Steps

1. Log in to the PowerFlex Installer:
   a. Point your browser to this URL: https://*<GW_Server_IP>*

      where *<GW_Server_IP>* is the IP address of the server where you installed the PowerFlex Gateway package.
   b. Accept the certificate warning; alternatively, install your own certificate for the Tomcat server.
   c. Enter the default user name, **admin**, and the password defined when the PowerFlex Gateway was prepared, and then click **Login**.
2. Upload installation packages:
   a. At the top of the **Welcome** window, click **Packages**.

      You may need to re-authenticate.
   b. From the **Manage Installation Packages** page, click **Browse**, and browse to the extracted PowerFlex component packages. Select all the required files, and then click **Open**.

      Select these packages for each operating system type that will be used in your PowerFlex system:

      - MDM
      - SDS
      - SDC
      - LIA

- RFcache XCACHE package (optional; required for Medium Granularity Storage Pool acceleration)
- VASAPROVIDER package (optional; required for VMware vVols)
- SDR (optional; required for the replication feature)

c. Click **Upload**.

The uploaded installation packages appear in the packages table.

d. Click **Proceed to Install**.

Results

The PowerFlex software packages have been uploaded to the PowerFlex Installer, in preparation for deployment. In the next task, you will use the PowerFlex Installer to install the software on the nodes.

Initiate the wizard and run the installation

Start installation of the PowerFlex software packages, using the PowerFlex Installer wizard. The installation performs the following phases: query, upload, install, and configure. At the end of each phase, you must approve continuation to the next phase.

Prerequisites

Ensure that the software packages were successfully uploaded to the PowerFlex Installer.

Steps

1. In the PowerFlex Installer **Provide Installation Topology** screen, click the **Installation Wizard** drop-down arrow, and select an MDM cluster type:
   - 3-node cluster (default)

     Configures a primary MDM, a secondary MDM, and a tiebreaker MDM.

   - 5-node cluster

     Configures a primary MDM, two secondary MDMs, and two tiebreaker MDMs.

   The **Topology** section is populated with nodes, according to the MDM cluster chosen in the previous step. Each node is also defined as an SDS and an SDC.
2. Type a new MDM password in the **password** box, and type it again in the **confirm password** box.

   The MDM password is a password that is configured for the MDM during the installation. It must meet the following criteria:

   - Between 6 and 31 ASCII-printable characters
   - Includes at least 3 of the following groups: [a-z], [A-Z], [0-9], special chars (!@#$ …)
   - Contains no white spaces
3. For **LIA Authentication**, select **Native** for local authentication, or **LDAP** for LDAP authentication.
   - If you selected **Native**, for **LIA Password**, type a password in the **password** box, and type it again in the **confirm password** box.

     The LIA password is a new password that will be used to authenticate communication between the PowerFlex Installer and the Light Installation Agent (LIA). It must meet the same criteria as the MDM password, as listed in the previous step.

   - If you selected **LDAP**, you can add the credentials for one LDAP server. After deployment, up to seven more LDAP servers can be added. For more information, see the "Security" section of the *Configure and Customize Dell PowerFlex*.

     Enter the required information in the additional fields that are displayed:

     - **LDAP Server URI**
     - **LDAP Group** and **LDAP BaseDN**
     - **LDAP User Name** and **LDAP Password**
4. Review the end user license agreement and select the check box to accept the license agreement.

   For complete information on licensing, see the "Licensing" section of this guide.
5. In the **Topology** section, enter server information:

a. For each node, click on the appropriate cells to edit its IP address, select its host operating system, and enter an MDM name and password.
   - For Linux, type the password of the root user.
   - To see the passwords, select **Show passwords while editing**.

b. To add more hosts, click **Clone host**.

   The assigned IP address is derived from the IP address of the previous host. Each cloned host is designated to act as an SDS and an SDC.

6. At the bottom of the screen, click **Start Installation**.

   NOTE: A post-installation notice appears, that outlines the steps that are required after the wizard completes the installation. The content of this notice will be repeated at the end of the process, or you can view it by clicking **Post installation instructions** from the **Monitor** screen.

7. Click the **Monitor** tab.

   The **Install - query** screen appears.

   In the query stage, the PowerFlex Installer validates that there are no previously installed PowerFlex components on any of the requested installation nodes.

   You can change the display by selecting the following:

   - **Auto Refresh** (default)
   - **Hide Completed Successfully**
   - **Show All Phases**

   At the end of each phase, a status message is displayed.

   If an error message is displayed during the process, you can retry the failed operations (using the **Retry failed** button), continue to the next phase, or abort the installation.

   NOTE: Once you proceed to the next stage, you will not be able to use the PowerFlex Installer to retry failed tasks.

8. When the query phase is complete, close the status message, and at the top right side of the screen, click **Start upload phase**.

   The **Install - upload** screen appears, displaying which PowerFlex packages are being uploaded to each server, and the status of each command.

9. When the previous phase is complete, close the status message, and at the top right side of the screen, click **Start install phase**.

   The **Install - install** screen appears, displaying the status of the installation commands.

10. When all Install command rows are in Completed status, click **Start configure phase**.

    The **Install - configure** screen displays configuration progress.

11. When all processes are finished, click **Mark operation completed**.

    The PowerFlex system installation is complete! The wizard installation creates one Protection Domain and one Storage Pool, both named "default". You will use these when you enable storage.

    The post-installation notes appear, directing you to the steps necessary to start using your storage.

Next steps

- Install the PowerFlex presentation server.
- Perform the post-installation tasks described in this guide.
- After you finish the tasks described in this guide, perform the mandatory and optional post-deployment configuration tasks described in the *Configure and Customize PowerFlex Guide*. In particular, add storage devices to the SDS nodes, and then create and map volumes.

Customizable installation using CSV file—hyper-converged system

This section describes how to use the PowerFlex Installer and a CSV topology file to get a fully customized hyper-converged PowerFlex system up and running.

Prerequisites

Ensure that you have performed all the preparation tasks before using the PowerFlex Installer, and in particular, that you have prepared the CSV file.

About this task

The PowerFlex Installer guides you through the following tasks:

Steps

1. Setting up the installation.
2. Running the installation.

   After installation is complete, a number of post-installation tasks must be performed, including PowerFlex presentation server installation (required for the PowerFlex GUI), configuration and license activation.

Log in and upload installation packages

Log in to the PowerFlex Installer and upload the PowerFlex installation packages.

Prerequisites

Ensure that you have network access to the software packages that you downloaded and extracted as part of the preparation tasks.

Ensure that you know the password that was defined for the PowerFlex Gateway.

Steps

1. Log in to the PowerFlex Installer:
   a. Point your browser to this URL: https://*<GW_Server_IP>*

      where *<GW_Server_IP>* is the IP address of the server where you installed the PowerFlex Gateway package.
   b. Accept the certificate warning; alternatively, install your own certificate for the Tomcat server.
   c. Enter the default user name, **admin**, and the password defined when the PowerFlex Gateway was prepared, and then click **Login**.
2. Upload installation packages:
   a. At the top of the **Welcome** window, click **Packages**.

      You may need to re-authenticate.
   b. From the **Manage Installation Packages** page, click **Browse**, and browse to the extracted PowerFlex component packages. Select all the required files, and then click **Open**.

      Select these packages for each operating system type that will be used in your PowerFlex system:

      - MDM
      - SDS
      - SDC
      - LIA
      - RFcache XCACHE package (optional; required for Medium Granularity Storage Pool acceleration)
      - VASAPROVIDER package (optional; required for VMware vVols)
      - SDR (optional; required for the replication feature)
   c. Click **Upload**.

      The uploaded installation packages appear in the packages table.

d. Click **Proceed to Install**.

Results

The PowerFlex software packages have been uploaded to the PowerFlex Installer, in preparation for deployment. In the next task, you will use the PowerFlex Installer to install the software on the nodes.

Upload the CSV file

Upload the prepared CSV file to the PowerFlex Installer.

Prerequisites

If you have not created the CSV file yet, you can download a template by clicking **Minimal** or **Complete** on the PowerFlex Installer **Provide Installation Topology** page. Then, prepare the file as explained in the "Preparation for deployment" section of this guide.

Steps

1. On the **Provide Installation Topology** page, click **Browse**, browse to the location of the installation CSV file, select the file, and click **Open**.
2. For a new installation, select **New installation**.

   To extend an existing installation, click the down-arrow and select **Add to existing sys**.
3. Click **Upload Installation CSV**.

   After successfully uploading the file, the **Installation Configuration** page appears.

   You can expand or contract the lists in the **Topology** section.

Results

You are now ready to start the installation.

Start the customized installation

Start the customized installation process for a PowerFlex system.

Prerequisites

- Ensure that the software packages and CSV file have been successfully uploaded to the PowerFlex Installer.
- If LIA authentication will be LDAP-based, configure all items required for LDAP support that are described in the *Dell PowerFlex User Roles and LDAP Usage Technical Notes*.
- If LIA authentication will be LDAP-based, and the LDAP service is Linux-based, ensure that the `gatewayUser.properties` file is configured with the parameters for `lia.ldap.initial.usernameDnFormat` and `lia.ldap.initial.searchFilterFormat`. Ensure that the same CN (user), OU (organizational unit) and DC (domain) parameters are configured in that file, and in the appropriate step in this procedure.

Steps

1. On the **Installation Configuration** page, **Credentials Configuration** section, for **MDM Password**, enter a new MDM password in the **password** field, and enter it again in the **confirm password** field.

   The MDM password is a password that is configured for the MDM during the installation. It must meet the following criteria:

   - Between 6 and 31 ASCII-printable characters
   - Includes at least 3 of the following groups: [a-z], [A-Z], [0-9], special chars (!@#$ …)
   - Contains no white spaces
2. For **LIA Authentication**, select **Native** for local authentication, or **LDAP** for LDAP authentication.

   NOTE: When using the PowerFlex Installer to extend (rather than to install) a system, enter the LIA credentials that you entered during the initial installation.

- If you selected **Native**, for **LIA Password**, enter a password in the **password** field, and enter it again in the **confirm password** field.

  The LIA password is a new password that will be used to authenticate communication between the PowerFlex Installer and the Light Installation Agent (LIA). It must meet the same criteria as the MDM password, as listed in the previous step.

- If you selected **LDAP**, you can add the credentials for one LDAP server. After deployment, up to seven more LDAP servers can be added. For more information, see the *Dell PowerFlex User Roles and LDAP Usage Technical Notes*.

  Enter the required information in the additional fields that are displayed:

  - **LDAP Server URI**
  - **LDAP Group** and **LDAP BaseDN**
  - **LDAP User Name** and **LDAP Password**

Example for a Windows-based LDAP server:

- LDAP Server URI: ldaps://win12r2-dc.ldaps.local
- LDAP Group: CN=SIO_GRP_1,OU=SIO_OU_1,dc=ldaps,dc=local
- LDAPBaseDN: DC=ldaps,DC=local
- LDAP User Name: sio_1@ldaps.local
- LDAP User Password: P$assword123

Example for a Linux-based LDAP server:

- LDAP Server URI: ldaps://server.ldap.com
- LDAP Group: cn=scaleio_users,ou=People,dc=ldap,dc=com
- LDAPBaseDN: dc=ldap,dc=com
- LDAP User Name: sio_1@ldap.com
- LDAP User Password: P$assword123

3. Review the end user license agreement and select the check box to accept the license agreement.

   For complete information about licensing, see the *Licensing* section of this guide.

4. Advanced options can be configured now. The following list describes the available options. Click **Set advanced options** to access these options and configure them.

Table 21. Advanced options

Option type	Option	Description
Skip options	Skip upload	Do not upload packages. You can use this if the packages were already uploaded.
	Skip installation	Do not install packages. You can use this if the packages were already installed.
	Skip configuration	Do not configure PowerFlex components. You can use this option if you only want to upload and install packages now.
	Skip Linux devices validation	Do not validate Linux device names. Use case: during the PowerFlex Installer validate stage, errors in the entering of Linux device names (such as `/dev/sdbb` instead of `/dev/sdb`) will prevent the continuation of the installation. This can be a problem if you want to add valid devices that do not meet the `/proc/partitions` standard (such as aliases and logical names). Therefore, when this option is selected, the validate stage will be completed, but the affected node will show a completed (warning) status. You can click **Details** to see

Table 21. Advanced options (continued)

Option type	Option	Description
		the error, and you can choose to proceed with the installation. Continue with the remainder of the installation procedures.
Configuration options	Enable alert service and create lockbox	Enable the alert service required for SNMP and Secure Remote Services reporting. This also creates and configures the lockbox. The lockbox is also required for CloudIQ reporting. This is the best practice method for creating the lockbox. **NOTE:** It is also possible to manually create the lockbox later, but the manual method involves security considerations. If you select this option, the **Traps Receiver IP/Hostname** field is displayed. Enter up to two (comma-separated) IP addresses or host names of the SNMP trap receiver servers.
Security options	Disable secure communication with the MDMs	Disable the need for secure communication mode between management clients and the MDM cluster members. **NOTE:** Non-secure communication has security implications, and should be used with caution. For more information about this mode, see "Use SCLI in non-secure mode" in the *Configure and Customize PowerFlex Guide*.
	Disable secure communication with the LIAs	Disable the need for secure communication mode between management clients and the LIAs. **NOTE:** Non-secure communication has security implications, and should be used with caution. For more information about this mode, see "Use SCLI in non-secure mode" in the *Configure and Customize PowerFlex Guide*.
	Disable authentication between internal components	Disable the need for authentication between SDSs and MDMs.
	Use only the following trusted IP addresses for LIA (in addition to this gateway IP address)	Limit which PowerFlex Gateway can communicate with the LIAs. If you select this option, enter a comma-separated list of the IP addresses to trust (up to 32 IP addresses, including the IP address of this PowerFlex Gateway). The PowerFlex Gateway IP address is automatically added to the list and does

Table 21. Advanced options (continued)

Option type	Option	Description
		not have to be explicitly mentioned. If the PowerFlex Gateway uses multiple IP addresses, enter all the addresses.

5. You can use the PowerFlex Installer to configure Syslog event reporting. You can also configure these features after installation, using the CLI. To configure Syslog reporting now, select **Configure the way Syslog events are sent**, and enter the following parameters:
   a. **Syslog Server**
      The host name or IP address of the syslog server where the messages should be sent. Enter up to two server names or addresses, comma-separated.
   b. **Port**
      The port of the syslog server (default 1468)
   c. **Syslog Facility**
      The facility level (default: Local0)
6. Review the displayed information, by scrolling through the tables, and by hovering over the blue hyperlinks. To view the password that will be configured for a specific component, click **show**.
   If you have not done so yet, keep a record of all the passwords that you have assigned to the system components.
7. Scroll down to the **Storage Pool Configuration** section of the table, select values for the following items. If the CSV file contained values in the **Storage Pool Configuration** section, the values will already appear here. These values are mandatory.
   - The expected **Device Media Type**
   - The **External Acceleration** type (if used):
     - none—No devices are accelerated by a non-PowerFlex read or write cache
     - read—All devices are accelerated by a non-PowerFlex read cache
     - write—All devices are accelerated by a non-PowerFlex write cache
     - read_and_write—All devices are accelerated by both non-PowerFlex read and write cache

     This input is required in order to prevent the generation of false alerts for media type mismatches. For example, if an HDD device is added which the SDS perceives as being too fast to fit the HDD criteria, alerts might be generated. External acceleration/caching is explained in the *Getting to Know Dell PowerFlex Guide*.
   - **Data Layout** type (use with caution; data layout type cannot be changed after Storage Pool creation)
   - If Fine Granularity data layout is selected, select a **Fine Granularity ACCP (Acceleration Pool)**, and a **Compression Method**
   - If Medium Granularity data layout is selected, the **Zero Padding** option is available for selection. Use with caution, because the zero padding setting can only be changed in Storage Pools without any storage devices.
   - **Replication Journal Capacity Percentage** must be defined if SDRs are deployed in the system. For more information about calculating the appropriate value for your system, refer to the *Getting to Know Dell PowerFlex Guide*.

   NOTE: Fine Granularity can only be selected if the Storage Pool contains NVDIMMs and SSDs.
8. Click **Start Installation**.
9. Select the **Monitor** tab.

Results

Installation begins. Proceed to the next task to perform the remaining mandatory steps in the installation process.

Complete the customized installation—hyper-converged system

Complete the installation of the PowerFlex software packages, using the PowerFlex Installer. The installation performs the following phases: query, upload, install, and configure. At the end of each phase, you must approve continuation to the next phase.

Steps

1. At the top of the screen, click the **Monitor** tab.

   The **Install - query** screen appears.

   In the query stage, the PowerFlex Installer validates that there are no previously installed PowerFlex components on any of the requested nodes.

   When extending an existing system, the query phase does expect to find currently-installed nodes.

   You can change the display by selecting the following:

   - **Auto Refresh** (default)
   - **Hide Completed Successfully**
   - **Show All Phases**

   At the end of each phase, a status message is displayed.

   If an error message is displayed during the process, you can try to resolve the problem and then retry the failed operations (with the **Retry failed** button), continue to the next phase, or abort the installation.

   NOTE: Once you proceed to the next stage, you will not be able to use the PowerFlex Installer to retry failed tasks.
2. When the query phase is complete, close the status message, and at the top right side of the screen, click **Start upload phase**.

   The **Install - upload** screen appears, displaying which PowerFlex packages are being uploaded to each server, and the status of each command.
3. When the upload phase is complete, close the status message, and at the top right side of the screen, click **Start install phase**.

   The **Install - install** screen appears, displaying the status of the installation commands.
4. When the install phase is complete, close the status message, and at the top right side of the screen, click **Start configure phase**.

   The **Install - configure** screen appears, displaying the status of the configuration commands.
5. When all processes are finished, close the status message, and click **Mark operation completed**.

   The PowerFlex system installation is complete!

   NOTE: Marking the operation completed signals to the PowerFlex Installer that it can now be used for other installations, upgrades, and so on.

   The post-installation notes appear, directing you to the steps necessary to start using your storage.

Next steps

- Install the PowerFlex GUI.
- Perform the post-installation tasks described in this guide.
- After you have finished the tasks in this guide, perform the mandatory and optional post-deployment configuration tasks described in the *Configure and Customize PowerFlex Guide*.

Customizable installation using CSV file—two-layer system

This section describes how to use the PowerFlex Installer and a CSV topology file to get a fully customized two-layer PowerFlex system up and running. The backend (MDM and SDS components) will be installed on Linux-based servers. The frontend servers (running the SDC component) can run on any operating system supported by this version of PowerFlex.

Prerequisites

Ensure that you have performed all the preparation tasks before using the PowerFlex Installer, and in particular, that you have prepared the CSV file.

About this task

- For frontend Linux or Windows-based servers, SDCs can be installed using the CSV file, or manually after deploying the backend
- For frontend ESXi-based servers, the SDCs are installed after deploying the backend, either using the vSphere PowerFlex plug-in, or manually

This section is specific to PowerFlex deployments. For VxFlex Ready Node or VxRack Node 100 Series systems, refer to your product's *Hardware Configuration and Operating System Installation Guide* for step-by-step installation instructions.

The PowerFlex Installer guides you through the following tasks:

Steps

1. Set up the installation.
2. Run the installation.
3. For ESXi frontend servers, register the PowerFlex plug-in and install SDC.
4. For Linux or Windows-based frontend servers where SDCs were not installed using the CSV file, install the SDC component.

   After installation is complete, a number of post-installation tasks must be performed, including PowerFlex presentation server installation (required for the PowerFlex GUI), configuration, and license activation.

Log in and upload installation packages

Log in to the PowerFlex Installer and upload the PowerFlex installation packages.

Prerequisites

Ensure that you have network access to the software packages that you downloaded and extracted as part of the preparation tasks.

Ensure that you know the password that was defined for the PowerFlex Gateway.

Steps

1. Log in to the PowerFlex Installer:
   a. Point your browser to this URL: https://*<GW_Server_IP>*

      where *<GW_Server_IP>* is the IP address of the server where you installed the PowerFlex Gateway package.
   b. Accept the certificate warning; alternatively, install your own certificate for the Tomcat server.
   c. Enter the default user name, **admin**, and the password defined when the PowerFlex Gateway was prepared, and then click **Login**.
2. Upload installation packages:
   a. At the top of the **Welcome** window, click **Packages**.

      You may need to re-authenticate.
   b. From the **Manage Installation Packages** page, click **Browse**, and browse to the extracted PowerFlex component packages. Select all the required files, and then click **Open**.

      Select these packages for each operating system type that will be used in your PowerFlex system:

      - MDM
      - SDS
      - SDC
      - LIA
      - RFcache XCACHE package (optional; required for Medium Granularity Storage Pool acceleration)
      - VASAPROVIDER package (optional; required for VMware vVols)
      - SDR (optional; required for the replication feature)
   c. Click **Upload**.

The uploaded installation packages appear in the packages table.

d. Click **Proceed to Install**.

Results

The PowerFlex software packages have been uploaded to the PowerFlex Installer, in preparation for deployment. In the next task, you will use the PowerFlex Installer to install the software on the nodes.

Upload the CSV file

Upload the prepared CSV file to the PowerFlex Installer.

Prerequisites

If you have not created the CSV file yet, you can download a template by clicking **Minimal** or **Complete** on the PowerFlex Installer **Provide Installation Topology** page. Then, prepare the file as explained in the "Preparation for deployment" section of this guide.

Steps

1. On the **Provide Installation Topology** page, click **Browse**, browse to the location of the installation CSV file, select the file, and click **Open**.
2. For a new installation, select **New installation**.
   To extend an existing installation, click the down-arrow and select **Add to existing sys**.
3. Click **Upload Installation CSV**.
   After successfully uploading the file, the **Installation Configuration** page appears.
   You can expand or contract the lists in the **Topology** section.

Results

You are now ready to start the installation.

Start the customized installation

Start the customized installation process for a PowerFlex system.

Prerequisites

- Ensure that the software packages and CSV file have been successfully uploaded to the PowerFlex Installer.
- If LIA authentication will be LDAP-based, configure all items required for LDAP support that are described in the *Dell PowerFlex User Roles and LDAP Usage Technical Notes*.
- If LIA authentication will be LDAP-based, and the LDAP service is Linux-based, ensure that the `gatewayUser.properties` file is configured with the parameters for `lia.ldap.initial.usernameDnFormat` and `lia.ldap.initial.searchFilterFormat`. Ensure that the same CN (user), OU (organizational unit) and DC (domain) parameters are configured in that file, and in the appropriate step in this procedure.

Steps

1. On the **Installation Configuration** page, **Credentials Configuration** section, for **MDM Password**, enter a new MDM password in the **password** field, and enter it again in the **confirm password** field.
   The MDM password is a password that is configured for the MDM during the installation. It must meet the following criteria:
   - Between 6 and 31 ASCII-printable characters
   - Includes at least 3 of the following groups: [a-z], [A-Z], [0-9], special chars (!@#$ …)
   - Contains no white spaces
2. For **LIA Authentication**, select **Native** for local authentication, or **LDAP** for LDAP authentication.

NOTE: When using the PowerFlex Installer to extend (rather than to install) a system, enter the LIA credentials that you entered during the initial installation.

- If you selected **Native**, for **LIA Password**, enter a password in the **password** field, and enter it again in the **confirm password** field.

  The LIA password is a new password that will be used to authenticate communication between the PowerFlex Installer and the Light Installation Agent (LIA). It must meet the same criteria as the MDM password, as listed in the previous step.

- If you selected **LDAP**, you can add the credentials for one LDAP server. After deployment, up to seven more LDAP servers can be added. For more information, see the *Dell PowerFlex User Roles and LDAP Usage Technical Notes*.

  Enter the required information in the additional fields that are displayed:

  - **LDAP Server URI**
  - **LDAP Group** and **LDAP BaseDN**
  - **LDAP User Name** and **LDAP Password**

Example for a Windows-based LDAP server:

- LDAP Server URI: ldaps://win12r2-dc.ldaps.local
- LDAP Group: CN=SIO_GRP_1,OU=SIO_OU_1,dc=ldaps,dc=local
- LDAPBaseDN: DC=ldaps,DC=local
- LDAP User Name: sio_1@ldaps.local
- LDAP User Password: P$assword123

Example for a Linux-based LDAP server:

- LDAP Server URI: ldaps://server.ldap.com
- LDAP Group: cn=scaleio_users,ou=People,dc=ldap,dc=com
- LDAPBaseDN: dc=ldap,dc=com
- LDAP User Name: sio_1@ldap.com
- LDAP User Password: P$assword123

3. Review the end user license agreement and select the check box to accept the license agreement.

   For complete information about licensing, see the *Licensing* section of this guide.

4. Advanced options can be configured now. The following list describes the available options. Click **Set advanced options** to access these options and configure them.

Table 22. Advanced options

Option type	Option	Description
Skip options	Skip upload	Do not upload packages. You can use this if the packages were already uploaded.
	Skip installation	Do not install packages. You can use this if the packages were already installed.
	Skip configuration	Do not configure PowerFlex components. You can use this option if you only want to upload and install packages now.
	Skip Linux devices validation	Do not validate Linux device names. Use case: during the PowerFlex Installer validate stage, errors in the entering of Linux device names (such as `/dev/sdbb` instead of `/dev/sdb`) will prevent the continuation of the installation. This can be a problem if you want to add valid devices that do not meet the `/proc/partitions` standard (such as aliases and logical names). Therefore, when this option

Table 22. Advanced options (continued)

Option type	Option	Description
		is selected, the validate stage will be completed, but the affected node will show a completed (warning) status. You can click **Details** to see the error, and you can choose to proceed with the installation. Continue with the remainder of the installation procedures.
Configuration options	Enable alert service and create lockbox	Enable the alert service required for SNMP and Secure Remote Services reporting. This also creates and configures the lockbox. The lockbox is also required for CloudIQ reporting. This is the best practice method for creating the lockbox. **NOTE:** It is also possible to manually create the lockbox later, but the manual method involves security considerations. If you select this option, the **Traps Receiver IP/Hostname** field is displayed. Enter up to two (comma-separated) IP addresses or host names of the SNMP trap receiver servers.
Security options	Disable secure communication with the MDMs	Disable the need for secure communication mode between management clients and the MDM cluster members. **NOTE:** Non-secure communication has security implications, and should be used with caution. For more information about this mode, see "Use SCLI in non-secure mode" in the *Configure and Customize PowerFlex Guide*.
	Disable secure communication with the LIAs	Disable the need for secure communication mode between management clients and the LIAs. **NOTE:** Non-secure communication has security implications, and should be used with caution. For more information about this mode, see "Use SCLI in non-secure mode" in the *Configure and Customize PowerFlex Guide*.
	Disable authentication between internal components	Disable the need for authentication between SDSs and MDMs.
	Use only the following trusted IP addresses for LIA (in addition to this gateway IP address)	Limit which PowerFlex Gateway can communicate with the LIAs. If you select this option, enter a comma-separated list of the IP addresses to trust (up to 32 IP

Table 22. Advanced options (continued)

Option type	Option	Description
		addresses, including the IP address of this PowerFlex Gateway). The PowerFlex Gateway IP address is automatically added to the list and does not have to be explicitly mentioned. If the PowerFlex Gateway uses multiple IP addresses, enter all the addresses.

5. You can use the PowerFlex Installer to configure Syslog event reporting. You can also configure these features after installation, using the CLI. To configure Syslog reporting now, select **Configure the way Syslog events are sent**, and enter the following parameters:
   a. **Syslog Server**
      The host name or IP address of the syslog server where the messages should be sent. Enter up to two server names or addresses, comma-separated.
   b. **Port**
      The port of the syslog server (default 1468)
   c. **Syslog Facility**
      The facility level (default: Local0)
6. Review the displayed information, by scrolling through the tables, and by hovering over the blue hyperlinks. To view the password that will be configured for a specific component, click **show**.
   If you have not done so yet, keep a record of all the passwords that you have assigned to the system components.
7. Scroll down to the **Storage Pool Configuration** section of the table, select values for the following items. If the CSV file contained values in the **Storage Pool Configuration** section, the values will already appear here. These values are mandatory.
   - The expected **Device Media Type**
   - The **External Acceleration** type (if used):
     - none—No devices are accelerated by a non-PowerFlex read or write cache
     - read—All devices are accelerated by a non-PowerFlex read cache
     - write—All devices are accelerated by a non-PowerFlex write cache
     - read_and_write—All devices are accelerated by both non-PowerFlex read and write cache

     This input is required in order to prevent the generation of false alerts for media type mismatches. For example, if an HDD device is added which the SDS perceives as being too fast to fit the HDD criteria, alerts might be generated. External acceleration/caching is explained in the *Getting to Know Dell PowerFlex Guide*.
   - **Data Layout** type (use with caution; data layout type cannot be changed after Storage Pool creation)
   - If Fine Granularity data layout is selected, select a **Fine Granularity ACCP (Acceleration Pool)**, and a **Compression Method**
   - If Medium Granularity data layout is selected, the **Zero Padding** option is available for selection. Use with caution, because the zero padding setting can only be changed in Storage Pools without any storage devices.
   - **Replication Journal Capacity Percentage** must be defined if SDRs are deployed in the system. For more information about calculating the appropriate value for your system, refer to the *Getting to Know Dell PowerFlex Guide*.

   NOTE: Fine Granularity can only be selected if the Storage Pool contains NVDIMMs and SSDs.
8. Click **Start Installation**.
9. Select the **Monitor** tab.

Results

Installation begins. Proceed to the next task to perform the remaining mandatory steps in the installation process.

Complete the customized installation—two-layer system backend

Complete the installation of the PowerFlex software packages using the PowerFlex Installer. The installation performs the following phases: query, upload, install, and configure.

Steps

1. Select the **Monitor** tab.

   The **Install - query** page appears.

   In the query stage, the PowerFlex Installer validates that there are no previously installed PowerFlex components on any of the requested nodes.

   When extending an existing system, the query phase does expect to find currently-installed nodes.

   You can change the display by selecting the following:

   - **Auto Refresh** (default)
   - **Hide Completed Successfully**
   - **Show All Phases**

   At the end of each phase, a status message appears.

   If an error message is displayed during the process, you can either try to resolve the problem and then retry the failed operations (with the **Retry failed** button), continue to the next phase, or abort the installation.

   CAUTION: Once you proceed to the next stage, you will not be able to use the PowerFlex Installer to retry failed tasks.
2. When the query phase is complete, close the status message, and click **Start upload phase**.

   The **Install - upload** page appears, displaying which PowerFlex packages are being uploaded to each server, and the status of each command.
3. When the upload phase is complete, close the status message, and click **Start install phase**.

   The **Install - install** page appears, displaying the status of the installation commands.
4. When the install phase is complete, close the status message, and click **Start configure phase**.

   The **Install - configure** page appears, displaying the status of the configuration commands.
5. When all processes are finished, close the status message, and click **Mark operation completed**.

   The PowerFlex system installation is complete!

   NOTE: Marking the operation completed signals to the PowerFlex Installer that it can now be used for other installations, upgrades, and so on.

   The post-installation notes appear, directing you to the steps necessary to start using your storage.

Next steps

- Install the PowerFlex presentation server.
- If SDCs were not installed during the process described above, install SDCs on the frontend servers. For ESXi servers, register the vSphere PowerFlex plug-in first, to enable installation of SDCs on the ESXi servers.
- If SDCs were installed during the process described above, perform the post-installation tasks described in this guide.
- After you have finished the tasks in this guide, perform the mandatory and optional post-deployment configuration tasks described in the *Configure and Customize PowerFlex Guide*.

Install SDC on frontend servers

If an SDC was not installed on frontend servers during initial deployment using the CSV file, follow the appropriate instructions for your frontend servers' operating system.

Consult the Support Matrix to ensure that the intended SDC host's operating system is compatible for use with your PowerFlex system.

The following methods can be used for SDC installation:

- ESXi:
  - vSphere PowerFlex plug-in (supported for vSphere version 6.7 or earlier)
  - Command line (esxcli)
- Linux:
  - PowerFlex Installer procedure for adding nodes to an existing system (not applicable for hLinux). For more information, see the *Configure and Customize PowerFlex Guide*.
  - Command line
- AIX: command line
- Windows:
  - PowerFlex Installer procedure for adding nodes to an existing system (not applicable for hLinux). For more information, see the *Configure and Customize PowerFlex Guide*.
  - Command line

Install SDC on ESXi servers

Install SDC on ESXi servers using either command line esxcli commands, or the vSphere PowerFlex plug-in.

Use of the PowerFlex plug-in requires registration of the PowerFlex plug-in on a vCenter.

Register the vSphere PowerFlex plug-in

Use PowerCLI for VMware to register the PowerFlex plug-in on a vCenter.

Prerequisites

NOTE: Adobe Flash Player reached end-of-life on December 31st, 2020. To use the vSphere PowerFlex plug-in from the Flash-based vSphere web client, apply a workaround for browsers as explained in the following knowledge base article:

Impact on PowerFlex vSphere Plugin after Adobe Flash End of Life

- Ensure that earlier versions of the PowerFlex plug-in have been cleaned from the vCenter.
- Ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the host storing the PowerFlex plug-in file.
- Obtain the vCenter credentials.
- Ensure that you have downloaded and extracted the required software package from the Customer Support site for your SDC's operating system.

NOTE: Only ESXi v6.x-based systems can be installed and managed using the vSphere PowerFlex plug-in. ESXi v7 is not supported.

About this task

To use your own host, see "Register the vSphere PowerFlex plug-in manually".

Steps

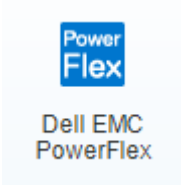
1. Copy the following file to your designated host (preferably your vCenter), that has PowerCLI for VMware installed:
   `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip`
2. Extract the contents of the ZIP file.
3. Using PowerCLI for VMware, set to `Run as administrator`, run the following script:

   ```
   VxFlexOSPluginSetup-3.6-X.<build>.ps1
   ```

   a. Type the vCenter name or IP address, user name, and password.
   You can also use this script to unregister a PowerFlex plug-in or to create an SVM template.
   b. For `Choose mode`, select option `1, Register PowerFlex plug-in`.
   c. Read the upgrade notice, and type `y` to continue.
   d. For `Select Registration Mode`, select `Standard` (simplified, using embedded Tomcat).
   This step may take a few minutes.

NOTE: You can use the `Advanced` option to install the PowerFlex plug-in using a PowerFlex Gateway from a previous installation, or using your own web service. In either case, you must place this version's `plugin.zip` file (`EMC-ScaleIO-vSphere-web-plugin-3.6-X.<build>.zip`) in your resources folder before running the installation. To use a previous PowerFlex Gateway, the resource folder is `PowerFlex Gateway installation folder\webapps\root\resources`.

e. If necessary, accept the thumbprint.

4. Log in to the vSphere web client.
   If you are already logged in, log out, then log in again.
5. Return to the PowerCLI window, and press ENTER to return to the script menu and complete the PowerFlex plug-in installation.
6. From the vSphere **Home** tab, verify that the PowerFlex icon 

is visible in the **Inventories** section.

Results

The PowerFlex plug-in is registered. If the PowerFlex icon is missing, the vCenter server failed to register the PowerFlex plug-in, due to one of the following reasons:

- Connectivity problem between the vSphere web client server and the host storing the PowerFlex plug-in (for example, network / firewall etc.).

  Resolution: Verify that there is communication between the vSphere web client server and the host storing the PowerFlex plug-in.

- URL problem when using an external web server.

  Resolution: Verify that the URL is https:// and is pointing to the correct web server IP address (PowerFlex Gateway).

For information about how to use the log to troubleshoot problems that may arise, see "Troubleshooting PowerFlex plug-in registration issues".

Install SDC on an ESXi server using the vSphere PowerFlex plug-in

Install SDC on ESXi servers using the vSphere PowerFlex plug-in.

Steps

1. Prepare the ESXi hosts:
   a. Log in to the vSphere web client.
   b. Open the PowerFlex plug-in window by clicking the **PowerFlex** icon.
   c. From the **Basic Tasks** section of the PowerFlex page, click **Pre-Deployment Actions**.
   The **Pre-Deployment Actions** page appears.
   d. Drill-down on the vCenter to display connected ESXi hosts.
   e. For every ESXi host that might be part of this PowerFlex system (even if only in the future):
      i. Select **Install SDC**.
      ii. Type **ESXi Root Password**.

      NOTE: Do not select **Convert ISO**. This feature is in preparation for a future release of VxFlex Ready Node.

   f. Click **Run**.
   The status appears in the dialog box.
   g. When finished, click **Finish**.
   h. You must restart the ESXi hosts before proceeding.
2. Register the system:
   a. From the PowerFlex plug-in page, click **Register PowerFlex system**.

b. Enter the credentials of the system.
c. Click **OK**.

3. Deploy PowerFlex:
   a. From the PowerFlex plug-in page, click **Deploy PowerFlex environment**.
   The PowerFlex deployment wizard is displayed.
   b. In the **Select Installation** page, select **Add servers to a registered PowerFlex system**, and select the system that you want to extend.
   c. Continue with the deployment steps, adding the new SDC nodes.
   You can skip steps that do not need to be changed. When adding components, the wizard adjusts the displayed pages to options that are relevant to the current PowerFlex system.
4. Complete the deployment.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install the SDC on an ESXi-based server and connect it to PowerFlex using esxcli

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host
- The required installation software package for your SDC's operating system (available from the zipped software packages that is downloaded from the Customer Support site)
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID needs to conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

The following procedure explains how to manually install an external SDC on an ESXi-based using esxcli in command line.

NOTE: This procedure requires two server reboots.

Steps

1. On the ESXi-based server on which you are installing the SDC, set the acceptance level:

```
esxcli software acceptance set --level PartnerSupported
```

2. Install the SDC:

Option	Description
ESXi 6.x	`esxcli software vib install -d <path_to_zip_file>` where *<path_to_zip_file>* is the absolute path to the SDC ZIP file. **NOTE:** The SDC offline ESXi 6.0 bundle is the same for all ESXi 6.x versions.

Option	Description
ESXi 7	`esxcli software component apply -d <path_to_zip_file>` where *<path_to_zip_file>* is the absolute path to the SDC ZIP file.

3. Get permission to reboot the ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot is required, as described in this procedure.

4. Set the IP address of the MDM:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
   IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
   ```

   where

   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a "," symbol.
     - Separate multiple MDM clusters with the "+" symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one:

       ```
       10.20.30.40,50.60.70.80+11.22.33.44
       ```

   - *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
   - Optional: use `IoctlMdmPasswordStr` for SDC authentication to the MDM for access to the PowerFlex system.

   CAUTION: A few applications require an advanced setting in ESXi for specific vital product data (VPD) pages. For more information about changing `blkScsiUnitSerialLenByStandard` parameter on an ESXi host for PowerFlex volumes, see Procedure to change parameter blkScsiUnitSerialLenByStandard on ESXi for PowerFlex volumes. Perform this procedure only if your applications require this setting. This is not a common setting.

5. Back up the parameters:

   ```
   /bin/auto-backup.sh
   ```

6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

   ```
   vmkload_mod -l | grep scini
   ```

   Output similar to the following indicates that the driver is loaded:

   ```
   scini 2 808
   ```

   The SDC component is installed on the ESXi-based server.

Results

The SDC is installed on the ESXi-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install SDC on a Linux-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Customer Support site)
- Secure Boot is not supported; ensure that Secure Boot is disabled in the BIOS

About this task

The following procedure explains how to manually install an external SDC on a Linux-based server. External SDCs can be deployed by the PowerFlex Installer. For more information about adding components to an existing system, see the *Deploy Dell PowerFlex Guide*.

The steps in the procedure below are relevant for Linux RHEL/CentOS/Oracle Linux operating systems. Deployment on other operating systems require the modifications described in the following table:

Table 23. Modifications for deployment on other operating systems

OS	Modifications required
Ubuntu/OL	Before installing PowerFlex, ensure that you have followed the required preparation procedures relating to various types of Linux operating systems. ● PowerFlex component packages are delivered as TAR files. Before installing, perform the following: 1. Untar all the packages: `tar -xvf <tar_file>` This yields SIOB files. 2. Extract the DEB from the SIOB files: `siob_extract <siob_file>` 3. The DEB file is in the following format: `EMC-ScaleIO-sdc-3.6-X.<build>.<operating_system>.X.X.x86_64.deb` You will use the extracted DEB files for the installation. ● Some commands are a bit different, noted where applicable.

Steps

1. Install the GPG key on every server on which SDC will be installed. From the PowerFlex installation folder, run the following command on every server:

```
rpm --import RPM-GPG-KEY-ScaleIO
```

2. In command line, install the SDC:
   - RHEL/CentOS /Oracle Linux/SLES

```
MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
```

   - Ubuntu (These files must be extracted before use, as described above.)

```
MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
```

   where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located
3. To verify that the driver is loaded, the following command can be used:

- SLES 15:

```
service sdc status
```

- All other Linux flavors:

```
service scini status
```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that were added to the system.

Install SDC on an AIX-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. These instructions are relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Customer Support site)

About this task

The following procedure explains how to manually install an external SDC on an AIX-based server. The PowerFlex Installer cannot be used.

Steps

1. In command line, install the SDC:

```
MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
```

   where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.

     The SDC package is in a format similar to this: `EMC-ScaleIO-sdc-3.6-X.<build>.aix7.aix7.2.ppc.rpm`

2. Optional: For SDC authentication to the MDM for access, in the command line, enter:

```
chdev -l scini -a "mdm_password=<ip>-<password>"
```

Results

The SDC is installed on the AIX-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the recommended post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install SDC on a Windows-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Customer Support site)

About this task

The following procedure explains how to manually install an external SDC on a Windows-based server. Alternatively, you can install the external SDC using the PowerFlex Installer. For more information about adding system components to an existing system, refer to the *Deploy Dell PowerFlex Guide*.

NOTE: This procedure requires a server reboot.

Steps

1. On the Windows-based server on which you are installing the SDC, run the following command in command line:

   ```
   msiexec /i <SDC_PATH>.msi MDM_IP=<LIST_VIP_MDM_IPS>
   ```

   where
   - *<SDC_PATH>* is the path where the SDC installation package is located
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not add tiebreaker addresses here.
2. Get permission to reboot the Windows-based server, and perform a reboot to load the SDC driver on the server.

Results

The SDC is installed on the Windows-based server, and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install the PowerFlex presentation server

Install the PowerFlex presentation server.

Prerequisites

- Ensure that the workstation complies with the PowerFlex system requirements.
- Ensure that you have the PowerFlex presentation server installation file. It is included in the PowerFlex complete software package that you downloaded from PowerFlex Software Drivers & Downloads page.

NOTE: It is not recommended to install the PowerFlex presentation server on the same host as the PowerFlex Gateway. If you must install them on the same host, the PowerFlex Gateway must be installed before the PowerFlex presentation server. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway checks that port 8443 is available, although it does not use this port.

Steps

1. Install the PowerFlex presentation server on a server:

- Linux:

```
rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
```

- Debian (run with administrator privileges):

```
sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system, and configure users. For more information, refer to the *Configure and Customize Dell PowerFlex*.

SDC (scini) driver configuration for SELinux

SELinux may prevent the scini driver from starting on RHEL-based SDC nodes.

About this task

If SELinux is enabled and enforced, it can prevent the SDC driver (scini) from starting on RHEL-based nodes.

NOTE: To remove the SELinux module, run: `semodule -r scini`

Perform the following procedure to prevent this problem:

Steps

1. On the SDC, use a text editor to create two text files, in any directory:
   a. Create a file called `scini.fe`, and add the following text to it:

```
/usr/bin/emc/scaleio/(.*).ko  system_u:object_r:modules_object_t:s0
/bin/emc/scaleio/(.*).ko  system_u:object_r:modules_object_t:s0
```

   b. Create a file called `scini.te`, and add the following text to it:

```
module scini 1.0;

require {
        type insmod_t;
        type modules_object_t;
        class system module_load;
}
```

2. Build and install the SELinux module, by running the following commands:

```
checkmodule -M -m scini.te -o scini.mod
semodule_package -o scini.pp -m scini.mod -f scini.fe
semodule -i scini.pp
restorecon -R -v /bin/emc/scaleio/
```

3. To check the current configuration, run the command:

```
sestatus
```

   Output similar to the following should be displayed, showing that *SELinux status* is "enabled", and *Current mode* is "enforcing":

```
SELinus status: enabled
SELinuxfs mount: /sys/fs/selinux
SELinux root directory: /etc/selinux
Loaded policy name: targeted
Current mode: enforcing
Mode from config file: enforcing
Policy MLS status: enabled
Policy deny_unknown status: allowed
```

```
Memory protection checking: actual (secure)
Max kernel policy version: 31
```

4. To check that the scini module is configured in SELinux, run the command:

```
semodule --list=full | grep scini
```

Output similar to the following should be displayed:

```
400    scini                pp
```

5. Start the scini service:

```
service scini start
```

5

Add components to a PowerFlex system—Linux

Add components to an existing Linux-based PowerFlex system.

Preparation for adding components to a system

Prepare items required for adding components to an existing Linux-based PowerFlex system.

Prerequisites

- Ensure that the new nodes meet the requirements listed in "Deployment checklist-Linux"
- Prepare new nodes that are based on SLES, Oracle Linux and Ubuntu as described in the "Preparation for deployment" section
- For new Fine Granularity Storage Pool hosts, prepare their NVDIMMs as DAX devices, as described in the "Preparation for deployment" section
- Ensure that you have the CSV file that was used to deploy the existing PowerFlex system.
- The secure communication mode of components must match the mode of the system to which the components are being added. If they do not match, you must either change the mode of the components to be added, or change the mode of the system so that they match.

About this task

Add and configure new components to existing or new servers using this method, but do not edit or remove existing components or configurations, such as storage devices, acceleration devices, Protection Domains, or MDM roles.

These procedures can also be used to add and configure new Storage Pools. However, do not modify or remove existing Storage Pools.

Steps

1. Make a copy of the CSV configuration file.
2. Add the details of the new components and Storage Pools to the CSV file.

   For information about the fields in the CSV file, see "Prepare the CSV file."
3. Save the CSV file for future use.

Add components to an existing system

Add components to an existing system, using the CSV file and the PowerFlex Gateway.

Prerequisites

- Ensure that you have the URL and the login credentials for the PowerFlex Installer
- Ensure that you have the IP address or host name of the primary MDM and password
- For native LIA authentication, ensure that you have the LIA password
- For LDAP LIA authentication, ensure that you have the LDAP Server URI, LDAP Group, LDAP BaseDN, LDAP UserName and LDAP password
- If you are adding component types that were not installed previously, ensure that you have the installation packages for those components
- Ensure that you have the CSV file containing the details of the components that you want to add to the system

Steps

1. Point your browser to the URL of the PowerFlex Installer, and log in using your credentials.
2. For components that were not installed previously, upload installation packages:
   a. At the top of the **Welcome** window, click **Packages**.

      You may need to re-authenticate.

   b. From the **Manage Installation Packages** page, click **Browse**, and browse to the extracted PowerFlex component packages. Select all the required files, and then click **Open**.

      Select these packages for each operating system type that will be used in your PowerFlex system:

      - MDM
      - SDS
      - SDC
      - LIA
      - RFcache XCACHE package (optional; required for Medium Granularity Storage Pool acceleration)
      - VASAPROVIDER package (optional; required for VMware vVols)
      - SDR (optional; required for the replication feature)

   c. Click **Upload**.

      The uploaded installation packages appear in the packages table.

   d. Click **Proceed to Install**.

   The PowerFlex software packages have been uploaded to the PowerFlex Installer, in preparation for deployment
3. On the **Provide Installation Topology** page, click **Browse**, browse to the location of the installation CSV file, select the file, and click **Open**.
4. Next to **New installation**, click the down-arrow and select **Add to existing sys**.
5. Click **Upload Installation CSV**.
   After successful upload of the CSV file, the **Installation Configuration** tab appears.
6. Enter the following:
   a. For **MDM Credentials**, enter the IP address or host name of the system's primary MDM, and its password.
   b. For **LIA Authentication**, select the LIA authentication mode used in your system (Native or LDAP).
      - If native authentication is used for LIA, enter the password in the **password** and **confirm password** boxes.
      - If LDAP authentication is used for LIA, for **LDAP Authentication**, enter values in the **LDAP Server URI**, **LDAP Group**, **LDAP BaseDN**, **LDAP UserName** and **LDAP password** boxes.
7. Review the displayed information, by scrolling through the tables, and by hovering over the blue hyperlinks. To view the password that will be configured for a specific component, click **show**.

   If you have not done so yet, keep a record of all the passwords that you have assigned to the system components.
8. If new Storage Pools were added to the CSV file, configure them as described in the following step. Do not change any configurations of existing Storage Pools. If no Storage Pools were added, skip the next step.
9. Scroll down to the **Storage Pool Configuration** section of the table, select values for the following items. If the CSV file contained values in the **Storage Pool Configuration** section, the values will already appear here. These values are mandatory.
   - The expected **Device Media Type**
   - The **External Acceleration** type (if used):
     - none—No devices are accelerated by a non-PowerFlex read or write cache
     - read—All devices are accelerated by a non-PowerFlex read cache
     - write—All devices are accelerated by a non-PowerFlex write cache
     - read_and_write—All devices are accelerated by both non-PowerFlex read and write cache

     This input is required in order to prevent the generation of false alerts for media type mismatches. For example, if an HDD device is added which the SDS perceives as being too fast to fit the HDD criteria, alerts might be generated. External acceleration/caching is explained in the *Getting to Know Dell PowerFlex Guide*.

   - **Data Layout** type (use with caution; data layout type cannot be changed after Storage Pool creation)
   - If Fine Granularity data layout is selected, select a **Fine Granularity ACCP (Acceleration Pool)**, and a **Compression Method**
   - If Medium Granularity data layout is selected, the **Zero Padding** option is available for selection. Use with caution, because the zero padding setting can only be changed in Storage Pools without any storage devices.

- **Replication Journal Capacity Percentage** must be defined if SDRs are deployed in the system. For more information about calculating the appropriate value for your system, refer to the *Getting to Know Dell PowerFlex Guide*.

NOTE: Fine Granularity can only be selected if the Storage Pool contains NVDIMMs and SSDs.

10. Click **Start Installation**.
11. At the top of the screen, click the **Monitor** tab.

    The **Install - query** screen appears.

    In the query stage, the PowerFlex Installer validates that there are no previously installed PowerFlex components on any of the requested nodes.

    When extending an existing system, the query phase does expect to find currently-installed nodes.

    You can change the display by selecting the following:

    - **Auto Refresh** (default)
    - **Hide Completed Successfully**
    - **Show All Phases**

    At the end of each phase, a status message is displayed.

    If an error message is displayed during the process, you can try to resolve the problem and then retry the failed operations (with the **Retry failed** button), continue to the next phase, or abort the installation.

    NOTE: Once you proceed to the next stage, you will not be able to use the PowerFlex Installer to retry failed tasks.
12. When the query phase is complete, close the status message, and at the top right side of the screen, click **Start upload phase**.

    The **Install - upload** screen appears, displaying which PowerFlex packages are being uploaded to each server, and the status of each command.
13. When the upload phase is complete, close the status message, and at the top right side of the screen, click **Start install phase**.

    The **Install - install** screen appears, displaying the status of the installation commands.
14. When the install phase is complete, close the status message, and at the top right side of the screen, click **Start configure phase**.

    The **Install - configure** screen appears, displaying the status of the configuration commands.
15. When all processes are finished, close the status message, and click **Mark operation completed**.

    The PowerFlex system installation is complete!

    NOTE: Marking the operation completed signals to the PowerFlex Installer that it can now be used for other installations, upgrades, and so on.

    The post-installation notes appear, directing you to the steps necessary to start using your storage.
16. For two-layer systems, if SDCs were not deployed in the previous steps, install them now. For more information, see "Install SDC on frontend servers."

II

VMware ESXi

Deployment procedures for VMware ESXi-based systems.

6

Overview—deployment steps

This section summarizes deployment options and steps for VMware ESXi-based environments.

Deployment workflow—ESXi

The PowerFlex deployment workflow for ESXi-based systems is described in this summary. The vSphere PowerFlex plug-in deployment wizard is used to deploy and configure PowerFlex components on ESXi servers automatically, from one workstation. Only ESXi v6.x-based systems can be installed using this method. ESXi v7 is not supported.

Although RDM architecture is supported, DirectPath architecture is the recommended best practice for RAID and SAS Controller managed drives.

For two-layer systems where MDM and SDS components are installed on Linux servers, and SDC components are installed on ESXi-based servers, follow the procedures described for Linux two-layer systems.

For VxFlex Ready Node systems, refer to your product's *Hardware Configuration and Operating System Installation Guide* and *Deployment Guide* for step-by-step installation instructions.

The main steps in the PowerFlex deployment workflow are:

1. Prepare the environment.
2. Register the PowerFlex plug-in.
3. Upload the OVA template.
4. Deploy PowerFlex.
5. For DirectPath mode deployments, add storage/acceleration devices.
6. For some two-layer system configurations, install SDCs on external servers.
7. Install the PowerFlex presentation server.
8. Enable storage.
9. Perform mandatory, recommended, and optional post-deployment tasks.
10. Move the system to production.

NOTE: It is highly recommended to use the PowerFlex plug-in deployment wizard for deployment and provisioning. If the need arises, you can perform these tasks manually, or in a combination with the wizard.

Deployment checklist—ESXi

Checklist for deployment requirements in ESXi-based environments.

The following table provides a checklist of items required for PowerFlex deployment.

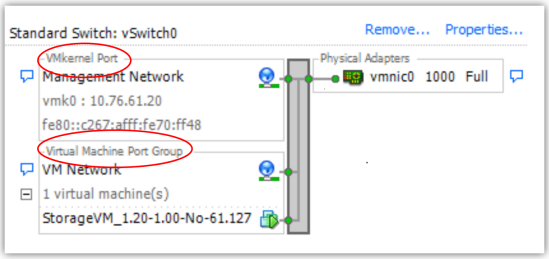
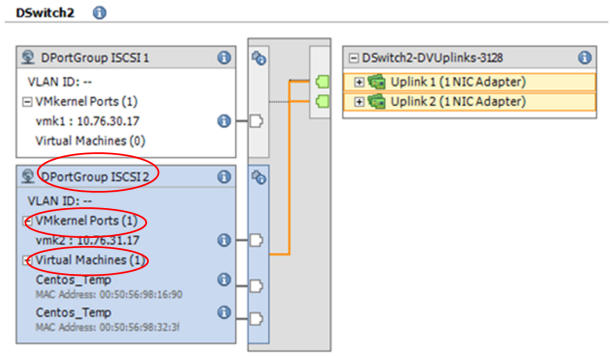
Table 24. Pre-deployment checklist

Item	Checked
General	
All servers intended for hosting PowerFlex components and the PowerFlex Gateway meet the system requirements described in the *Getting to Know Dell PowerFlex Guide*. **NOTE:** The Dell Technologies-supplied hardware configurations satisfy all the hardware requirements.	-
All hosts must have unique hostnames.	-
Prepare passwords for the PowerFlex Gateway, MDM, and LIA. These passwords will be required for future access to system components for upgrade and certain configuration procedures. Store them in a secure location.	-

Table 24. Pre-deployment checklist (continued)

Item	Checked
Servers that contained PowerFlex components from other systems must be cleaned before deployment of a new system. Use operating system tools to remove all those PowerFlex components.	-
Familiarize yourself with the ESX vStorage APIs for Array Integration (VAAI) features supported by the system. For more information, see the "Architecture" section in *Getting to Know PowerFlex*.	-
To use secure authentication mode, ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system.	-
Gateway	
The server that hosts the PowerFlex Gateway has adequate memory to run the PowerFlex Installer and any other applications. Refer to the *Getting to Know Dell PowerFlex Guide* for PowerFlex Gateway server requirements.	-
The PowerFlex Gateway must not be installed on a server that hosts either the SDC component, or on an SDS component that uses RFcache acceleration feature. **NOTE:** In hyper-converged infrastructure setups, the PowerFlex Gateway component is installed on a separate storage VM (automatic when deployment is done using the vSphere PowerFlex plug-in).	-
ESXi	
The host from which you run the PowerShell (.ps1) script, where the vSphere PowerFlex plug-in will be stored, meets the following prerequisites: • Runs on Windows, with 64-bit Oracle JRE 1.8 build 291 or OpenJDK JDK 8 build 292.b10 installed • PowerCLI from VMware (not Windows PowerShell) is installed. PowerCLI version should match the version of the vCenter (for example, use PowerCLI v6.5 with vCenter v6.7). • Has incoming and outgoing communication access to the vCenter **NOTE:** If your vCenter runs on Windows OS, it is recommended to use it.	-
All ESXi hosts selected to have either an MDM, tiebreaker, or SDS component installed on them, must have a defined local datastore, with a minimum of 16 GB free space (to be used for the storage virtual machine (SVM)). If the ESXi host is only being used as an SDC, there is no need for this datastore. **NOTE:** If you are planning to host the SVM template on one of the nodes, you will need an additional 16 GB of space. If you are planning to deploy the PowerFlex Gateway SVM on a node that will also host another SVM, you will need an additional 16 GB of space.	-
Storage/acceleration devices	
Configure disks so they are visible to the operating system according to operating system and vendor hardware guidelines.	-
Plan your system so that storage VMs (SVMs) with SDS and MDM roles reside on the local datastore, because moving them after deployment is not supported.	-
If you are planning to use DirectPath, ensure that you enable the relevant setting in the Server BIOS (in Dell servers, refer to SR-IOV setting).	-
A minimum of three devices must be added to each SDS, that all meet the following prerequisites: • A minimum of 240 GB available storage capacity. • The devices must be free of partitions. • If a device is part of a datastore, before adding the device, you must either remove the datastore, or use the PowerFlex plug-in **Advanced settings** option to enable VMDK creation. • If the device has the ESXi operating system on it, you must use the PowerFlex plug-in **Advanced settings** option to enable VMDK creation. **NOTE:** Use of VMDK-based disks for storage devices is not recommended, and should be used only if no other option is possible.	-
In the following migration or re-installation scenarios, you should contact customer support before deploying: • If a datastore was previously used for RDM, and you want to use VMDK devices now. • If a datastore was previously used for VMDK, and you want to use RDM devices now.	-

Table 24. Pre-deployment checklist (continued)

Item	Checked
On nodes configured with NVDIMMs, ensure that the ndctl and daxio packages are installed: ndctl package—for example: `ndctl-64.1-2.el7.x86_64.rpm` daxio package—for example: `daxio-1.5.1-2.1.el7.x86_64.rpm`	-
Networking	
If you are using a firewall to protect your network, and need to enable ports and IP addresses, refer to the *PowerFlex Security Configuration Guide*.	-
The management network on all of the ESXi hosts that are part of the PowerFlex system must have the following items configured: ● Virtual Machine Port Group. (The name must be the same on all the ESXi hosts.)  ● When using distributed switches, the vDS must have the following items configured: ○ VMkernel port (necessary only if using a single network) ○ dvPortGroup for virtual machines  	-
If using only a single network (management), manually configure the following: ● vSwitch ● VMkernel Port ● Virtual Machine Port Group ● VMkernel Port Binding	-

7

Preparation for deployment

Prepare the software packages, register the vSphere PowerFlex plug-in, upload the OVA file, and prepare the ESXi servers.

NOTE: The OVA file upload is not necessary for VxFlex Ready Node servers.

Download installation packages

Download and extract the complete software package for this version.

About this task

Steps

1. From the Online Support site (Dell Support for ScaleIO Drivers), download the complete software for this version.
2. Extract the packages from the downloaded file.
3. From the `PowerFlex <version> Complete VMware SW` directory, save the following software artifacts. These artifacts must be saved on the server that will be used to register the vSphere PowerFlex plug-in and to upload the OVA template.

   NOTE: PowerCLI from VMware must also be installed on this server.

   a. PowerFlex <version> ESX
   b. PowerFlex <version> vSphere plug-in

      The PowerFlex plug-in will be used to deploy PowerFlex on the servers in the system.

      NOTE: The SDC package for ESXi 7 cannot be installed in an automated process by the PowerFlex plug-in.

4. From the `PowerFlex <version> Windows` directory, save the software artifacts for the PowerFlex GUI.

   The GUI will be installed and used after deployment to configure some of the system features.

Preparation of NVDIMMs for use in PowerFlex in ESXi environments

Prepare your environment for use of NVDIMMs as acceleration devices in Fine Granularity storage pools.

Prerequisites for adding an NVDIMM to an ESXi 6.7 system

Meet the following prerequisites before adding an NVDIMM to an ESXi 6.7 based PowerFlex system.

Before you begin adding an NVDIMM to an ESXi 6.7 based PowerFlex system, ensure the following:

- You are using VMware version 6.7 for the ESXi host and the vCenter server.
- The VM version of your SVM is version 14 or higher.
- The firmware of the NVDIMM is updated to the version supported by your server. For example, for compatible Dell EMC PowerEdge servers, use NVDIMM firmware version 9324 or higher.
- The ESXi host can detect the NVDIMM.
- You calculate required NVDIMM capacity, according to number of FG devices and their size.
- There is enough NVDIMM capacity on the ESXi host.

- The appropriate VMware licensing is required for persistent memory support; an Enterprise Plus license might be required. To check whether your license supports persistent memory, refer to VMware Compare vSphere Editions and Features.

Calculate required NVDIMM and RAM capacity for FG SDS

Use the following formulas to calculate the necessary capacity for NVDIMM and RAM.

About this task

NOTE: When adding NVDIMM capacity to the SVM, the complete capacity of the NVDIMM must be allocated in a single device when adding to the SVM. For example, 16 GB*2 of NVDIMM is added as 31 GB (rounded down, since VMware uses some of the capacity for datastore management).

Steps

1. Calculate the required NVDIMM and RAM capacity using the following table, where *X* is the total capacity of SDS devices used for the FG storage pool, and *Y* is the total RAM needed:

Table 25. NVDIMM and RAM capacity for FG capacity

FG capacity	Required NVDIMM memory size (GB)	Required RAM capacity (rounded)	Additional memory for MDM, LIA, and SVM operating system	Total RAM (rounded)
49 TB	32 GB NVDIMM (in SVM it is 31 GB)	>36 GB	MDM: 6.5 GB LIA: 0.35 GB Operating system Base: 1 GB Buffer: 2 GB CloudLink 4 GB	50 GB
49 TB < *X* < 96 TB	64 GB NVDIMM (in SVM it os 62 GB)	>41 GB <59 GB		49 GB < *Y* < 73 GB
96 TB < *X* < 128 TB (122.88 is the actual limit)	96 GB NVDIMM (in SVM it is 93 GB)	>64.5 GB <72 GB		86 GB

2. Alternatively, you can calculate NVDIMM capacity and RAM capacity using the following formulas:

   NOTE: The calculation is in binary MB, GB, and TB.

   - *NVDIMM_capacity_in_GiB* = ((100 * *Number_Of_Drives*) + (700 * Capacity_in_TiB))/1024
   - *RAM_capacity_in_GiB* = 10 + ((100 * *Number_Of_Drives*) + (550 * Capacity_in_TiB))/1024

Verify that the ESXi host can detect the NVDIMM

Before adding an NVDIMM to an ESXi-based PowerFlex system, verify that the ESXi host can detect the NVDIMM.

Prerequisites

Steps

1. Login to the vCenter.
2. Select your ESXi host.
3. Navigate to the **Summary** tab.
4. In the **Hardware** section, verify that the required amount of **Persistent Memory** is listed.

What to do next for NVDIMMs

The remaining activities required for adding NVDIMMs to ESXi-based systems should be performed after system deployment. For more information, see the *Configure and Customize Dell PowerFlex*.

Register the vSphere PowerFlex plug-in

Use PowerCLI for VMware to register the PowerFlex plug-in on a vCenter.

Prerequisites

NOTE: Adobe Flash Player reached end-of-life on December 31st, 2020. To use the vSphere PowerFlex plug-in from the Flash-based vSphere web client, apply a workaround for browsers as explained in the following knowledge base article:

Impact on PowerFlex vSphere Plugin after Adobe Flash End of Life

- Ensure that earlier versions of the PowerFlex plug-in have been cleaned from the vCenter.
- Ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the host storing the PowerFlex plug-in file.
- Obtain the vCenter credentials.
- Ensure that you have downloaded and extracted the required software package from the Customer Support site for your SDC's operating system.

NOTE: Only ESXi v6.x-based systems can be installed and managed using the vSphere PowerFlex plug-in. ESXi v7 is not supported.

About this task

To use your own host, see "Register the vSphere PowerFlex plug-in manually".

Steps

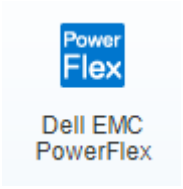
1. Copy the following file to your designated host (preferably your vCenter), that has PowerCLI for VMware installed: `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip`
2. Extract the contents of the ZIP file.
3. Using PowerCLI for VMware, set to `Run as administrator`, run the following script:

   ```
   VxFlexOSPluginSetup-3.6-X.<build>.ps1
   ```

   a. Type the vCenter name or IP address, user name, and password.
   You can also use this script to unregister a PowerFlex plug-in or to create an SVM template.
   b. For `Choose mode`, select option `1, Register PowerFlex plug-in`.
   c. Read the upgrade notice, and type `y` to continue.
   d. For `Select Registration Mode`, select `Standard` (simplified, using embedded Tomcat).
   This step may take a few minutes.

   NOTE: You can use the `Advanced` option to install the PowerFlex plug-in using a PowerFlex Gateway from a previous installation, or using your own web service. In either case, you must place this version's `plugin.zip` file (`EMC-ScaleIO-vSphere-web-plugin-3.6-X.<build>.zip`) in your resources folder before running the installation. To use a previous PowerFlex Gateway, the resource folder is `PowerFlex Gateway installation folder\webapps\root\resources`.

   e. If necessary, accept the thumbprint.
4. Log in to the vSphere web client.
   If you are already logged in, log out, then log in again.
5. Return to the PowerCLI window, and press ENTER to return to the script menu and complete the PowerFlex plug-in installation.

6. From the vSphere **Home** tab, verify that the PowerFlex icon 

is visible in the **Inventories** section.

Results

The PowerFlex plug-in is registered. If the PowerFlex icon is missing, the vCenter server failed to register the PowerFlex plug-in, due to one of the following reasons:

- Connectivity problem between the vSphere web client server and the host storing the PowerFlex plug-in (for example, network / firewall etc.).

  Resolution: Verify that there is communication between the vSphere web client server and the host storing the PowerFlex plug-in.

- URL problem when using an external web server.

  Resolution: Verify that the URL is https:// and is pointing to the correct web server IP address (PowerFlex Gateway).

For information about how to use the log to troubleshoot problems that may arise, see "Troubleshooting PowerFlex plug-in registration issues".

Register the PowerFlex plug-in manually

This topic describes an advanced way to use PowerCLI to register the vSphere PowerFlex plug-in on a vCenter.

Prerequisites

Before you begin, ensure that there is communication between the vSphere web client server and the web server storing the PowerFlex plug-in.

About this task

The PowerFlex plug-in is provided as a ZIP file that can be downloaded by the vSphere web client servers in your environment. The ZIP file can be downloaded directly from the Online Support site, or, if the web servers do not have internet access, from a file server. The `EMC-ScaleIO-vSphere-web-plugin-3.6-X.<build>.zip` file is included with the latest gateway rpm (for example, `EMC-ScaleIO-gateway-3.5-1000.175.x86_64.rpm` already includes `EMC-ScaleIO-vSphere-web-plugin-3.5.1000.175.zip`), so there is no need to manually upload the zip file to the PowerFlex Gateway.

Steps

1. You can upload the ZIP file to an HTTP or an HTTPS server.

   If you are uploading the ZIP file to an HTTP server, perform the following:

   a. On the computer where the vSphere Web client is installed, locate the `webclient.properties` file.

   Table 26. Location of `webclient.properties` file

vCenter	Operating system	Path to file
6.x	Windows	C:\ProgramData\VMware\vCenterServer\cfg\vsphere-client
	Linux	/etc/vmware/vsphere-client/

   b. Add the following line to the file:

   ```
   allowHttp=true
   ```

   c. Restart the VMware vSphere Web Client service.

2. Using PowerCLI for VMware, and set to **Run as administrator**, run `Set-ExecutionPolicy AllSigned`.
3. Close PowerCLI, and reopen it, set as **Run as administrator**.
4. Extract `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip`
5. Upload `EMC-ScaleIO-vSphere-web-plugin-3.6-X.<build>.zip` to the HTTP/HTTPS server.

6. From PowerCLI, run the following script: `VxFlexOSPluginSetup-3.6-X.<build>.ps1`
   a. Enter the vCenter name or IP address, user name, and password.
   b. For **Select Mode**, choose option `1, Register VxFlex OS plug-in`.
   c. For **Select Registration Mode**, choose `Advanced`.
   d. Enter the full URL of the plug-in ZIP file.

      If the ZIP file is located on a PowerFlex Gateway, enter the following URL: https://<*VxFlex_OS_GATEWAY_IP_ADDRESS*>.

      If you are using a web server, enter the web server's complete URL of the PowerFlex plug-in ZIP file (see the example in Step 7).

   e. If necessary, accept the thumbprint.
7. Log out, and log back in to the vSphere web client.

   Example for HTTP server:

```
.\registerVxFlexOSPlugin.ps1 -vcenter 10.103.109.16
    -userUrl "http://10.76.60.14/sample/ScaleIO-vSphere-web-plugin-1.30.0.160.zip"
```

   Example for HTTPS server:

   .\registerVxFlexOSPlugin.ps1 -vcenter 10.103.109.16

```
-userUrl
"https://10.76.61.139/sample/ScaleIO-vSphere-web-plugin-1.30.0.160.zip"
-thumbprint CA:66:49:D0:CE:D9:8C:A0:D0:93:E3:83:DE:59:25:5F:79:E1:53:B6
-adminEmail test.email@emc.com
```

   The script registers the plug-in and the following message appears:

```
Registering VxFlexOS extension...
```

```
The extension was registered successfully
```

PowerFlex plug-in Advanced Settings options

This topic describes the advanced settings options that you can use to configure the vSphere PowerFlex plug-in deployment settings.

To access these settings, on the PowerFlex screen, click **Advanced settings**.

The **Advanced Settings** dialog box appears.

Enable the features you want before starting the PowerFlex deployment, or, if you are in the middle of a deployment, you can click **Cancel**, then make the changes. When you restart the deployment, you can pick up from where you left off.

- **Enable VMDK creation**

  Enables the addition of devices as VMDK, as opposed to RDM.

  NOTE: Use of VMDK-based disks for storage devices is not recommended, and should be used only if no other option is possible.

- **Force RDM on device with unknown RDM support**

  Enables non-SCSI controller devices to be added as RDM.

  NOTE: Do not enable this option if the device does not support SCSI Inquiry Vital Data Product (VPD) page code 0x83.

- **Allow the take over of devices with existing signature**

  Enables the reuse of devices that were part of a previous PowerFlex deployment, and were not removed properly. This setting will revert to not selected after each use, and must be reactivated if required again.

- **Allow using non-local datastores for PowerFlex Gateway**

Enable deployment of the PowerFlex Gateway SVM on non-local datastores (such as remote storage), in case of insufficient space on the local datastore, or if no datastore can be found. Another advantage of having the PowerFlex Gateway available at all times is that even when an ESXi node is rebooted, the PowerFlex Gateway can vMotion like any other VM residing on a non-local datastore.

- **Parallelism limit**

  Enables the increase of the parallelism limit (default: 100), thus speeding up the deployment, which can be useful in deployment of a very large system (several hundred nodes). This is dependent on the processing power of the vCenter.

Troubleshoot PowerFlex plug-in registration issues

You can use logs to assist in troubleshooting problems that may occur during registration of the vSphere PowerFlex plug-in.

Search for issues in the log files. To find relevant log entries, search for `scaleio` in the log file. You can also use the troubleshooting procedure to completely remove and clean-up the PowerFlex plug-in files, and then reregister the PowerFlex plug-in. In PowerFlex plug-in downgrade scenarios, complete removal and clean-up procedures, followed by vCenter restart, are mandatory.

The vSphere web client (Virgo) logs are located in the following directories:

Table 27. vCenter operating system paths

vCenter	Operating system	Path to file
6.x	Windows	C:\ProgramData\VMware\vCenterServer\logs\vsphere-client\logs
	Linux	/var/log/vmware/vsphere-client/logs

Other relevant logs:

Table 28. vCenter operating system paths

vCenter	Operating system	Path to file
6.x	Windows	Plug-in logs and other files: C:\Users\vspherewebclientsvc\AppData\Roaming\VMware\scaleio Plug-in location: C:\ProgramData\VMware\vCenterServer\cfg\vsphere-client\vc-packages\vsphere-client-serenity
	Linux	Plug-in logs and other files: /etc/vmware/vsphere-client/vc-packages/scaleio Plug-in location: /etc/vmware/vsphere-client/vc-packages/vsphere-client-serenity

For more information about log collection, refer to *PowerFlex Log Collection Technical Notes*

Troubleshooting procedure for PowerFlex plug-in registration issues

To solve PowerFlex plug-in registration issues, perform a complete removal and cleanup, and then reregister the PowerFlex plug-in.

About this task

Completely clean your system of the PowerFlex plug-in in order to solve registration issues.

Steps

1. Close the existing PowerCLI sessions and open a new one (as administrator).
2. Using the PowerCLI script, unregister the PowerFlex plug-in.
3. Stop the vSphere web client service.

On Linux, for example:

```
service-control --stop vsphere-client
```

4. Delete the contents of the folder where the PowerFlex plug-in is stored.
5. Delete the contents of the PowerFlex plug-in's logs folder, or the folder itself.
   On Linux, for example: `/etc/vmware/vsphere-client/vc-packages/scaleio`
6. Clean the Virgo logs folder:
   On Linux, for example: `/var/log/vmware/vsphere-client/logs`
7. Start the vSphere web client service.
   For example, on Linux:

```
service-control --start vsphere-client
```

8. Clear your web browser's cache/cookies (or use a different web browser).
9. Using the PS1 script in PowerCLI, register the PowerFlex plug-in via PowerCLI remember

   NOTE: Do not to press ENTER until you perform the log-in into your vSphere web client (that completes the registration).

10. Once you see the PowerFlex plug-in's icon in the vSphere web client, you can press ENTER in the PowerCLI session.

Upload the OVA template

Use PowerCLI for VMware to upload the OVA template to a vCenter.

Prerequisites

- Ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the host storing the vSphere PowerFlex plug-in file.
- Ensure that the server with PowerCLI for VMware can run the PowerFlex plug-in setup script.
- Obtain the vCenter credentials.
- Ensure that you have the required software packages that were downloaded from the Customer Support site.

Steps

1. Copy the following file to your designated host (preferably your vCenter) which has PowerCLI for VMware installed:
   `ScaleIOVM_2nics_3.6.X.ova`
2. Using PowerCLI, set to `Run as administrator`, run the following script:

```
PowerFlexPluginSetup-3.6-<build>.X.ps1
```

   a. Enter the vCenter name or IP address, user name, and password.
   b. From the PowerCLI script, select **3** to `Create SVM template`.
   c. Enter the parameters described in the following table. (If you are already logged in, some of these parameters may not be necessary.)

Table 29. vCenter configuration parameters

Parameter	Description
vcenter	vCenter name or IP address
user name	vCenter user name
password	vCenter password
datacenter	The name of the data center where the datastore that will store the template resides
ova_path	The path of the SVM's OVA

Table 29. vCenter configuration parameters (continued)

Parameter	Description
datastores	A list of datastores, up to eight, on which the templates will be created. Press ENTER to stop specifying datastores. **NOTE:** Each template requires approximately 8 GB of space in the datastore.

NOTE: For best results, enter a local (not shared) datastore for each ESXi server.

For faster, parallel, deployment in large-scale environments, you can use the OVA to create SVM templates on as many as eight datastores. To do so, type the datastore names, and when you are done, leave the next line blank. The following example shows how to enter two datastores:

```
datastores[0]: datastore1
datastores[1]: datastore1 (1)
datastores[2]:
```

The upload procedure can take several minutes, during which time a temporary SVM is created, the templates are created, and then the temporary SVM is deleted.

When each template is created, a message similar to the following appears:

```
The template VxFlexOS SVM Template (v3.6.X) was successfully created
```

d. When the process is complete, type **4** to exit the PowerFlex plug-in script.

Add or replace an OVA file

If you have already used the OVA file to create a template, you cannot create another template with the same name in the same datacenter.

Either remove the original template first, or use the `VxFlexOSPluginSetup-3.6-<build>.X.ps1` script, option #3, to assign a different name to the new template.

You can also upload the OVA manually using VMware OVA upload tools, however this method is not recommended. Configure the networks manually after deployment, or during the wizard workflow. For more information, see VMware documentation.

Access the vSphere PowerFlex plug-in

Access the PowerFlex plug-in from the vSphere Web Client home screen.

Steps

- On the vSphere Web Client **Home** tab, click the **Dell Technologies PowerFlex** icon.
  The PowerFlex screen is displayed.

Prepare the ESXi hosts

Use the vSphere PowerFlex plug-in to prepare the ESXi hosts for PowerFlex deployment.

About this task

This procedure is a prerequisite for deploying a new PowerFlex system, and in some cases for extending an existing system.

Preparing the ESXi hosts includes the following:

- Install SDC on the host.

This is required for every ESXi host that will be included in the system, and it is strongly recommended to enable this for every host that might be part of the system in the future.

- Configure DirectPath

  Enables the SVM to take control of ESXi devices. If the host has multiple controllers, you must configure DirectPath on that host manually, not with the PowerFlex plug-in. After that is done, you can use the PowerFlex plug-in to deploy PowerFlex on that host and to add devices to it.

  When DirectPath is configured, all servers in the system must use DirectPath. For VxFlex Ready Node systems, this can be configured with the PowerFlexPowerFlex plug-in. For other servers, use the vSphere client to configure each ESXi host manually.

Steps

1. From the **Basic tasks** section of the Dell Technologies PowerFlex screen, click **Pre-Deployment Actions**.

   The **Pre-Deployment Actions** screen appears, showing all the ESXi hosts on all the connected vCenters.

2. Select the ESXi hosts, and select the settings required for each.

   NOTE:

   For the **Install SDC** option, it is highly recommended to select all ESXi hosts that may be included in an ESXi system, even if only in the future.

3. Type the ESX root password for each host.

   All changes or entries made at the vCenter level will be applied to all servers in the cluster.

   The **Select the SDC driver file you want to install** dialog box is displayed.

4. Select the required SDC driver file, and click **OK**.
   The SDC driver is installed on the selected nodes.
5. Click **Run**.

   The status appears in the dialog.

6. When finished, click **Finish**.
7. Restart each ESXi host.

   NOTE: You must restart the ESXi hosts before proceeding.

   After rebooting, a RAID controller that was configured with DirectPath will be displayed in the vSphere client **Configure** tab, on the **DirectPath I/O PCI Devices Available to VMs** screen.

8. When using all-SSD chassis, perform this step.

   Sometimes the ESXi nodes in an all-SSD chassis do not identify the node disks as Standard Parallel SCSI devices. If so, you must enable the **Force RDM on device with unknown RDM support** option.

   a. In the **VxFlex OS** screen, click **Advanced Settings** to display the settings options.
   b. Select the **Enable RDMs on non Parallel SCSI Controller** option and click **OK**.

Results

After finishing this task, the results of your selections are displayed after reopening the **Pre-Deployment Actions** screen.

Next steps

Proceed with the PowerFlex deployment.

Configure ESXi security settings

Before deploying PowerFlex systems, you must configure some VMware security features. This procedure describes how to perform the security configurations using the vSphere web client.

Prerequisites

Ensure that you have:

- IP address of VMware vCenter server
- User name and password for accessing the vCenter with the vSphere Web Client

About this task

NOTE: Ensure that the remote connection is established within 1 to 2 minutes. Otherwise, lockdown mode is enabled on the ESXi servers, preventing you from subsequent remote connectivity. When in lockdown mode, you can connect to the ESXi servers locally.

Steps

1. Log in to the vSphere web client.
   The **vSphere Web Client** window is displayed with **Navigator** and **Home** panes. The **Navigator** pane displays network entities as nodes, and the **Home** pane displays network entities as icons.
2. In the **Navigator** pane, select **Home** > **Hosts and Clusters**.
   Preconfigured hosts, clusters, and datacenters are displayed as navigation tree nodes, in the **Navigator** pane.
3. Configure the security-related parameters:
   a. In the **Navigator** pane, select an ESXi server from a host, cluster, or datacenter node .
   b. From the **Manage** tab, select **Settings** > **System** > **Advanced System Settings**.
   A list of system-defined parameters with predefined parameter values is displayed.
   c. In the **Search** field, find the following security parameters:
      - `Set Security.AccountLockFailures` = *0*

        The parameter value defines the maximum number of failed login attempts allowed, before locking out the user's account. The zero (0) value disables account locking.

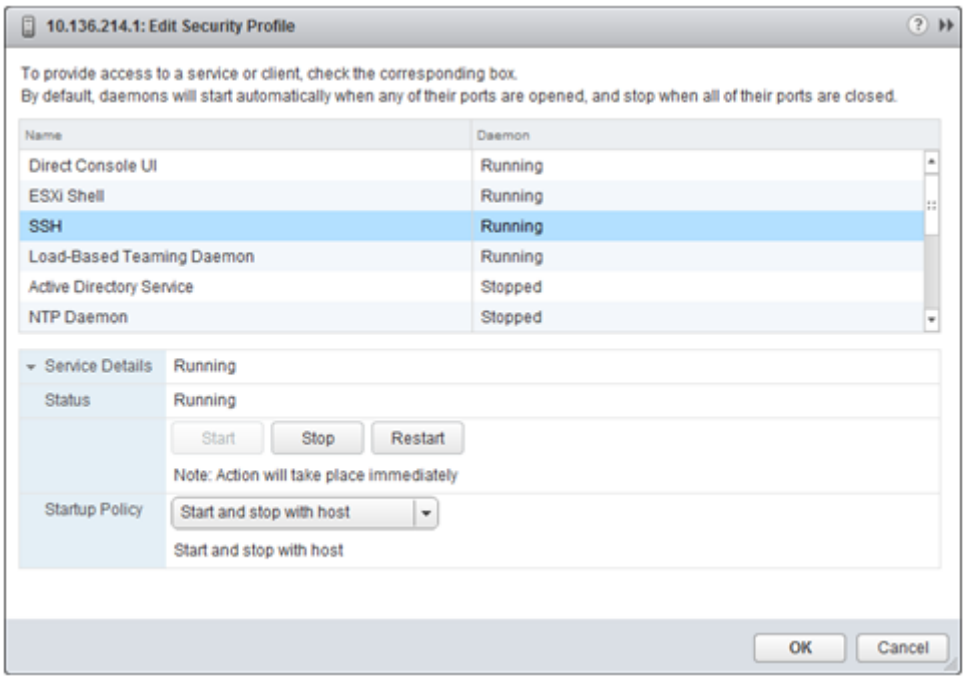
        NOTE: To disable the locking out of user accounts, the parameter value must be set as zero (0).

      - `Set Security.AccountUnlockTime` = *1* (in seconds)

        The parameter value defines the duration in seconds to lock out a user's account, after exceeding the maximum number of allowed failed login attempts.

        NOTE: For ease of operation, it is recommended that the parameter value be set as 1 or 2 seconds.

   d. If required, click **Edit** and update the security parameter values.
4. Configure the `Lockdown Mode` parameter:
   a. From the **Manage** tab, select **Settings** > **System** > **Security Profile**.
   b. Scroll to the **Lockdown Mode** pane and verify the value of the `Lockdown Mode` parameter.
   c. Verify that `Lockdown Mode` is set to **Disabled**.
   If Lockdown mode is not disabled, click **Edit** and change the parameter value to **Disabled**.
5. Configure the SSH parameter:
   a. From the **Manage** tab, select **Settings** > **System** > **Security Profile**.
   b. Scroll to the **Services** pane and verify the SSH option.



c. SSH must be running. If SSH is stopped, change its status to **Running**:
   i. In the **Services** pane, click **Edit**.
   ii. In the **Edit Security Profile** dialog boc, select the SSH parameter.

   The SSH status is *Stopped*.

   iii. Click **Start**.

   The SSH status changes to *Running*.

   iv. Click **OK**

   SSH status changes to *Running*.

6. Restart the management service or restart the server.

Clean the PowerFlex VMware environment

For two-layer systems, or ESXi-based systems, clean the PowerFlex VMware environment in order to perform a clean installation. This procedure is recommended for cases when previously defined networks will be used again, or when PowerFlex components such as SDCs that ran on other networks exist in the environment.

Prerequisites

NOTE: This procedure is only required when PowerFlex plug-in installation failed.

Before you begin, unmap and delete any PowerFlex volumes in your system.

If necessary, unregister PowerFlex from within the vSphere PowerFlex plug-in, and delete all the PowerFlex SVMs.

Steps

1. Close the existing PowerCLI sessions, set PowerCLI to **Run as administrator**, and then open a new session.
2. Using the PS1 script, unregister the PowerFlex plug-in.
3. Stop the vSphere web client service:
   VC Linux: `service-control --stop vsphere-client`
4. Delete the contents of the PowerFlex plug-in folder.
   The vSphere web client (Virgo) plug-in folders are located at:

Table 30. Location of vSphere web client plug-in folders

vCenter	Operating system	Path to file
6.x	Windows	C:\ProgramData\VMware\vCenterServer\cfg\vsphere-client\vc-packages\vsphere-client-serenity
	Linux	/etc/vmware/vsphere-client/vc-packages/vsphere-client-serenity

5. Delete the `scaleio` folder or its contents.
   The `scaleio` folders are located at:

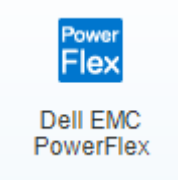
Table 31. Location of `scaleio` folder

vCenter	Operating system	Path to file
6.x	Windows	C:\Users\vspherewebclientsvc\AppData\Roaming\VMware\scaleio
	Linux	/etc/vmware/vsphere-client/vc-packages/scaleio

6. Start the vSphere web client service:
   VC Linux: `service-control --start vsphere-client`
7. Clear your web browser's cache and cookies, or open a different web browser.
8. Using the PS1 script in PowerCLI, register the PowerFlex plug-in.

   NOTE: Do not press ENTER at this point.

9. After you have logged in to the vSphere web client to complete the registration and you see the PowerFlex icon

   , press ENTER in the PowerCLI session.
   This stops the embedded Tomcat server.
10. If necessary, remove the SDC module parameters and VIB from the ESXi servers:
    a. Connect to each ESXi server via SSH.
    b. Run:

```
~ # esxcli system module parameters set -m scini -p ""
~ # esxcli software vib remove -n scaleio-sdc-esx6.x
```

    c. Reboot each ESXi server.
11. Proceed to the procedures for re-registering the PowerFlex plug-in, and then deploy PowerFlex.

8

Deploy PowerFlex using the vSphere PowerFlex plug-in

Deploy PowerFlex using an automated workflow.

Install the PowerFlex presentation server

Install the PowerFlex presentation server.

Prerequisites

- Ensure that the workstation complies with the PowerFlex system requirements.
- Ensure that you have the PowerFlex presentation server installation file. It is included in the PowerFlex complete software package that you downloaded from PowerFlex Software Drivers & Downloads page.

Steps

1. Install the PowerFlex presentation server on a server:
   - Linux:

     ```
     rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
     ```

   - Debian (run with administrator privileges):

     ```
     sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
     ```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system, and configure users. For more information, refer to the *Configure and Customize Dell PowerFlex*.

Deploy PowerFlex with DirectPath device management

Use the vSphere PowerFlex plug-in deployment wizard to deploy PowerFlex when devices are managed by VMware DirectPath I/O.

Prerequisites

- Ensure that all ESXi servers to be added to the system have been pre-configured to enable DirectPath.
- Configure advanced installation options (optional):
  - Allow devices that were used in other PowerFlex systems to be taken over for this system.
  - Allow the use of non-local datastores for the PowerFlex Gateway.
  - Increase the parallelism limit.

To access these settings, click **Advanced settings** on the PowerFlex plug-in's **PowerFlex** screen.

NOTE: Only ESXi v6.x-based systems can be installed using the vSphere PowerFlex plug-in. ESXi v7 is not supported. Use manual deployment procedures for ESXi 7.

Steps

1. From the **Basic tasks** section of the screen, click **Deploy PowerFlex environment**.

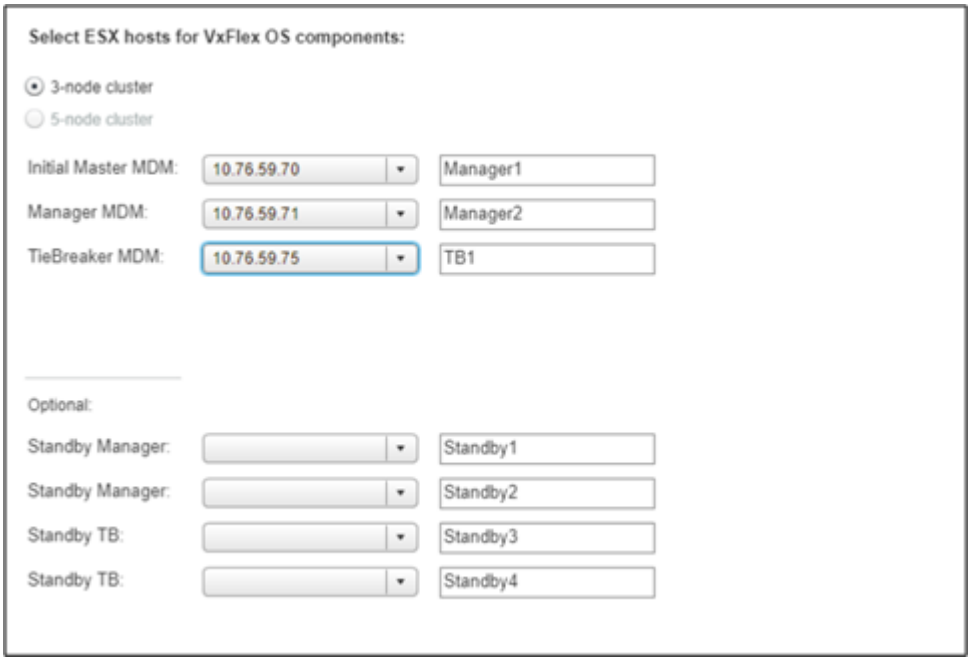
The PowerFlex PowerFlex plug-in deployment wizard begins. If you exited the previous deployment before completion, you can return from where you left off.

NOTE: The PowerFlex plug-in deployment wizard operates on the assumption that you are using the provided PowerFlex OVA template to create the PowerFlex virtual machines.

2. In the **Select Installation** screen, start the deployment of a new system:
   a. Select **Create new PowerFlex system**.
   b. Review and approve the license terms.
   c. Click **Next**.
3. In the **Create New System** screen, type the following, then click **Next**:
   - **System Name**: Type a unique name for this system.
   - **Admin Password**: Type and confirm a password for the PowerFlex admin user. The password must meet the listed criteria.
4. In the **Add ESXi Hosts to Cluster** screen, select the ESXi hosts to add as part of the system:
   a. Select the vCenter on which to deploy the PowerFlex system.
      The vCenter information is populated in the lower part of the screen.
   b. Select **Set up a DirectPath-based system**.
   c. Expand the vCenter, select the ESXi hosts to add to the PowerFlex system, then click **Next**.

      NOTE: To configure PowerFlex, you must select a minimum of three ESXi hosts. ESXi hosts that do not have the SDC installed, or hosts for which DirectPath was configured before deployment, but DirectPath was not selected in the previous step, will not be available.

   The **Select Management Components** screen appears:



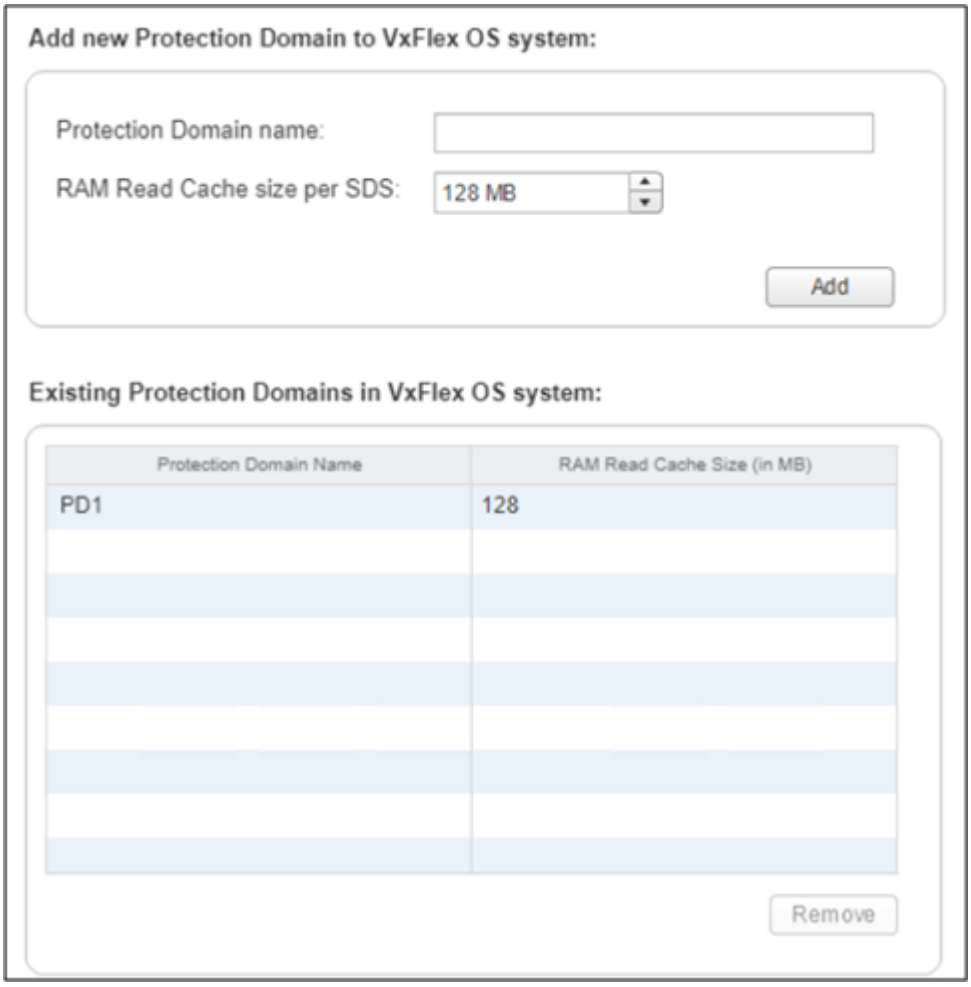
5. Configure the management components:
   a. Select to deploy a 3-node or 5-node cluster.
      The next fields on this screen will change, depending on your choice.
   b. Select an ESXi server to serve for each of the MDM cluster roles.
      You can give a name to the MDM servers, such as Manager1, and so on.
   c. Select ESXi servers to serve as Standby Manager and tiebreaker roles (optional).
   d. Click **Next**.

   The **Configure Performance, Sizing, and Syslog** screen appears.
6. Configure the following settings (optional), then click **Next**:
   - To remove the high-performance profile from components, clear their check boxes.
   - To configure the allocation of storage virtual machine (SVM) RAM, select from the following:

- To use default RAM allocation, select **Standard size**.
- To use custom settings, select **Custom size**, and type the maximum capacity and maximum number of volumes.

- To configure syslog reporting, select **Configure syslog**, and type the syslog server, port (default: 1468), and facility (default: 0).
- To configure DNS servers, type their details.

The **Configure Protection Domains** screen appears:



You can create (or remove) Protection Domains (PD). You must create at least one PD.

7. Create a Protection Domain:
   a. Enter the following information:
      - **Protection Domain name**: It is recommended to use a meaningful name.
      - **Read RAM Cache size per SDS**: Minimum 128 MB (You can increase this for your environment needs.)

        NOTE: If you enter a RMcache size that is over the limit, the memory calculator sets the size according to the calculation that is performed on each host during deployment.

   b. Click **Add**.
      The added PDs appear in the lower section of the screen, together with the existing PDs. To remove a newly created PD, select it and click **Remove**.
   c. To create an additional PD, repeat this step.
   d. Click **Next**.

   The **Configure Acceleration Pools** screen appears. In this screen, you can create one or more Acceleration Pools, which is used to accelerate storage.
8. Create an Acceleration Pool:
   a. Enter the Acceleration Pool name.
   b. Select the Protection Domain to which the Acceleration Pool belongs.
   c. Click **Add**, and then **Next**.

The **Create a new Storage Pool** screen appears.

In the **Configure Storage Pools** screen, you can create (or remove) Storage Pools (SP). You must create at least one SP.

9. Create a Storage Pool:
   a. Type the **Storage Pool name**: It is recommended to use meaningful names.
   b. Select the PD to which to add the SP.
   c. Select the expected **Device Media Type** of the devices that will be added to the SP.
   d. Select the **External Acceleration** type (if used):
      - none—No devices are accelerated by a non-PowerFlex read or write cache
      - read—All devices are accelerated by a non-PowerFlex read cache
      - write—All devices are accelerated by a non-PowerFlex write cache
      - read_and_write—All devices are accelerated by both non-PowerFlex read and write cache
   e. To enable zero padding, select **Enable zero padding**. Zero padding must be enabled for using the background scanner in data comparison mode.
   f. To enable Read Flash cache, select **Enable RFcache**.
   g. Click **Add**.
   The added SPs appear in the lower section of the screen, together with the existing PDs. To remove a newly created SP, select it and click **Remove**.
   h. To create additional SPs, repeat this step.
   i. Click **Next**.

   The **Create Fault Sets** screen appears. You can use this screen to create Fault Sets (optional).

   NOTE: When defining Fault Sets, you must follow the guidelines described in the "Architecture" section of the *Getting to Know PowerFlex* guide. Failure to do so may prevent creation of volumes.

10. Create a Fault Set (optional):
    a. Type the **Fault Set name**. It is recommended to use meaningful names.
    b. Select to which PD to add the Fault Set.
    c. Click **Add**
    Added Fault Sets appear in the lower section of the screen, inside the folder of the parent PD. You can remove a newly created Fault Set by selecting it and clicking **Remove**.
    d. Repeat to create additional Fault Sets (a minimum of three is required), then click **Next**.

    The **Add SDSs** screen appears.

11. Configure the following for every ESXi host or SVM, then click **Next**:
    a. For every SVM in a DirectPath deployment, you must select **SDS** and assign a **Protection Domain**.

    NOTE: To make the same selections for every ESXi in a cluster, you can make your selections per cluster or datacenter.

    b. You can select a **Fault Set** (optional).
    c. Type the **Total Planned Capacity** for the SDS, even if it is only being partially populated now. This sets the memory allocation for the SVM, thus avoiding manual memory calculations.

    If you do not configure this now, you can set the SVM memory later manually, as described in "SVM manual memory allocation."
    d. Click **Next**.

    Adding devices to SDS is done after the deployment is complete.

    The **Add SDCs** screen appears.

12. Configure the SDCs:
    a. For each ESXi to be added as an SDC:
       i. Select the **SDC** check box.
       ii. Type the ESXi root password.

       NOTE: To show the entered ESXi passwords, select **Show passwords**.

    b. Choose whether to enable or disable the LUN comparison for ESXi hosts.

    In general, in environments where the SDC is installed on ESXi and also on physical hosts, you should set this to **Disable**.

    NOTE: Before disabling LUN comparison, consult your environment administrator.

    c. Click **Next**.

The **Configure Upgrade Components** dialog box appears.

13. Configure the PowerFlex Gateway and Lightweight Installation Agent (LIA):
    a. Select an ESXi to host the PowerFlex Gateway SVM.

       A unique SVM will be created for the PowerFlex Gateway.

       If the previously-selected ESXi servers do not have sufficient free space (on any datastore) to contain the PowerFlex SVM template, an SVM, and the PowerFlex Gateway SVM, the option to select an ESXi host in this step will not be available. It will be done automatically.

    b. Type and confirm a password for the PowerFlex Gateway administrative user.
    c. Type and confirm a password for the LIA.

       The password must be the same across all SVMs in the system.
    d. Click **Next**.

       NOTE: You can only move to the next step if the passwords meet the listed criteria, and if the confirmation passwords match the entered passwords.

    The **Select OVA Template** screen appears.
14. Configure templates:
    a. Select the template to use to create the SVM.

       The default is **PowerFlex SVM Template**. If you uploaded a template to multiple datastores, you can select them all, for faster deployment.

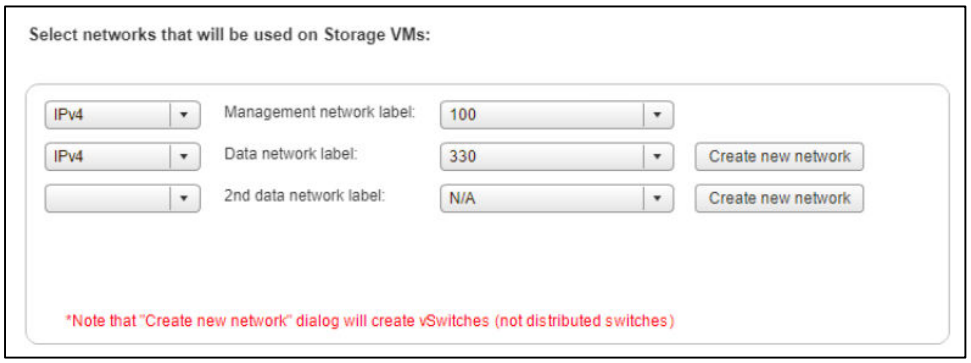
       If the PowerFlex Gateway selection was performed automatically in the previous step (indicating insufficient space), you must choose at least two templates in this step, one of which will be converted to the PowerFlex Gateway SVM.

       After selecting the templates, the deployment wizard will automatically select one of the ESXis with the templates to host the PowerFlex Gateway, and during deployment will convert the template to a VM for the PowerFlex Gateway (instead of cloning the template).

       NOTE: If you select a custom template, ensure that it is compatible with the PowerFlex plug-in and the PowerFlex MDM.

    b. Type and confirm a new root password that will be used for all SVMs to be created.
    c. Click **Next**.

    The **Configure Networks** screen appears:



15. Select the network configuration. You can select an existing (simple or distributed) network, or select **Create a new network**.

    The **Create a new network** command is only relevant for a regular vSwitch, and not for a distributed vSwitch.

    You can use a single network for management and data transfer or separate networks. Separating the networks is recommended for security and increased efficiency. You can select one data network or two data networks.

    The management network, used to connect and manage the SVMs, is normally connected to the client management network, a 1 GB network.

    The data network is internal, enabling communication between the PowerFlex components, and is recommended to be at least a 10 GB network.

    For high availability and performance, it is recommended to have two data networks.

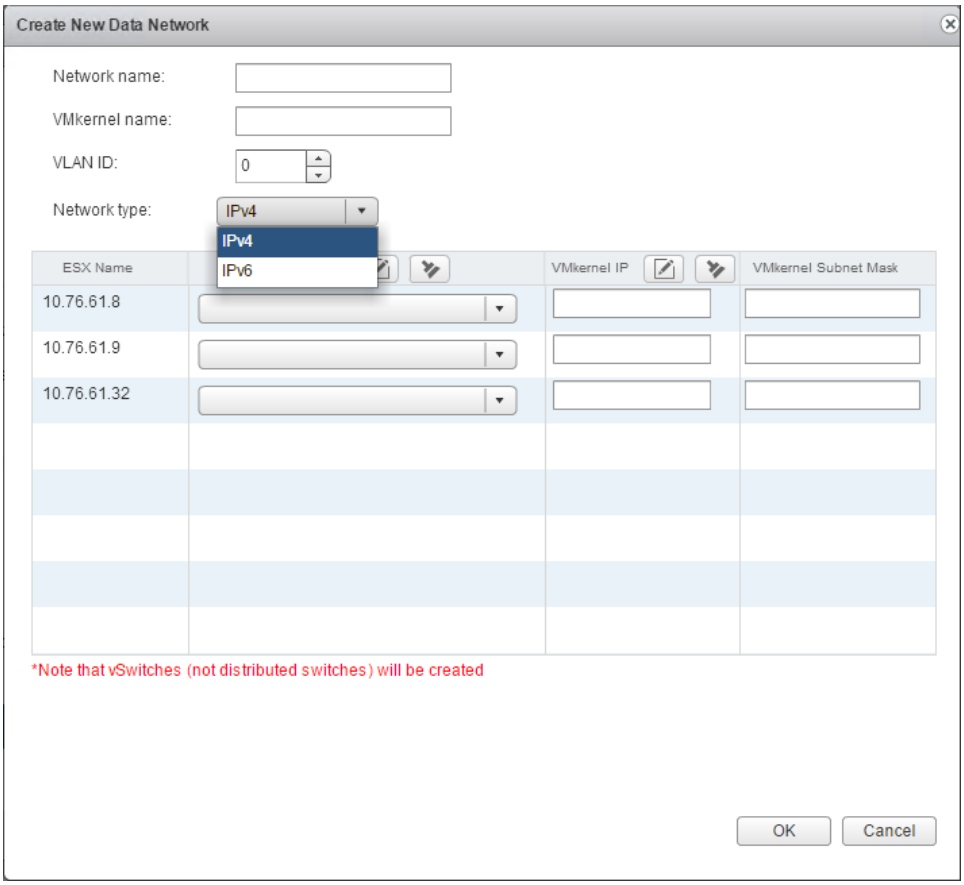
NOTE: The selected networks must have communication with all of the system nodes. In some cases, while the wizard does verify that the network names match, this does not guarantee communication, as the VLAN IDs may have been manually altered.

a. To use one network, select a protocol (IPv4 or IPv6) and a management network, click **Next** and proceed with SVM configuration.

   For best results, it is highly recommended to use the plug-in to create the data networks, as opposed to creating them manually.

b. To use separate networks, select a protocol (IPv4 or IPv6) for the management network label and one or two data network labels. If the data network already exists (such as a customer pre-configured distributed switch or a simple vswitch), select it from the dropdown box. Otherwise, configure the data network by clicking **Create new network**.

   The **Create New Data Network** screen appears:



c. Configure the networks:

- **Network name**: The name of the VMware network
- **VMkernel name**: The name of the VMkernel (used to support multipathing)
- **VLAN ID**: the network ID
- **Network type**: IPv4 or IPv6
- For each ESXi, select a **Data NIC**, a **VMkernel IP**, and a **VmKernel Subnet Mask**.

  NOTE: You can click  to auto-fill the values for **Data NIC** and **VMkernel IP**.

d. Click **OK**.

   The data network is created.

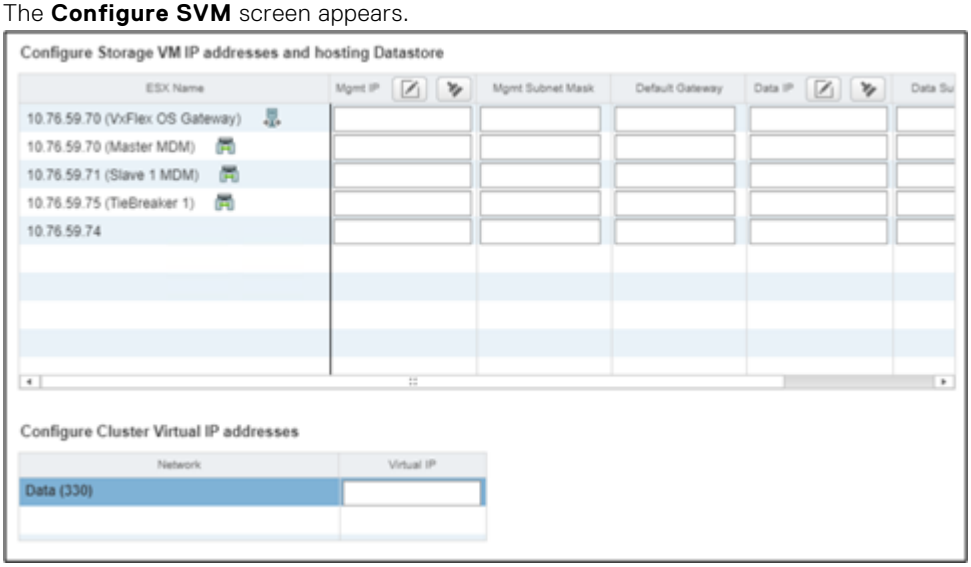
   The wizard automatically configures the following for the data network:

- vSwitch
- VMkernel Port
- Virtual Machine Port Group

- VMkernel Port Binding

e. Click **Next**.

The **Configure SVM** screen appears.



16. Configure all the SVMs:
    a. Enter the IP address, subnet mask, and default gateway for the management network, then the data network.
    b. Enter the Cluster Virtual IP address for each network interface.
    c. Select a datastore, or allow automatic selection.
    d. Configure the cluster's virtual IP addresses by entering the virtual IP address for each data network.
    e. Click **Next**.

    Icons indicate the role that the server plays in the PowerFlex system. You can select  to auto-fill the values for IP addresses.

    The **Review Summary** screen appears.
17. Review the configuration.

    Click **Finish** to begin deployment or **Back** to make changes.
18. Type the vCenter user name and password, then click **OK** to begin the deployment.

    The **Deployment Progress** screen appears.

    During the deployment process you can view progress, pause the deployment, and view logs.

    To pause the deployment, click **Pause**. Steps that are already in progress will pause when they are complete.

    After pausing, select one of the following options:

    - **Continue deployment** to continue.
    - **Abort** to abort the deployment process.
    - **Cancel and Rollback entire deployment** to roll back all deployment activities (rollback cannot be canceled once started).
    - **Rollback failed tasks** to roll back only the tasks that failed (rollback cannot be canceled once started).
19. When the deployment is complete, click **Finish**.

    If a task failed, click **Continue deployment** to try again.

Next steps

- After deployment is complete, set all SVMs to start automatically with the system. Do not set SVMs under the VMware resource-pool feature.
- Use the PowerFlex plug-in to add devices to the SDSs.
- Perform the post-installation tasks described in this guide.
- Perform mandatory and optional configuration tasks in the *Configure and Customize PowerFlex Guide*.

NOTE: When using DirectPath, also known as PCI-passthrough on the SVM, the following alert appears in vCenter: "Virtual Machine memory Usage". This is a VMware limitation and cannot be resolved. Refer to the VMware KB for more details: Memory usage alarm triggers for certain types of Virtual Machines in ESXi.

Add devices to an SDS—PowerFlex plug-in

Use the vSphere PowerFlex plug-in to add devices to an SDS in PowerFlex.

About this task

In a DirectPath-based system, you add devices only after the deployment. In an RDM/VMDK-based PowerFlex, you can add devices during and after the deployment.

You can add a Storage device or an Acceleration device to an SDS or to all SDSs in the system. For a Storage device, you must define the Storage Pool and Media Type for each device you add. For Acceleration device, you must select the Acceleration Pool (which may be RFCache or NVDIMM) for each device added. For Fine Granularity (FG), an Acceleration device must be configured with NVDIMM. For more information on FG, see "Data layout" in the *Getting to Know Guide*.

All data on added devices will be erased.

NOTE: If you intend to enable zero padding on a Storage Pool, you must do so before you add any devices to the Storage Pool.

Steps

1. From the **SDSs** screen of the PowerFlex plug-in, select one of the following:
   - Right-click a specific SDS, and then select **Add devices to a single SDS**.
   - Right-click any SDS, and then select **Add devices to PowerFlex system**.

   The **Add Device** dialog box is displayed. All devices that can be attached to the selected SDS are listed. For the system view, all SDSs are listed, and you can choose devices to add for each SDS. It may take a few moments to load the list of devices from the vCenter.
2. Add devices:
   - One-at-a-time:
     a. Select whether the device should be used for storage or to provide acceleration.
     b. Select the Storage Pool to which the devices should be assigned.
     c. To enable the use of devices that may have been part of a previous PowerFlex system, select **Allow the take over of devices with existing signature**.
     d. Click **OK**.
   - All devices on a server at once:
     a. Click **Select all devices**.
     b. Select whether to use the devices for storage or to provide acceleration.
     c. Select the Storage Pool to which the devices should be assigned.
     d. To enable the use of devices that may have been part of a previous PowerFlex system, select **Allow the take over of devices with existing signature**.
     e. Click **Assign**.
3. Confirm the action by typing the PowerFlex password.
4. When the add operation is complete, click **Close**.

Results

The devices are added.

Deploy PowerFlex with RDM/VMDK device management

Use the VMware deployment wizard to deploy PowerFlex when devices are configured as RDM or VMDK. Note that DirectPath architecture is the recommended best practice for RAID and SAS Controller managed drives.

Prerequisites

Configure advanced installation options (optional):

- Ensure that all ESXi servers to be added to the system were pre-configured with the pre-deployment steps.
- Configure advanced installation options (optional):
  - Enable creation of VMDK.

    NOTE: Use of VMDK-based disks for storage devices is not recommended, and should be used only if no other option is possible.

  - Force RDM on device with unknown RDM support.
  - Allow the taking over of devices that were used in other PowerFlex systems.
  - Allow the use of non-local datastores for the PowerFlex Gateway.
  - Increase the parallelism limit.

To access these settings, click **Advanced settings** on the **PowerFlex** screen.

NOTE: Only ESXi v6.x-based systems can be installed using the vSphere PowerFlex plug-in. ESXi v7 is not supported. Use manual deployment procedures for ESXi 7.

About this task

For two-layer systems where only the SDCs are deployed on ESXi servers, follow the deployment procedures for two-layer systems.

Steps

1. From the **Basic tasks** section of the screen, click **Deploy PowerFlex environment**.

   The PowerFlex VMware deployment wizard begins. If you exited the previous deployment before completion, you can return from where you left off.

   NOTE: When you use the deployment wizard, it is assumed that you are using the provided PowerFlex OVA template to create the PowerFlex virtual machines.

2. In the **Select Installation type** screen, start the deployment of a new system:
   a. Select **Create new PowerFlex system**.
   b. Review and approve the license terms.
   c. Click **Next**.
3. In the **Create new system** screen, enter the following, then click **Next**:
   - **System Name**: Enter a unique name for this system.
   - **Admin Password**: Enter and confirm a password for the PowerFlex admin user. The password must meet the listed criteria.
4. In the **Add ESX Hosts to Cluster** screen, select the ESXi hosts to add as part of the system:
   a. Select the vCenter on which to deploy the PowerFlex system.
   The vCenter information is populated in the lower part of the screen.
   b. Expand the vCenter, select the ESXi hosts to add to the PowerFlex system, then click **Next**.

   NOTE: To configure PowerFlex, you must select a minimum of three ESXi hosts. ESXi hosts that do not have the SDC installed, or hosts for which DirectPath was configured before deployment, but DirectPath was not selected in the previous step, will not be available.

   The **Select management components** screen appears.
5. Configure the management components:

a. Select an option to deploy either a 3-node or 5-node cluster.

   The next fields on this screen will change, depending on your choice.

b. Select an ESXi server to serve for each of the MDM cluster roles.

   You can give a name to the MDM servers, such as Manager1, and so on.

c. Select ESXi servers to serve as Standby Manager and tiebreaker roles (optional).

d. Click **Next**.

The **Configure Performance, Sizing, and Syslog** screen appears.

6. Configure the following settings (optional), then click **Next**:
   - To configure components with compact performance profiles, clear the high-performance option.
   - To configure the allocation of SVM RAM, select from the following:
     - To use default RAM allocation, select **Standard size**.
     - To use custom settings, select **Custom size**, and enter the maximum capacity and maximum number of volumes.
   - To configure syslog reporting, select **Configure syslog**, and enter the syslog server, port (default: 1468), and facility (default: 0).
   - To configure DNS servers, enter their details.

   The **Configure Protection Domains** screen appears.

   You can create (or remove) Protection Domains (PD). You must create at least one PD.

7. Create a Protection Domain:
   a. Enter the following information:
      - **Protection Domain name**: It is recommended to use a meaningful name.
      - **Read RAM Cache size per SDS**: Minimum 128 MB (You can increase this for your environment needs.)

   b. Click **Add**.
   The added PDs appear in the lower section of the screen, together with the existing PDs. To remove a newly created PD, select it and click **Remove**.

   c. To create an additional PD, repeat this step.

   d. Click **Next**.

   The **Configure Acceleration Pool** screen appears. In this screen, you can create an Acceleration Pool, which will be used to accelerate storage.

8. Create an Acceleration Pool:
   a. Enter the Acceleration Pool name.

   b. Select the Protection Domain to which the Acceleration Pool belongs.

   c. Click **Add**, and then **Next**.

   The **Create a new Storage Pool** screen appears.

   In the **Configure Storage Pools** screen, you can create (or remove) Storage Pools (SP). You must create at least one SP.

9. Create a Storage Pool:
   a. Enter the **Storage Pool name**: It is recommended to use meaningful names.

   b. Select to which PD to add the SP.

   c. Select the expected **Device Media Type** for the SP (hard drive or SSD).

   d. Select the **External Acceleration** type (if used):
      - none—No devices are accelerated by a non-PowerFlex read or write cache
      - read—All devices are accelerated by a non-PowerFlex read cache
      - write—All devices are accelerated by a non-PowerFlex write cache
      - read_and_write—All devices are accelerated by both non-PowerFlex read and write cache

      This input is required in order to prevent the generation of false alerts for media type mismatches. For example, if an hard drive device is added which the SDS perceives as being too fast to fit the hard drive criteria, alerts might be generated. External acceleration/caching is explained in the *Getting to Know Dell PowerFlex Guide*.

   e. To enable zero padding, select **Enable zero padding**. Zero padding must be enabled for using the background scanner in data comparison mode.

   f. To enable Read Flash cache, select the **Enable RFcache** checkbox.

   g. Click **Add**.

      This input is required in order to prevent the generation of false alerts for media type mismatches. For example, if an hard drive device is added which the SDS perceives as being too fast to fit the hard drive criteria, alerts might be generated. External acceleration/caching is explained in the *Getting to Know Dell PowerFlex Guide*.

The added SPs appear in the lower section of the screen, together with the existing PDs. To remove a newly created SP, select it and click **Remove**.

h. To create additional SPs, repeat this step.

i. Click **Next**.

The **Create Fault Sets** screen appears. You can use this screen to create Fault Sets (optional).

NOTE: When defining Fault Sets, you must follow the Fault Set guidelines described in the *Getting to Know PowerFlex* guide. Failure to do so may prevent the creation of volumes.

10. Create a Fault Set (optional):

a. Enter the **Fault Set name**. It is recommended to use meaningful names.

b. Select to which PD to add the Fault Set.

c. Click **Add**
Added Fault Sets appear in the lower section of the screen, inside the folder of the parent PD. You can remove a newly created Fault Set by selecting it and clicking **Remove**.

d. Repeat these steps to create additional Fault Sets (minimum of three), then click **Next**.

The **Add SDSs** screen appears.

11. Configure the following for every ESXi host or SVM:

a. Select the corresponding **SDS** checkbox to assign an SDS role.

NOTE: To make the same selections for every ESXi in a cluster, you can make your selections per cluster or datacenter.

b. If the node is an SDS, assign a **Protection Domain**.

c. You can select a **Fault Set** (optional).

d. Click **Next**.

The **Add devices to SDSs** screen appears, showing the clusters.

This screen has the following tabs:

- **Information** - shows the selected ESXi and cluster.
- **Assign devices** - select hosts and assign devices.
- **Replicate selection** - replicate device selections from one host to others.

NOTE: This can be useful if your ESXis have identical attached devices. For example, if you select an SSD device for the source ESXi, and then replicate this selection to the targets, the deployment wizard can automatically select all other SSD devices on the target SDSs.

Device matching is performed based on the device runtime name.

To replicate device selections, all the following conditions must be met:

- The number of devices on each ESXi must be identical.
- Source and target devices must be identical in the following ways: a) both are SSD or non-SSD, b) both have datastores on them or do not, c) both are roughly the same size (within 20%), and d) both are connected using a RAID controller or directly attached.
- At least one of the following conditions must be met: a) both SDSs are in the same Protection Domain, b) both SDSs are in different Protection Domains, but with the same list of Storage Pools, or c) the target SDS is in a Protection Domain with only one Storage Pool.

12. On the **Information** tab, select an ESXi host from a cluster, then click **Assign devices**.

The **Assign devices** tab appears.

This screen shows the devices whose free space can be added to the selected ESXi host/SDS. You should balance the capacity over the selected SDS.

13. To assign a device's space to an SDS, perform the following:

a. In the **Use for** dropdown, select **Storage** or **Acceleration**.

b. In the **Pool Name** dropdown, select the Storage Pool (SP) to which to assign the device.

NOTE: If the selected SP has RFcache enabled, you must select at least one RFcache device for that SDS node.

NOTE: You can select all available devices by clicking **Select all devices**, and selecting their use and Storage Pool.

NOTE: If you selected to create VMDK (before the deployment), the following options appear:

- **Create VMDK**. Select this for all relevant devices.
- **Select all available devices**. Click this to select all devices with a VMFS, and with unused capacity that can be added to the PowerFlex system.

c. Click **Assign**.

14. To replicate selections to other SDSs, perform the following:
    a. Select the **Replicate selection** tab.
    b. Select the ESXi whose device selection you want to replicate.
    This is the source ESXi.
    c. Select the target ESXis to which to replicate the selection of the source ESXi.
    d. Click **Copy configuration**.
    The results are displayed in the right pane of the screen.
15. When you have selected devices for all SDSs, click **Next**.

    NOTE: You must select at least one device for each SDS.

    The **Add SDCs** screen appears.
16. Configure the SDCs:
    a. For each ESXi to be added as an SDC:
        i. Select the **SDC** checkbox.
        ii. Enter the ESXi root password.

        NOTE: To show the entered ESXi passwords, select **Show passwords**.

    b. Choose whether to enable or disable LUN number comparison for ESXi hosts.

        In general, in environments where the SDC is installed on ESXi and also on physical hosts, you should set this to **Disable**.

        NOTE: Before disabling LUN number comparison, consult your environment administrator.

    c. Click **Next**.

    The **Configure Upgrade Components** dialog box appears.
17. Configure the PowerFlex Gateway and LIA:
    a. Select an ESXi to host the PowerFlex Gateway Storage virtual machine (SVM).
    A unique SVM is created for the PowerFlex Gateway.

    If the previously-selected ESXi servers do not have sufficient free space (on any datastore) to contain the PowerFlex SVM template, an SVM, and the PowerFlex Gateway SVM, you will not have an option to select an ESXi in this step. It will be done automatically.

    b. Enter and confirm a password for the PowerFlex Gateway administrative user.
    c. Enter and confirm a password for the LIA.
    The password must be the same across all SVMs in the system.
    d. Click **Next**.

        NOTE: You can only move forward if the passwords meet the listed criteria, and if the confirmation passwords match the entered passwords.

    The **Select OVA Template** screen appears:
18. Configure templates:
    a. Select the template to use to create the PowerFlex virtual machines (SVM).
    The default is **PowerFlex SVM Template**. If you uploaded a template to multiple datastores, you can select them all, for faster deployment.

    If the PowerFlex Gateway selection was performed automatically in the previous step (indicating insufficient space), you must choose at least two templates in this step, one of which will be converted to the PowerFlex Gateway SVM.

    After selecting the templates, the deployment wizard will automatically select one of the ESXis with the templates to host the PowerFlex Gateway, and during deployment will convert the template to a VM for the PowerFlex Gateway (instead of cloning the template).

NOTE: If you select a custom template, ensure that it is compatible with the PowerFlex plug-in and the PowerFlex MDM.

b. Enter and confirm a new root password that will be used for all SVMs to be created.
c. Click **Next**.

The **Configure Networks** screen appears:

19. Select the network configuration. You can select an existing (simple or distributed) network, or select **Create a new network**.

The **Create a new network** command is only relevant for a regular vSwitch, and not for a distributed vSwitch.

You can use a single network for management and data transfer, or separate networks. Separate networks are recommended for security and increased efficiency. You can select one data network, or two.

The management network, used to connect and manage the SVMs, is normally connected to the client management network, a 1 GB network.

The data network is internal, enabling communication between the PowerFlex components, and is recommended to be at least a 10 GB network.

For high availability and performance, it is recommended to have two data networks.

NOTE: The selected networks must have communication with all of the system nodes. In some cases, while the wizard does verify that the network names match, this does not guarantee communication, as the VLAN IDs may have been manually altered.

a. To use one network, select a protocol (IPv4 or IPv6), and a management network, then proceed to the next step, configuring the SVMs.

For best results, it is highly recommended to use the PowerFlex plug-in to create the data networks, as opposed to creating them manually.

b. To use separate networks, select a protocol (IPv4 or IPv6) for the management network label, and one or two data network labels. If the data network already exists (such as a customer pre-configured distributed switch or a simple vswitch), select it from the drop-down box. Otherwise, configure the data network by clicking **Create new network**.

The **Create New Data Network** screen appears.

c. Configure the networks:

NOTE: You can click  to auto-fill the values for **Data NIC** and **VMkernel IP**.

- **Network name**: The name of the VMware network
- **VMkernel name**: The name of the VMkernel (used to support multipathing)
- **VLAN ID**: the network ID
- **Network type**: IPv4 or IPv6
- For each ESXi, select a **Data NIC**, a **VMkernel IP**, and a **VmKernel Subnet Mask**.

d. Click **OK**.

The data network is created.

The wizard will automatically configure the following for the data network:

- vSwitch
- VMkernel Port
- Virtual Machine Port Group
- VMkernel Port Binding

e. Click **Next**.

The **Configure SVM** screen appears.

20. Configure all the SVMs:

NOTE: You can click  to auto-fill a range of values for IP addresses, subnet mask and default gateway.

a. Enter the IP address, subnet mask, and default gateway for the management network, then the data network.

b. Enter the Cluster Virtual IP address for each network interface.
c. You can select a datastore, or allow automatic selection.
d. Configure the cluster's virtual IP addresses by entering the virtual IP address for each data network.
e. Click **Next**.

Icons indicate the role that the server plays in the PowerFlex system.

The **Review Summary** screen appears.

21. Review the configuration.

Click **Finish** to begin deployment or **Back** to make changes.

22. Enter the vCenter user name and password, then click **OK** to begin the deployment.

The **Deployment Progress** screen appears.

During the deployment process you can view progress, pause the deployment, and view logs.

To pause the deployment, click **Pause**. Steps that are already in progress will pause after they are completed.

After pausing, select one of the following options:

- **Continue deployment** to continue.
- **Abort** to abort the deployment process.
- **Cancel and Rollback entire deployment** to roll back all deployment activities (rollback cannot be canceled once started).
- **Rollback failed tasks** to roll back only the tasks that failed (rollback cannot be canceled once started).

23. When the deployment is complete, click **Finish**.

If a task failed, click **Continue deployment** to try again.

Next steps

- After deployment is complete, set all SVMs to start automatically with the system. Do not set SVMs under the VMware resource-pool feature.
- Perform the post-installation tasks described in this guide.

Roll back deployment using the vSphere PowerFlex plug-in

In VMware environments where deployment fails due to incorrect input or failed devices, you can roll back the deployment, fix or change the specific deployment setting, and restart deployment using the vSphere PowerFlex plug-in deployment wizard.

Steps

1. When deployment fails, click **Cancel and roll back entire deployment**.
2. In the **Rollback settings** dialog box, select whether you want to remove the SDC configuration.
   - If you select **Remove SDCs configuration (requires a manual restart of the ESXis)**, the SVMs are deleted, the SDC MDM IP address configurations are removed from the ESXis, and you must manually restart the ESXis.
   - If you do not select **Remove SDCs configuration (requires a manual restart of the ESXis)**, the SVMs are deleted, the SDC MDM IP address configurations are not removed from the ESXis, and you do not need to manually restart the ESXis.
3. When the deployment roll back is complete, select **Finish deployment and return to wizard**.

   If you select **Finish**, you will not be returned to the wizard and you will have to start the deployment from the beginning; your deployment details will not be saved. After confirming that you want to finish the deployment and return to the wizard, the last screen in the deployment wizard is displayed.
4. Click **Back** or **Next** to navigate backwards or forwards through the deployment wizard.

   Change the settings that caused the initial deployment to fail.
5. Click **Finish** to finish using the wizard and to begin deployment.

9

Manually deploy PowerFlex on ESXi servers

Manually deploy a PowerFlex system on ESXi servers. When possible, it is recommended to use the automated workflow, in preference to manual procedures.

Manual Deployment of VxFlex OS on ESXi Servers

You can deploy VxFlex OS on ESXi servers manually, but when possible, it is highly recommended to deploy using the vSphere PowerFlex plug-in deployment wizard.

This section describes tasks that are necessary in cases where automated deployment is not possible or desirable, such as for scripting.

In addition to these procedures, familiarize yourself with the ESX vStorage APIs for Array Integration (VAAI) features supported by the system, as described in the Architecture chapter of the *Getting to Know Dell PowerFlex Guide*.

After manual deployment on ESXi, you must perform manual memory allocation on the SVM, as described in the *Configure and Customize Dell PowerFlex*, "SVM manual memory allocation."

Before starting, obtain the IP addresses for all of the nodes to be installed, and an additional IP address for the MDM cluster, as described in the following table:

Table 32. IP address table for manual deployment on ESXi servers

Number of networks	IP addresses required
1 network (management network)	• 1 for vmnic • 1 for SVM
2 networks	• 2 for vmnic (1 for mgmt, 1 for data) • 2 for SVM (1 for mgmt, 1 for data)
3 networks	• 3 for vmnic (1 for mgmt, 2 for data) • 3 for SVM (1 for mgmt, 2 for data)

Deploying the PowerFlex storage virtual machine (SVM)

This topic describes how to deploy the SVM and prepare it for installation of PowerFlex. This is not necessary for SDCs installed on ESXi, but all other components are installed on Storage virtual machines (SVM).

About this task

NOTE: Before deployment, ensure that networks for each ESXi to be part of the system have a properly defined Port Group and VMkernel.

To deploy the SVM on the ESXi, perform the following:

Steps

1. Download (and extract, if necessary) the files for your operating system.

   NOTE: The SDC offline ESXi 6.0 bundle is the same for all ESXi 6.x versions.

You can download all files from the Complete Software download for your software version, available from Online Support (PowerFlex Software Drivers & Downloads page).

To install with a customized Java configuration, see the *Deploy Dell PowerFlex* , "PowerFlex Gateway installation with a custom Java configuration."

2. Use the vSphere web client to connect to a vCenter.
3. From the main window, select **Hosts and Clusters**.
4. From the left panel, select the ESXi on which you would like to deploy the OVA. Right-click it, and select **Deploy OVF Template**.

   The **Deploy OVF Template** wizard appears.
5. Enter the full path to the OVA, and click **Next**.
6. Assign a unique name to the VM, and select a local (DAS) datastore to deploy the OVA, accept remaining default values in the wizard, and click **Finish**.
7. After the OVA is deployed, click the **VMs and Templates** tab.
8. Locate the VM that you deployed, right-click it, select **Template** and then **Convert to Template**.
9. Locate the new template, right-click it, and select **New VM from this Template…**.
10. Enter a unique name for the VM, and click **Next**.
11. Select the ESXi server on which you wish to deploy the SVM, and click **Next**.
12. Select a local (DAS) datastore for the SVM, and click **Finish**.
13. Repeat steps 10-13 for each ESXi server on which you wish to deploy an SVM.
14. On each ESXi physical host, configure the network and NTP (and DNS, if necessary).
    a. Using the console, start the SVM, and log in.

       The default user name is `root`, and the default password is `admin`.
    b. Configure the network:
       - IP address
       - Gateway address
       - DNS
       - NTP server

       NOTE: When using the OVA, the installation packages can be found in /root/install/.
15. On each SVM in the system, add the local disk devices as RDM devices.

    Perform this task by changing the SVM settings.
16. Configure each ESXi host to enable *Start and Stop Virtual Machines with the system*, and set the SVM to do *Automatic Startup*.
17. Install the GPG key. From the VxFlex OS installation folder, run `rpm --import RPM-GPG-KEY-ScaleIO` on every server.
18. On each SVM in the system, install the relevant PowerFlex components, using the procedures for manual deployment on SVMs.

    The packages are located under `/root/install`

    NOTE: After manual deployment on ESXi, you must perform manual memory allocation on the SVM, as described in the *Configure and Customize Dell PowerFlex*, "SVM manual memory allocation."

Upgrade the hardware version of a Storage VM

Use VMware tools to upgrade the hardware version of PowerFlex SVMs.

Prerequisites

- Upgrade VMware tools.

About this task

If the current hardware version is already the most recent version, the upgrade options in the VMware tools will not be available, and this procedure is not necessary.

Steps

1. Start the vSphere Web Client and log in to the vCenter Server
2. Power off the virtual machine.
3. Right-click the virtual machine and select the menu option to upgrade virtual hardware.
   In the vSphere Web Client, the option is **Compatibility** > **Upgrade VM Compatibility**.
4. Click **Yes** to continue.

Results

The virtual hardware is upgraded to the latest supported version.

Configure the UUID on virtual machines

To configure the virtual machine, you must enable the UUID attribute on every virtual machine.

About this task

Perform the following procedure:

Steps

1. Start the vSphere Client, and log in to a vCenter Server.
2. From the main window, select **Virtual Machines and Templates**.
3. From the left panel, right-click the virtual machine, and select **Power** > **Power Off**.
   The virtual machine powers off.
4. Right-click the virtual machine, and click **Edit Settings**.
5. From the **VM Options** tab, expand **Advanced**, and click **Edit Configuration**.
6. In the **Name** column, enter `disk.EnableUUID`.
7. In the **Value** column, enter `TRUE`.
8. Click **Add**, and **OK**.
9. Power on the virtual machine.
10. Repeat this procedure for every SVM in the VxFlex OS system.

Optimize the guest operating system performance

The configuration of the guest operating system affects the performance of the system and must be optimized for use with PowerFlex.

About this task:

The Storage VM (SVM) is optimized. If the SVM that is hosting the SDS and MDM roles is not supplied by PowerFlex, refer to the following guidelines and procedures.

Modify the device scheduler (Linux guests only)

The following command modifies the I/O scheduler of the devices. It should be run for each device that exposes a VxFlex OS volume inside the guest virtual machine.

Command

Device scheduler command

```
echo noop > /sys/block/<device_name>/queue/scheduler
```

Example

```
echo noop > /sys/block/sds/queue/scheduler
```

Modify the SCSI controller type of the guest virtual machine

To configure the SCSI controller type to use a Paravirtual SCSI (PVSCSI) adapter, perform the following steps for each guest using VxFlex OS storage.

About this task

NOTE: Before performing the following steps, consult VMware documentation to verify that your guest operating system supports Paravirtual SCSI adapters.

Steps

1. Power off the guest virtual machine.
2. Open the vSphere client, and right click on the guest.
3. Select **Edit Settings**.
4. Select **SCSI Controller 0**.
5. Click **Change Type**.
6. Select VMware **Paravirtual**.
7. Click **OK**.

Manually install PowerFlex components on the SVMs

Use the CentOS-based PowerFlex packages to install MDM and SDS components on the SVMs.

The SVMs that you installed manually run the CentOS operating system. Obtain the MDM and SDS packages for CentOS-based systems, and install them on the SVMs.

Creating the MDM cluster

You can create the MDM cluster in 5-node, 3-node, or single node mode.

You can create the MDM cluster in the following modes:

- 3-node cluster

  One primary MDM, one secondary MDM, and one tiebreaker (TB). This cluster can sustain a single point of failure.

- 5-node cluster

  One primary MDM, two secondary MDMs, and two tiebreakers. This extended cluster can sustain two points of failure.

- Single-node cluster

  One primary MDM. Single-node cluster has only one copy of the repository, thus it cannot sustain a failure. It is not recommended to use Single Mode in production systems, except in temporary situations.

  An MDM can be installed on any server. For example, one MDM can run on an application server, while the second MDM can run on any other server, for example one of the SDS nodes.

  NOTE: For CLI commands, in cluster mode, use the IP address of the primary MDM or the virtual IP address.

  The following is the proper order for creating an MDM cluster:

Order of creating a cluster:

1. Install the MDMs with Manager role (two in a 3-node cluster, or three in a 5-node cluster).
2. Install the MDMs with TB role (one in a 3-node cluster, or two in a 5-node cluster).

3. Create a single-node cluster from one of the Manager MDMs.
4. Add secondary MDMs to the cluster, as Standbys.
5. Add tiebreakers to the cluster, as Standbys.
6. Switch the cluster to 3-node or 5-node mode, while designating the standby secondary and standby tiebreaker names in the cluster.

Proceed to the appropriate section that describes the cluster mode that you want to install.

3-node MDM cluster

3-node MDM cluster example

For this example for a 3-node cluster, use the following sample topology:

Table 33. Sample manual installation topology

Role	Management IP address	MDM IP address	Virtual IP address	Virtual IP NIC	MDM name	Cluster role
MDM 1	10.103.110.152	192.168.1.152	192.168.111.152	eth4	mdm152	primary
MDM 2	10.103.110.153	192.168.1.153	192.168.111.152	eth4	mdm153	secondary
TB 1	10.103.110.154	192.168.1.154	-	-	tb154	tiebreaker
SDS and SDC are installed on all nodes.						

Install the MDM packages (3-node cluster)

Install the PowerFlex MDM package for CentOS on SVMs that will be part of a 3-node cluster.

About this task

To install PowerFlex components, perform the following:

Steps

1. Copy the following file to all MDM and tiebreaker servers (MDM 1, MDM 2, and TB 1):
   a. `EMC-ScaleIO-mdm-3.6-<build>.X.elX.x86_64.rpm`
2. Install the MDM and configure the Manager role, by running the following command on MDM 1 and MDM 2:

```
MDM_ROLE_IS_MANAGER=1 rpm -i <mdm_path.rpm>
```

   NOTE: The default MDM credentials are:
   - user name: admin
   - password: admin

   For example:

```
MDM_ROLE_IS_MANAGER=1 rpm -i EMC-ScaleIO-mdm-3.6-<build>.X.el6.x86_64.rpm
```

3. Install the MDM on TB 1:

```
MDM_ROLE_IS_MANAGER=0 rpm -i <mdm_path.rpm>
```

   For example:

```
MDM_ROLE_IS_MANAGER=0 rpm -i EMC-ScaleIO-mdm-3.6-<build>.X.el6.x86_64.rpm
```

Create the 3-node MDM cluster

Create the 3-node MDM cluster.

Steps

1. Create the MDM single-node cluster on MDM 1, by running the following command on MDM 1:

   ```
   scli --create_mdm_cluster --master_mdm_ip <IP of MDM 1> --master_mdm_management_ip
   <Management IP of MDM 1> --master_mdm_name <Name to give MDM 1> [--cluster_virtual_ip
   <IP>]  [--master_mdm_virtual_ip_interface <INTF>] --accept_license
   ```

   Example:

   ```
   scli --create_mdm_cluster --master_mdm_ip 192.168.1.152 --master_mdm_management_ip
   10.103.110.152 --master_mdm_name mdm152 --cluster_virtual_ip 192.168.111.152 --
   master_mdm_virtual_ip_interface eth4 --accept_license
   ```

   Follow these rules:

   - Each MDM name must be unique. Ensure that each server on which MDM is installed has a unique hostname.
   - Each virtual IP address must be related to a single interface.
   - When listing multiple virtual IP/interface combinations, the list of interfaces must be in the same order as the list of virtual IP addresses.

2. If prompted, press Y to continue.
3. Log in and change the MDM password, by running the following commands on MDM 1:

   ```
   scli --login --username admin --password admin
   ```

   ```
   scli --set_password --old_password admin --new_password Scaleio123
   ```

   The login password is changed from admin to Scaleio123.

4. Log in with the new credentials:

   ```
   scli --login --username admin --password Scaleio123
   ```

5. Add MDM 2 to the MDM cluster as a Standby MDM, by running the following command on MDM 1:

   ```
   scli --add_standby_mdm --new_mdm_ip <IP of MDM 2> --mdm_role manager --
   new_mdm_management_ip <Management IP of MDM 2> --new_mdm_virtual_ip_interface <INTF>
   --new_mdm_name <Name to give MDM 2>
   ```

   Example:

   ```
   scli --add_standby_mdm --new_mdm_ip 192.168.1.153 --mdm_role manager
   --new_mdm_management_ip 10.103.110.153 --new_mdm_virtual_ip_interface eth4 --
   new_mdm_name mdm153
   ```

6. Add TB 1 to the MDM cluster as a tiebreaker, by running the following command on MDM 1:

   ```
   scli --add_standby_mdm --new_mdm_ip <IP of TB 1> --mdm_role tb --new_mdm_name <Name
   to give TB 1>
   ```

   Example:

   ```
   scli --add_standby_mdm --new_mdm_ip 192.168.1.154 --mdm_role tb --new_mdm_name tb154
   ```

7. (Optional) Query the cluster to verify that it's running in single mode, by running the following command on MDM 1:

   ```
   scli --query_cluster
   ```

   Output, similar to the following, is displayed:

```
Cluster: Mode: 1_node
```

8. Switch to 3-node cluster mode, by running the following command on MDM 1:

```
scli --switch_cluster_mode --cluster_mode 3_node --add_slave_mdm_name <MDM 2 name> --
add_tb_name <TB 1 name>
```

Example:

```
scli --switch_cluster_mode --cluster_mode 3_node --add_slave_mdm_name mdm153 --
add_tb_name tb154
```

9. (Optional) Query the cluster to verify that it is running in cluster mode, by running the following command on MDM 1:

```
scli --query_cluster
```

Output, similar to the following, is displayed:

```
# scli --query_cluster
Cluster:
    Mode: 3_node, State: Normal, Active: 3/3, Replicas: 2/2
    Virtual IPs: 192.168.111.152
Master MDM:
    Name: MDM152, ID: 0x7d76db0d2ad840b0
        IPs: 192.168.1.152, Management IPs: 10.103.110.152, Port: 9011, Virtual IP
interfaces: eth4
        Version: 2.0.13000
Slave MDMs:
    Name: MDM153, ID: 0x41788ea96626fe21
        IPs: 192.168.1.153, Management IPs: 10.103.110.153, Port: 9011, Virtual IP
interfaces: eth4
        Status: Normal, Version: 2.0.13000
Tiebreakers:
    Name: MDM154, ID: 0x3d8263b64fcf2642
        IPs: 192.168.1.154, Port: 9011
        Status: Normal, Version: 2.0.13000
```

5-node MDM cluster

5-node MDM cluster example

For this example for a 5-node cluster, use the following sample topology:

Table 34. Sample manual installation topology

Role	Management IP address	MDM IP address	Virtual IP address	Virtual IP NIC	MDM name	Cluster role
MDM 1	10.103.110.152	192.168.1.152	192.168.111.152	eth4	mdm152	primary
MDM 2	10.103.110.153	192.168.1.153	192.168.111.152	eth4	mdm153	secondary 1
MDM 3	10.103.110.154	192.168.1.154	192.168.111.152	eth4	mdm154	secondary 2
TB 1	10.103.110.155	192.168.1.155	-	-	tb155	tiebreaker 1
TB 2	10.103.110.156	192.168.1.156	-	-	tb156	tiebreaker 2
SDS and SDC are installed on all nodes.						

Install the MDM packages (5-node cluster)

Install the PowerFlex MDM package for CentOS on SVMs that will be part of a 5-node MDM cluster.

About this task

To install PowerFlex components, perform the following:

Steps

1. Copy the following file to all MDM and tiebreaker (TB) servers (MDM 1, 2, 3 and TB 1 and 2):
   a. `EMC-ScaleIO-mdm-3.6-<build>.X.elX.x86_64.rpm`
2. Install the MDM and configure the Manager role, by running the following commands on the primary (MDM 1) and secondary MDMs (MDM 2 and MDM 3):

```
MDM_ROLE_IS_MANAGER=1 rpm -i <mdm_path.rpm>
```

   NOTE: The default MDM credentials are:
   - user name: admin
   - password: admin

   For example:

```
MDM_ROLE_IS_MANAGER=1 rpm -i EMC-ScaleIO-mdm-3.6-<build>.X.el6.x86_64.rpm
```

3. Install the MDM on the tiebreaker MDMs (TB 1 and TB 2):

```
MDM_ROLE_IS_MANAGER=0 rpm -i <mdm_path.rpm>
```

   For example:

```
MDM_ROLE_IS_MANAGER=0 rpm -i EMC-ScaleIO-mdm-3.6-<build>.X.el6.x86_64.rpm
```

Create the 5-node MDM cluster

Create the 5-node cluster.

About this task

Steps

1. Create the MDM single-node cluster on MDM 1, by running the following command on MDM 1:

```
scli --create_mdm_cluster --master_mdm_ip <IP of MDM 1> --master_mdm_management_ip
<Management IP of MDM 1> --master_mdm_name <Name to give MDM 1> [--cluster_virtual_ip
<IP>]  [--master_mdm_virtual_ip_interface <INTF>] --accept_license
```

   Example:

```
scli --create_mdm_cluster --master_mdm_ip 192.168.1.152 --master_mdm_management_ip
10.103.110.152 --master_mdm_name mdm152 --cluster_virtual_ip 192.168.111.152 --
master_mdm_virtual_ip_interface eth4 --accept_license
```

   NOTE: Each MDM name must be unique. Ensure that each server on which MDM is installed has a unique hostname.

2. Log in and change the MDM password, by running the following commands on MDM 1:

```
scli --login --username admin --password admin
```

```
scli --set_password --old_password admin --new_password Scaleio123
```

The login password is changed from admin to Scaleio123.

3. Log in with the new credentials:

```
scli --login --username admin --password Scaleio123
```

4. Add the secondary MDMs (MDM 2 and MDM 3) to the MDM cluster as Standby MDMs, by running the following commands on MDM 1:

```
scli --add_standby_mdm --new_mdm_ip <IP of MDM X> --mdm_role manager --
new_mdm_management_ip <Management IP of MDM X> --new_mdm_virtual_ip_interface <INTF>
--new_mdm_name <Name to give MDM X>
```

Examples:

```
scli --add_standby_mdm --new_mdm_ip 192.168.1.153 --mdm_role manager
--new_mdm_management_ip 10.103.110.153 --new_mdm_virtual_ip_interface eth4 --
new_mdm_name mdm153
```

```
scli --add_standby_mdm --new_mdm_ip 192.168.1.154 --mdm_role manager --
new_mdm_management_ip 10.103.110.154 --new_mdm_name mdm154
```

5. Add TB 1 and TB 2 to the MDM cluster as tiebreakers, by running the following commands on MDM 1:

```
scli --add_standby_mdm --new_mdm_ip <IP of TB X> --mdm_role tb --new_mdm_name <Name
to give TB X>
```

Examples:

```
scli --add_standby_mdm --new_mdm_ip 192.168.1.155 --mdm_role tb --new_mdm_name tb155
```

```
scli --add_standby_mdm --new_mdm_ip 192.168.1.156 --mdm_role tb --new_mdm_name tb156
```

6. (Optional) Query the cluster to verify that it is running in single mode, by running the following command on MDM 1:

```
scli --query_cluster
```

Output, similar to the following, is displayed:

```
Cluster: Mode: 1_node
```

7. Switch to 5-node cluster mode, by running the following command on MDM 1:

```
scli --switch_cluster_mode --cluster_mode 5_node --add_slave_mdm_name <MDM 2
name>,<MDM 3 name> --add_tb_name <TB 1 name>,<TB 2 name>
```

Example:

```
scli --switch_cluster_mode --cluster_mode 5_node --add_slave_mdm_name mdm153,mdm154 --
add_tb_name tb155,tb156
```

8. (Optional) Query the cluster to verify that it is running in cluster mode, by running the following command on MDM 1:

```
scli --query_cluster
```

Output, similar to the following, is displayed:

```
# scli --query_cluster
Cluster:
Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
Virtual IPs: 192.168.111.152
Master MDM:
Name: mdm6, ID: 0x1cb79cb46ed13200
IPs: 192.168.247.6, 172.16.247.6, Management IPs: 10.136.247.6, Port: 9011, Virtual
IP interfaces: eth4
Version: 2.0.13000
Slave MDMs:
Name: mdm8, ID: 0x67d759a83ec90d42
IPs: 172.16.247.8, 192.168.247.8, Management IPs: 10.136.247.8, Port: 9011, Virtual
IP interfaces: eth4
Status: Normal, Version: 2.0.13000
Name: mdm7, ID: 0x4d9cb20c720f7481
IPs: 192.168.247.7, 172.16.247.7, Management IPs: 10.136.247.7, Port: 9011, Virtual
IP interfaces: eth4
Status: Normal, Version: 2.0.13000
Tiebreakers:
Name: mdm9, ID: 0x64bd85740ae9b2a4
IPs: 172.16.247.9, 192.168.247.9, Port: 9011
Status: Normal, Version: 2.0.13000
Name: mdm10, ID: 0x593656bc4ef59083
IPs: 172.16.247.10, 192.168.247.10, Port: 9011
Status: Normal, Version: 2.0.13000
```

Deploy the SDS component

Deploy the PowerFlex CentOS-based SDS package on all SVMs that will host SDSs.

About this task

To install the SDS package, perform the following steps.

Steps

1. Copy the SDS package `EMC-ScaleIO-sds-3.6-<build>.X.elX.x86_64.rpm` to all SVMs that will contribute storage devices to the PowerFlex system (in our example, all the SVMs).
2. On each SVM to which you copied the package, install the SDS using the command:

   ```
   rpm -i <sds_path.rpm>
   ```

   For example:

   ```
   rpm -i EMC-ScaleIO-sds-3.6-<build>.X.el6.x86_64.rpm
   ```

Manually install the RFcache package on SDSs

On SDSs where RFcache acceleration will be used for Medium Granularity Storage Pools, install the XCACHE package. This package must be installed on the SVM of every SDS where the RFcache feature will be used.

Prerequisites

Obtain the PowerFlex XCACHE package.

Steps

Install the XCACHE package on the SVM of every SDS that will provide RFcache acceleration.

```
rpm -i EMC-ScaleIO-xcache-3.6-<build>.X.elX.x86_64.rpm
```

Next steps

After deployment activities are complete, perform post-deployment tasks, and configure storage and the RFcache feature. For more information, refer to the *Configure and Customize Dell PowerFlex*.

Register the vSphere PowerFlex plug-in

Use PowerCLI for VMware to register the PowerFlex plug-in on a vCenter.

Prerequisites

NOTE: Adobe Flash Player reached end-of-life on December 31st, 2020. To use the vSphere PowerFlex plug-in from the Flash-based vSphere web client, apply a workaround for browsers as explained in the following knowledge base article:

Impact on PowerFlex vSphere Plugin after Adobe Flash End of Life

- Ensure that earlier versions of the PowerFlex plug-in have been cleaned from the vCenter.
- Ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the host storing the PowerFlex plug-in file.
- Obtain the vCenter credentials.
- Ensure that you have downloaded and extracted the required software package from the Customer Support site for your SDC's operating system.

NOTE: Only ESXi v6.x-based systems can be installed and managed using the vSphere PowerFlex plug-in. ESXi v7 is not supported.

About this task

To use your own host, see "Register the vSphere PowerFlex plug-in manually".

Steps

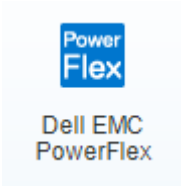
1. Copy the following file to your designated host (preferably your vCenter), that has PowerCLI for VMware installed: `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip`
2. Extract the contents of the ZIP file.
3. Using PowerCLI for VMware, set to `Run as administrator`, run the following script:

```
VxFlexOSPluginSetup-3.6-X.<build>.ps1
```

   a. Type the vCenter name or IP address, user name, and password.
   You can also use this script to unregister a PowerFlex plug-in or to create an SVM template.
   b. For `Choose mode`, select option `1, Register PowerFlex plug-in`.
   c. Read the upgrade notice, and type `y` to continue.
   d. For `Select Registration Mode`, select `Standard` (simplified, using embedded Tomcat).
   This step may take a few minutes.

   NOTE: You can use the `Advanced` option to install the PowerFlex plug-in using a PowerFlex Gateway from a previous installation, or using your own web service. In either case, you must place this version's `plugin.zip` file (`EMC-ScaleIO-vSphere-web-plugin-3.6-X.<build>.zip`) in your resources folder before running the installation. To use a previous PowerFlex Gateway, the resource folder is `PowerFlex Gateway installation folder\webapps\root\resources`.

   e. If necessary, accept the thumbprint.
4. Log in to the vSphere web client.
   If you are already logged in, log out, then log in again.
5. Return to the PowerCLI window, and press ENTER to return to the script menu and complete the PowerFlex plug-in installation.

6. From the vSphere **Home** tab, verify that the PowerFlex icon 

is visible in the **Inventories** section.

Results

The PowerFlex plug-in is registered. If the PowerFlex icon is missing, the vCenter server failed to register the PowerFlex plug-in, due to one of the following reasons:

- Connectivity problem between the vSphere web client server and the host storing the PowerFlex plug-in (for example, network / firewall etc.).

  Resolution: Verify that there is communication between the vSphere web client server and the host storing the PowerFlex plug-in.

- URL problem when using an external web server.

  Resolution: Verify that the URL is https:// and is pointing to the correct web server IP address (PowerFlex Gateway).

For information about how to use the log to troubleshoot problems that may arise, see "Troubleshooting PowerFlex plug-in registration issues".

Install SDC on an ESXi server using the vSphere PowerFlex plug-in

Install SDC on ESXi servers using the vSphere PowerFlex plug-in.

Steps

1. Prepare the ESXi hosts:
   a. Log in to the vSphere web client.
   b. Open the PowerFlex plug-in window by clicking the **PowerFlex** icon.
   c. From the **Basic Tasks** section of the PowerFlex page, click **Pre-Deployment Actions**.
   The **Pre-Deployment Actions** page appears.
   d. Drill-down on the vCenter to display connected ESXi hosts.
   e. For every ESXi host that might be part of this PowerFlex system (even if only in the future):
      i. Select **Install SDC**.
      ii. Type **ESXi Root Password**.

      NOTE: Do not select **Convert ISO**. This feature is in preparation for a future release of VxFlex Ready Node.

   f. Click **Run**.
   The status appears in the dialog box.
   g. When finished, click **Finish**.
   h. You must restart the ESXi hosts before proceeding.
2. Register the system:
   a. From the PowerFlex plug-in page, click **Register PowerFlex system**.
   b. Enter the credentials of the system.
   c. Click **OK**.
3. Deploy PowerFlex:
   a. From the PowerFlex plug-in page, click **Deploy PowerFlex environment**.
   The PowerFlex deployment wizard is displayed.
   b. In the **Select Installation** page, select **Add servers to a registered PowerFlex system**, and select the system that you want to extend.
   c. Continue with the deployment steps, adding the new SDC nodes.

You can skip steps that do not need to be changed. When adding components, the wizard adjusts the displayed pages to options that are relevant to the current PowerFlex system.

4. Complete the deployment.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install the SDC on an ESXi-based server and connect it to PowerFlex using esxcli

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host
- The required installation software package for your SDC's operating system (available from the zipped software packages that is downloaded from the Customer Support site)
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID needs to conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

The following procedure explains how to manually install an external SDC on an ESXi-based using esxcli in command line.

NOTE: This procedure requires two server reboots.

Steps

1. On the ESXi-based server on which you are installing the SDC, set the acceptance level:

```
esxcli software acceptance set --level PartnerSupported
```

2. Install the SDC:

Option	Description
ESXi 6.x	`esxcli software vib install -d <path_to_zip_file>` where *<path_to_zip_file>* is the absolute path to the SDC ZIP file. **NOTE:** The SDC offline ESXi 6.0 bundle is the same for all ESXi 6.x versions.
ESXi 7	`esxcli software component apply -d <path_to_zip_file>` where *<path_to_zip_file>* is the absolute path to the SDC ZIP file.

3. Get permission to reboot the ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot is required, as described in this procedure.

4. Set the IP address of the MDM:

```
esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
```

where

- *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
  - Separate the IP addresses of the same MDM cluster with a "," symbol.
  - Separate multiple MDM clusters with the "+" symbol.

    In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one:

    `10.20.30.40,50.60.70.80+11.22.33.44`
- *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
- Optional: use `IoctlMdmPasswordStr` for SDC authentication to the MDM for access to the PowerFlex system.

CAUTION: A few applications require an advanced setting in ESXi for specific vital product data (VPD) pages. For more information about changing `blkScsiUnitSerialLenByStandard` parameter on an ESXi host for PowerFlex volumes, see Procedure to change parameter blkScsiUnitSerialLenByStandard on ESXi for PowerFlex volumes. Perform this procedure only if your applications require this setting. This is not a common setting.

5. Back up the parameters:

```
/bin/auto-backup.sh
```

6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

```
vmkload_mod -l | grep scini
```

Output similar to the following indicates that the driver is loaded:

```
scini 2 808
```

The SDC component is installed on the ESXi-based server.

Results

The SDC is installed on the ESXi-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Manually remove a VIB file from an ESXi host

Remove a VIB file manually from an ESXi host.

Prerequisites

Obtain permission to reboot the ESXi host.

Steps

1. In an SSH session, check the version of VMware ESXi installed on the host by typing this command:

   ```
   esxcli system version get
   ```

   Example of output:

   ```
   Product: VMware ESXi
   Version: 5.5.0
   Build: Releasebuild-2068190
   Update: 2
   ```

   In the above example, the version is `5.5.0`.

2. Type the command that corresponds to the version installed on the host. For example:
   a. For ESXi 5.5:

      ```
      esxcli software vib remove --vibname scaleio-sdc-esx5.5
      ```

   b. For ESXi 6.0, 6.5, and 6.7:

      ```
      esxcli software vib remove --vibname scaleio-sdc-esx6.X
      ```

      Example of output:

      ```
      Removal Result
      Message: The update completed successfully, but the system needs to be rebooted for
      the changes to be effective.
      Reboot Required: true
      VIBs Installed:
      VIBs Removed: VMware_bootbank_scaleio-sdc-esx5.5_1.32-201.0
      VIBs Skipped:
      ```

3. Open the `esx.conf` file for editing by typing this command:

   ```
   vi /etc/vmware/esx.conf
   ```

   a. Search for the line containing the scini text (use key sequence /scini).
   b. Delete the line using either the <X> key or <D> to delete to full line.
   c. Save and exit using the key sequence: <ESC>+<:wq!> and <Enter>.
4. Reboot the server.

10

Add components to a PowerFlex system—ESXi

Add components to an existing ESXi-based PowerFlex system.

Preparation for adding components to a system

Prepare items required for adding components to an existing ESXi-based PowerFlex system.

Prerequisites

- Ensure that the new nodes meet the requirements listed in "Deployment checklist—ESXi"
- Obtain the vCenter login credentials

About this task

The preparation activities described below can be found in the "VMware ESXi > Preparation for deployment" section.

Steps

1. For new Fine Granularity Storage Pool hosts, prepare their NVDIMMs as DAX devices, as described in "Preparation for deployment."
2. Prepare the new ESXi hosts, as described in "Prepare the ESXi hosts."
3. Configure security settings, as described in "Configure ESXi security settings."
4. If the ESXis previously hosted PowerFlex components, perform the procedure "Clean the PowerFlex VMware environment" on those hosts.

Add components using the vSphere PowerFlex plug-in deployment wizard

You can add PowerFlex components to an existing system using the vSphere PowerFlex plug-in deployment wizard.

About this task

NOTE: The following procedure cannot be used to add an SDS component to an existing SVM. To do so, contact Dell Technologies Support.

Steps

1. Log in to vSphere and open the PowerFlex plug-in.
2. From the **Basic tasks** section of the screen, click **Deploy PowerFlex environment**.

   The PowerFlex plug-in deployment wizard is launched. If you exited the previous deployment before completion, you will be able to return from where you left off.

   NOTE: The deployment wizard operates on the assumption that you are using the provided PowerFlex OVA template to create the Storage Virtual Machines (SVMs).
3. In the **Select Installation** screen, select **Add servers to a registered PowerFlex system**, and select the system to which you want to add components.
4. Continue following the deployment steps to add the new nodes.

You can skip steps where no changes are required.

NOTE: When adding components, the wizard adjusts the displayed screens to options that are relevant to the current PowerFlex system.

5. Complete the deployment.

NOTE: After extending an existing SVM with a new PowerFlex role/component, you must perform manual memory allocation on the SVM, as described in "SVM manual memory location" in the *Deploy Dell PowerFlex* .

III

PowerFlex Installer REST API

PowerFlex Installer REST API deployment procedures

11

Deploying PowerFlex using the REST API

Deploy PowerFlex using the REST API.

Preparation for deployment using the REST API

Before you use the REST API to deploy a system, follow recommended preparation guidelines.

Refer to the guidelines in the deployment checklist and preparation tasks supplied in this guide, corresponding to operating system type.

REST API version

The REST API version should be added to the `Accept` parameter in the header field. For example: `Accept: application/json;version=1.0`.

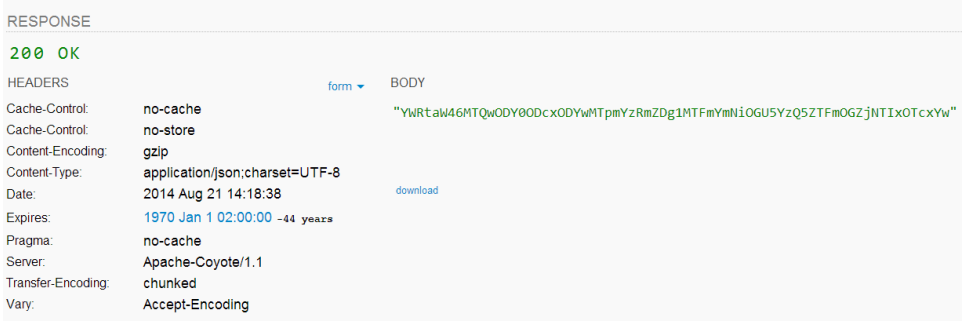
If no `Accept` header field is present, it is assumed that the client accepts all media types and versions.

HTTP authentication

To perform authentication, invoke the HTTP GET request, with the URI: `/api/login` , with `MDM user\password` passes in HTTP Basic authentication. A token is returned.

For example:

```
https://10.76.60.190:443/api/login
```



This token is valid for 8 hours from the time it was created, unless there has been no activity for 10 minutes, or if the client has sent a logout request.

HTTP token invalidation (logout)

The invalidation is achieved by invoking the HTTP GET request, with the URI: `/api/logout`.

The token mentioned above is the password in the HTTP Basic authentication. (The user is ignored and can be empty.)

For every REST API request that does not require the `gatewayAdminPassword`, the authentication is done by passing the token mentioned above as the password in the HTTP Basic authentication. (The user is ignored and can be empty.)

Requests that require the `gatewayAdminPassword` work similarly, with these exceptions:

- Instead of `/api/login`, you should invoke an HTTP GET request, `/api/gatewayLogin`, with `user: admin password: <gatewayAdminPassword>` in HTTP Basic authentication. A token is returned.
- Instead of invoking `/api/logout`, invoke `/api/gatewayLogout` with the token received when you logged in.

NOTE: Requests that require `gatewayAdminPassword` are:

GET:

```
/api/Configuration
/api/gatewayLogout
/api/getHostCertificate/{Mdm | Lia}
```

POST:

```
/api/updateConfiguration
/api/instances/System/action/createMdmCluster
/api/trustHostCertificate/{Mdm | Lia}
/api/gatewaySetSecureCommunication
```

Response fields

The order of the fields in the responses may change. More fields may be added in the future.

URIs

General URIs

POST (create) / GET all objects for a given type

`/api/types/{type}/instances`

GET by ID

`/api/instances/{type::id}`

POST a special action on an object

`/api/instances/{type::id}/action/{actionName}`

POST a special action on a given type

`/api/types/{type}/instances/action/{actionName}`

Get current API version

`/api/version`

GET all instances

`/api/instances/`

Response

- Every row in the Parent table of the object appears as a link in the response of a GET object:
  `/api/instances/{type::id}`
- Every row in the Relationships table of the object appears as a link in the response of a GET object:
  `/api/instances/{type::id}/relationships/{Relationship name}`

Table 35. Responses

Property	Type	Note
sessionTag	Long	-
isDirty	Boolean	The version on some objects changed while the MDM created the response.
System	List of System objects. See the System objects table.	-
lastSystemVersion	Long	-
protectionDomainList	List of Protection Domain objects. See the Protection Domain objects table.	-
lastProtectionDomainVersion	Long	-
sdsList	List of SDS objects. See the SDS objects table.	-
lastSdsVersion	Long	-
storagePoolList	List of Storage Pool objects. See the Storage Pool objects table.	-
lastStoragePoolVersion	Long	-
deviceList	List of Device objects. Device objects	-
lastDeviceVersion	Long	-
volumeList	List of Volume objects. See the Volume objects table.	-
lastVolumeVersion	Long	-
vTreeList	List of V-Tree objects. See the V-Tree objects table.	-
lastVTreeVersion	Long	-
sdcList	List of SDC objects. See the SDC Objects table.	-
lastSdcVersion	Long	-
faultSetList	List of Fault Set objects. See the Fault Set objects table.	-
lastFaultSetVersion	Long	-
accelerationPoolList	List of AccelerationPool objects. See the Acceleration Pool objects table.	-
lastAccelerationPoolVersion	Long	-
spSdslList	List of SpSds objects. See the SpSds objects table.	-
lastSpSdsVersion	Long	-

Table 35. Responses (continued)

Property	Type	Note
snapshotPolicyList	List of SnapshotPolicy objects. See the SnapshotPolicy objects table.	-
lastSnapshotPolicyVersion	Long	-

Change configuration of PowerFlex Gateway (POST)

`/api/updateConfiguration/`

Request

Must contain one or more of the following parameters:

- `mdmAddresses` – List of MDM IP addresses that the PowerFlex Gateway will use to connect to the MDM (for performance improvement, place the primary MDM IP addresses before the secondary ones)
- `mdmPort` – MDM port
- `gatewayAdminPassword` – Password for installation manager, get/update PowerFlex Gateway configuration, and add primary MDM
- `systemId` – ID of the system configured by `mdmAddresses`. If `systemId` is left empty, it will be populated on the first invocation of `/api/login`. At every `/api/login`, the system ID value is compared to the ID of the PowerFlex cluster (MDM). If the IDs are different (`mdmAddresses` points to a cluster that does not match the `systemId`), an error message is displayed, and all login tokens (`/api/login`, `/api/gatewayLogin`) are invalidated. None of the PowerFlex Gateway clients will be able to send requests to the MDM until one of the following happens:
  - `systemId` is configured to the correct value
  - `mdmAddresses` is configured to the correct IP addresses
  - `systemId` is set to an empty value. In this case, on the first invocation of `/api/login`, the value for `systemId` will be populated according to the ID of the matching `mdmAddresses`.

    If the value for `systemId` is different from the configured `systemId` (and the configured `systemId` is not empty), all login tokens (`/api/login`, `/api/gatewayLogin`) are invalidated.
- `snmpSamplingFrequency` – MDM sampling frequency in seconds. If sampling frequency is set to `0`, the SNMP trap sender will be disabled, and no sampling to the MDM will be performed.
- `snmpResendFrequency` – Resend frequency of SNMP traps, in minutes. If resend frequency is set to `0`, all traps will be sent in every sampling.
- `snmpTrapsReceiverIp` – SNMP trap receivers' IP addresses (supports up to two comma-separated or semi-colon-separated IP addresses)
- `snmpPort` – Port number used for SNMP traps

Example

```
{
"mdmAddresses":["10.76.60.150", "10.76.60.11"],
"mdmPort":"6611",
"gatewayAdminPassword":"1000:3ac8ab8129d71c9641e02ec809140c765f22459d78e4a5cc805f3b164d3b
fb1f:02e80453f62ba3caddc42db3818a4c0e084cd50f8f681da6c2b40dec8e92af94",
"systemId":"7f5d8fc72a3d7f3d" ,
"snmpSamplingFrequency":"30",
"snmpResendFrequency":"0",
"snmpTrapsReceiverIps":["10.76.60.190","10.76.60.191"],
"snmpPort":"162"
}
```

Get configuration of PowerFlex Gateway (GET)

`/api/Configuration/`

Response

Includes the following:

- `mdmAddresses` – List of primary MDM IP addresses that the PowerFlex Gateway will use to connect to the MDM
- `mdmUsername` – MDM username
- `mdmPort` – MDM port
- `systemId` – ID of the system, configured by `mdmAddresses`. (See detailed explanation for "Change configuration of PowerFlex Gateway (POST).")
- `snmpSamplingFrequency` – MDM sampling frequency in seconds
- `snmpResendFrequency` – Resend frequency of SNMP traps, in minutes
- `snmpTrapsReceiverIp` – SNMP trap receivers' IP addresses
- `snmpPort` – The port number used for SNMP traps
- `remoteSyslog` – The remote syslog servers configuration in the MDM
- `featuresEnableSnmp` – Indicates whether SNMP is enabled
- `cipherSuites` – List of names of cipher suites (as recognized by Java) that the SSL will use instead of the default list
- `featuresEnableIM` – Indicates whether the PowerFlex Installer is enabled
- `featuresEnableCollectAutoLogs` – Enable the Auto Collect Logs feature
- `bypassCertificateCheck` – When set to `TRUE`, when opening a connection, the gateway will trust any certificate during the SSL handshake
- `allowNonSecureCommunication`

Example


```
{
        "allowNonSecureCommunication": "false",
 "bypassCertificateCheck": "false",
        "cipherSuites": null,
        "featuresEnableCollectAutoLogs": "false",
        "featuresEnableIM": "true",
"snmpResendFrequency": "0",
"snmpSamplingFrequency": "30",
"snmpPort": "162",
"mdmPort":"6611",
"mdmUsername": "alertservice",
"remoteSyslog": [
        {
            "hostName": "10.76.60.100",
            "port": 1468,
            "facility": 16
        },
        {
            "hostName": "10.76.60.101",
            "port": 1468,
            "facility": 16
        }
],
"mdmAddresses":[
"10.76.60.150",
"10.76.60.135"
],
"systemId":"7f5d8fc72a3d7f3d",
"snmpTrapsReceiverIps": [
        "10.76.60.192"
    ]
}
```

Query selected statistics (POST)

`/api/instances/querySelectedStatistics`

Request

- Required parameters:

  `selectedStatisticsList` – list of objects containing:

  - `type` – object type (`System`, `ProtectionDomain`, `Sds`, `StoragePool`, `Device`, `Volume`, `VTree`, `Sdc`, `FaultSet`, `AccelerationPool`, `SpSds`, `SnapshotPolicy`
  - `properties` – list of properties to fetch
- The following parameters are not relevant to the System type and can be omitted:

  `ids` – list of objects ids

  or

  `allIds` – with empty value

Response

- The response contains a list of all requested types, the IDs, and properties, according to the request parameters.
- The System type does not contain an ID.
- Types for which the requested IDs do not exist in the cluster will appear with an empty {}. (See `FaultSet` in the example below.)
- Any ID (from the IDs parameters in the request) that does not exist in the cluster will not appear in the response. The following will appear in the `scaleio.log` file:

  `Got no statistics for <type>::<id>.`

Example

In the following example, `Sds::022beb2300000004` does not exist in the cluster, so it does not appear in the response. Instead, this will appear in the `scaleio.log` file:

```
POST https://localhost:8443/api/instances/querySelectedStatistics
```

Body:

```
{"selectedStatisticsList":[{"type":"Sds","ids":["df4bd899000000aa"],"properties":
["capacityInUseInKb","rfcachePoolNumOfDriverTheads","snapCapacityInUseInKb","thickCapacit
yInUseInKb","numOfDevices"]}]}
```

Response:

```
{
"Sds":{
"62b2b5c700000002":{
"snapCapacityInUseInKb": 1048576,
"rfcachePoolNumOfDriverTheads": 80,
"thickCapacityInUseInKb": 5595136,
"capacityInUseInKb": 5595136,
"numOfDevices": 1
}
}
}
```

Get MDM's or LIA's certificate (GET)

```
/api/getHostCertificate/{Mdm|Lia}?host={host ip}
```

The request can be sent for either MDM or LIA.

Example

```
/api/getHostCertificate/Mdm?host=10.76.60.10
```

Response

The host certificate in PEM encoding.

NOTE: The whole certificate should be saved into a `cer` file for the `trustHostCertificate` request.

For example:

```
-----BEGIN CERTIFICATE-----
MIIDcTCCAlmgAwIBAAIBATANBgkqhkiG9w0BAQUFADB8MQwwCgYDVQQqEwNNRE0xFzAVBgNVBAMTDmNlbnRvcy02L
TQtYWRpMRIwEAYDVQQHEwlIb3BraW50b24xFjAUBgNVBAgTDU1hc3NhY2h1c2V0dHMxCzAJBgNVBAYTAlVTMQwwCg
YDVQQKEwNFTUMxDDAKBgNVBAsTA0FTRDAeFw0xNTEyMTkwNzI4MTVaFw0yNTEyMTcwODI4MTVaMHwxDDAKBgNVBCo
TA01ETTEXMBUGA1UEAxMOY2VudG9zLTYtNC1hZGkxEjAQBgNVBAcTCUhvcGtpbnRvbjEWMBQGA1UECBMNTWFzc2Fj
aHVzZXR0czELMAkGA1UEBhMCVVMxDDAKBgNVBAoTA0VNQzEMMAoGA1UECxMDQVNEMIIBIjANBgkqhkiG9w0BAQEFA
AOCAQ8AMIIBCgKCAQEA4SMybaAEZjfBX9wLglr3wxYHOvID5Pe1Z26Pv8oIR/
MTOVa1Bw4A9px1MHHSIfkAfgRlLC24uebZXhbh0snBq+OL+SJPwEfbOVbif/saXL8RJFwm/VNg8KHUwjuq/
sJkKDjx9uSf0U+/9FzwvKVuM87xDj/rVvJgBYh6pH34q/XD5l8am/iEQr/
EnGZmIsa+VkCL0IeYKbkA3ZINfI4YsjSJ+qeu5e/
KMsNlHEvmhk1DdJbLayn9QkiS5Q9e8A40jjkb2e1Q71awoOlb6+8XXWWkpBhxAnRa9P8Pb1BfcNyUfXtrKuy+fRjw
4Gp+rw2MdoIDuMbO1+1sQaRvVPTYxwIDAQABMA0GCSqGSIb3DQEBBQUAA4IBAQCu1e/jBz63ZFDS+pFvZ3XI/
VMdn9NArM+8Cjn7Luar8oEVAYI6jqYYcZCk2jQyfuI1HP2jXqPTJo8CNyNT5S6rZ5ryHOiRjn/
K2pVC6kT497LY5lc3LjhXUdLjpWnW2jsGfM93cCkkrxu8wmkh9oo8WizOiRAyKmz02uTEuEok7GJBS/
DR6csnLo2YLUV6ZqeBN9jdzZbIY7SoFWya1K4xZmqhkAtnj1ynP3uoxTkd+wfDRmYeDv8l5eciLj2BXNuV8zXYWSC
yABZC//jvajNtSEXgUura3uh0YBIfbO/AZ980zUMwJBMBr06yw4tHnHRRYgfI3tnZOD4byaJOdHuq
-----END CERTIFICATE-----
```

Trust an MDM's or LIA's certificate (POST)

```
/api/trustHostCertificate/{Mdm|Lia}
```

For example:

```
/api/trustHostCertificate/Mdm
```

Request

- Content-type: multipart/form-data.
- The body should contain a part named "file" and the file containing the certificate to be trusted.

Set the PowerFlex Gateway to work with secured communications with the MDM (POST)

```
/api/gatewaySetSecureCommunication
```

The PowerFlex Gateway will be unable to connect to the MDM using non-secured communications.

Working with the PowerFlex Installer (IM) REST API

PowerFlex Installer is an orchestration engine that runs commands from different queues (nodes and MDM) at different phases, and labels them as a process such as an upgrade.

The PowerFlex Gateway REST API begins with the `/im/`prefix.

Login to the IM REST API must be done by running the following:

```
POST https://localhost/j_spring_security_check
Content: "j_username=admin&j_password=Password1&submit=Login"
```

An operation such as the above provides a JSESSIONID cookie in the response that should be used for identification of the session in all future requests. You must include the JSESSIONID for identification or you will be unable to run IM commands using the REST API. For example:

```
POST "Content-Type: application/json" "Cookie:
JSESSIONID=969F624A761937AE80E6CC9E91756B10" https://localhost/im/types/Command/
instances/actions/retry
```

Topology/configuration object

Most PowerFlex Gateway operations require a topology/configuration object as a parameter (payload) for the REST (HTTP) request in JSON format.

Such an object has the following format:

NOTE: Each field's functionality is marked with the `/* */` notation—it should not be used when generating such instance with JSON format `{"parameter":"value"}`.

/* target IP address to send SNMP traps while configuring the PowerFlex Gateway to become an SNMP sampler */

```
   "snmpIp": null,
```

/* system installation ID of the system to be used or configured */

```
      "installationId": null,
```

/* list of all MDM manager IP addresses */

```
      "mdmIPs": ["10.76.1.2", "10.76.1.1", "10.76.1.3"],
```

/* list of IP addresses from which LIA will only accept connections—if null, accept any connection */

```
   "safeIPsForLia": null,
```

/* password/username for MDM and LIA (and LDAP, depending on mode) */

```
      "mdmUser": null,
      "mdmPassword": null,
      "liaPassword": null,
      "liaLdapUsername": null,
      "liaLdapPassword": null,
```

/* LIA LDAP initial configuration */

```
   "liaLdapInitialMode": null,
   "liaLdapInitialUri": null,
      "liaLdapInitialBaseDn": null,
      "liaLdapInitialGroup": null,
      "liaLdapInitialObjectClassAttr": null,
      "liaLdapInitialUserIdAttrName": null,
      "liaLdapInitialMemberOfAttrName": null,
      "liaLdapInitialDisableRecursiveSearch": null,
      "liaLdapInitialUsernameDnFormat": null,
```

/* deprecated—not in use */

```
    "licenseKey": null,
```

/* deprecated—not in use */

```
    "licenseType": null,
```

/* boolean, indicating whether cluster should be set with high performance profile */

```
    "isClusterOptimized": false,
    "upgradeableByCurrentUser": false,
    "ignoredSdcs": null,
```

/* boolean indicating whether system is running */

```
    "upgradeRunning": false,
```

/* boolean indicating whether current MDM is a cluster (if running) */

```
    "clustered": false,
```

/* deprecated – not in use */

```
       "secondaryMdm": null,
```

/* deprecated – not in use */

```
       "tb": null,
```

/* Map of Protection Domain to Storage Pools */

```
       "pd2Sps": {
             "domain1": ["pool1",
             "pool2"]
       },
```

/* deprecated - not in use */

```
       "primaryMdm": null
```

/* deprecated - not in use */

```
       "callHomeConfiguration": null,
```

/* string describing parameters for remote syslog configuration */

```
       "remoteSyslogConfiguration": null,
```

/* string describing cluster (primary (Master)) version */

```
     "systemVersionName": "",
```

/* security settings: set whether the process can connect to non-secured LIA/MDM and whether PowerFlex components should be configured to be non-secured. (They are secured by default) */

```
    "securityConfiguration": {
       "allowNonSecureCommunicationWithMdm": false,
       "allowNonSecureCommunicationWithLia": false,
       "dontAutoApproveCertificates": false
       "disableNonMgmtComponentsAuth": false
```

/* cluster virtual IP address list */

```
       "virtualIps": ["10.76.60.203"],
```

/* enable management-client secure communication */

```
      "securityCommunicationEnabled": false,
      "masterMdm": {
            "node": {
                  "ostype": "linux",
                  "nodeName": null,
                  "nodeIPs": ["10.76.1.1"],
                  "domain": null,
                  "userName": "root",
                  "password": "Password1",
                  "liaLdapUsername": null,
                  "liaLdapPassword": null,
                  "liaPassword": null
            },
            "nodeInfo": null,
            "tunables": null,
            "rollbackVersion": null,
            "mdmIPs": ["10.76.60.195"],
            "name": "MDM1",
            "id": null,
            "ipForActor": null,
            "managementIPs": ["10.76.1.1"],
            /* virtual interfaces to be used by virtual ip */
            "virtIpIntfsList": ["eth1"]
      },
      "slaveMdmSet": [{
            "node": {
                  "ostype": "linux",
                  "nodeName": null,
                  "nodeIPs": ["10.76.1.3"],
                  "domain": null,
                  "userName": "root",
                  "password": "Password1",
                  "liaLdapUsername": null,
                  "liaLdapPassword": null,
                  "liaPassword": null
            },
```

/* internal data fields - should not be used */

```
            "nodeInfo": null,
            "tunables": null,
```

/* if rollback occurs during upgrade, set which version to rollback to*/

```
            "rollbackVersion": null,
            "mdmIPs": ["10.76.60.18"],
            "name": "MDM3",
```

/* unique MDM ID */

```
            "id": null,
            "ipForActor": null,
```

/* IP addresses to be used and configured for management */

```
            "managementIPs": ["10.76.1.3"],
            "virtIpIntfsList": ["eth1"]
      },
      {
            "node": {
                  "ostype": "linux",
                  "nodeName": null,
                  "nodeIPs": ["10.76.1.2"],
                  "domain": null,
                  "userName": "non_root(sudo)",
                  "password": "Password1",
                  "liaLdapUsername": null,
                  "liaLdapPassword": null,
                  "liaPassword": null
```

```
            },
            "nodeInfo": null,
            "tunables": null,
            "rollbackVersion": null,
            "mdmIPs": ["10.76.60.17"],
            "name": "MDM2",
            "id": null,
            "ipForActor": null,
            "managementIPs": ["10.76.1.2"],
            "virtIpIntfsList": ["eth1"]
       }],
```

/* similar to primary (Master) MDM */

```
      "tbSet": [{
            "node": {
                  "ostype": "linux",
                  "nodeName": null,
                  "nodeIPs": ["10.76.60.33"],
                  "domain": null,
                  "userName": "root",
                  "password": "Password1",
                  "liaLdapUsername": null,
                  "liaLdapPassword": null,
                  "liaPassword": null
            },
            "nodeInfo": null,
            "tunables": null,
            "rollbackVersion": null,
            "mdmIPs": ["10.76.60.33"],
            "name": "TB1",
            "id": null,
            "tbIPs": ["10.76.60.33"]
      },
      {
            "node": {
                  "ostype": "linux",
                  "nodeName": null,
                  "nodeIPs": ["10.76.60.19"],
                  "domain": null,
                  "userName": "root",
                  "password": "Password1",
                  "liaLdapUsername": null,
                  "liaLdapPassword": null,
                  "liaPassword": null
            },
      "nodeInfo": null,
      "tunables": null,
      "rollbackVersion": null,
      "mdmIPs": ["10.76.60.19"],
      "name": "TB2",
      "id": null,
      "tbIPs": ["10.76.60.19"]
      }],
```

/* similar to primary (Master) MDM */

```
      "standbyMdmSet": [{
            "node": {
         "ostype": "linux",
         "nodeName": null,
         "nodeIPs": ["10.76.1.5"],
         "domain": null,
         "userName": "root",
         "password": "Password1",
         "liaLdapUsername": null,
         "liaLdapPassword": null,
         "liaPassword": null
      },
      "nodeInfo": null,
      "tunables": null,
      "rollbackVersion": null,
```

```
      "mdmIPs": ["10.76.60.196"],
      "name": "MDMa",
      "id": null,
      "ipForActor": null,
      "managementIPs": ["10.76.1.5"],
      "virtIpIntfsList": ["eth1"]
      }],
```

/* similar to primary (Master) MDM */

```
      "standbyTbSet": [{
            "node": {
         "ostype": "linux",
         "nodeName": null,
         "nodeIPs": ["10.76.60.37"],
         "domain": null,
         "userName": "root",
         "password": "Password1",
         "liaLdapUsername": null,
         "liaLdapPassword": null,
         "liaPassword": null
      },
      "nodeInfo": null,
      "tunables": null,
      "rollbackVersion": null,
      "mdmIPs": ["10.76.60.37"],
      "name": "Tba",
      "id": null,
      "tbIPs": ["10.76.60.37"]
      "sdsList": [{
            "node": {
         "ostype": "linux",
         "nodeName": null,
         "nodeIPs": ["10.76.1.3"],
         "domain": null,
         "userName": "root",
         "password": "Password1",
         "liaLdapUsername": null,
         "liaLdapPassword": null,
         "liaPassword": null
            },
      "nodeInfo": null,
      "tunables": null,
      "rollbackVersion": null,
      "sdsName": "SDS_10.76.1.3",
```

/* Protection Domain name (required) and ID (if it already exists, this is optional) */

```
            "protectionDomain": "domain1",
            "protectionDomainId": null,
```

/* Fault Set ID - if defined */

```
      "faultSet": "fs2",
      "faultSetId": "0",
      "allIPs": ["10.76.60.18"],
      "sdsOnlyIPs": null,
      "sdcOnlyIPs": null,
      "devices": [{
         "devicePath": "\\\\.\\PhysicalDrive1",
         "storagePool": "pool1",
         "deviceName": null,
         "maxCapacityInKb": -1
            },
            {
         "devicePath": "\\\\.\\PhysicalDrive2",
         "storagePool": "pool2",
         "deviceName": null,
         "maxCapacityInKb": -1
      }],
      "rfCached": true,
      "rfCachedPools": ["pool1"],
```

```
      "rfCachedDevices": ["/dev/sdd"],
      "rfCacheType": "MEDIA_TYPE_SSD",
      "flashAccDevices": [],
      "nvdimmAccDevices": [{
         "devicePath": "/dev/dax0.1",
         "storagePool": "accp2",
         "deviceName": "/dev/dax0.1",
         "maxCapacityInKb": -1
      }],
      "useRmCache": false,
```

/* Boolean indicating whether SDS is set to high performance profile */

```
      "optimized": true,
      "packageNumber": 0,
      "optimizedNumOfIOBufs": 3,
      "port": 7072,
      "id": "0"
      },
      {
            "node": {
         "ostype": "linux",
         "nodeName": null,
         "nodeIPs": ["10.76.1.2"],
         "domain": null,
         "userName": "non_root(sudo)",
         "password": "Password1",
         "liaLdapUsername": null,
         "liaLdapPassword": null,
         "liaPassword": null
      },
      "nodeInfo": null,
      "tunables": null,
      "rollbackVersion": null,
      "sdsName": "SDS_10.76.1.2",
      "protectionDomain": "domain1",
      "protectionDomainId": null,
      "faultSet": "fs1",
      "faultSetId": "0",
      "allIPs": ["10.76.60.17"],
      "sdsOnlyIPs": null,
      "sdcOnlyIPs": null,
      "devices": [{
         "devicePath": "E:",
         "storagePool": "pool2",
         "deviceName": null,
         "maxCapacityInKb": -1
      }],
      "rfCached": true,
      "rfCachedPools": [],
      "rfCachedDevices": ["/dev/sdd"],
      "rfCacheType": "MEDIA_TYPE_SSD",
      "flashAccDevices": [],
      "nvdimmAccDevices": [{
         "devicePath": "/dev/dax0.0",
         "storagePool": "accp2",
         "deviceName": "/dev/dax0.0",
         "maxCapacityInKb": -1
      }],
      "useRmCache": false,
      "optimized": true,
      "packageNumber": 0,
      "optimizedNumOfIOBufs": 3,
      "port": 7072,
      "id": "0"
      },
      {
            "node": {
         "ostype": "linux",
         "nodeName": null,
         "nodeIPs": ["10.76.1.1"],
         "domain": null,
         "userName": "root",
```

```
            "password": "Password1",
            "liaLdapUsername": null,
            "liaLdapPassword": null,
            "liaPassword": null
         },
         "nodeInfo": null,
         "tunables": null,
         "rollbackVersion": null,
         "sdsName": "SDS_10.76.1.1",
         "protectionDomain": "domain1",
         "protectionDomainId": null,
         "faultSet": "fs1",
         "faultSetId": "0",
         "allIPs": ["10.76.60.195"],
         "sdsOnlyIPs": null,
         "sdcOnlyIPs": null,
         "devices": [{
            "devicePath": "/dev/sdb",
            "storagePool": "pool1",
            "deviceName": "sdb_Device",
            "maxCapacityInKb": -1
         },
         {
            "devicePath": "/dev/sdc",
            "storagePool": "pool1",
            "deviceName": "sdc_Device",
            "maxCapacityInKb": -1
         }],
         "rfCached": true,
         "rfCachedPools": ["pool1"],
         "rfCachedDevices": ["/dev/sdd"],
         "rfCacheType": "MEDIA_TYPE_SSD",
         "flashAccDevices": [],
         "nvdimmAccDevices": [{
            "devicePath": "/dev/dax0.0",
            "storagePool": "accp2",
            "deviceName": "/dev/dax0.0",
            "maxCapacityInKb": -1
         }],
         "useRmCache": false,
         "optimized": true,
         "packageNumber": 0,
         "optimizedNumOfIOBufs": 3,
         "port": 7072,
         "id": "0"
      },
      {
         "node": {
            "ostype": "linux",
            "nodeName": null,
            "nodeIPs": ["10.76.60.33"],
            "domain": null,
            "userName": "root",
            "password": "Password1",
            "liaLdapUsername": null,
            "liaLdapPassword": null,
            "liaPassword": null
         },
         "nodeInfo": null,
         "tunables": null,
         "rollbackVersion": null,
         "sdsName": "SDS1",
         "protectionDomain": "domain1",
         "protectionDomainId": null,
         "faultSet": "fs3",
         "faultSetId": "0",
         "allIPs": ["10.76.60.33"],
         "sdsOnlyIPs": ["10.76.4.4",
         "10.76.2.2"],
         "sdcOnlyIPs": ["10.76.3.3"],
         "devices": [{
            "devicePath": "/dev/sdb",
            "storagePool": "pool1",
```

```
            "deviceName": null,
            "maxCapacityInKb": -1
         },
         {
            "devicePath": "/dev/sdc",
            "storagePool": "pool2",
            "deviceName": null,
            "maxCapacityInKb": -1
         }],
         "rfCached": true,
         "rfCachedPools": ["pool1"],
         "rfCachedDevices": ["/dev/sdd"],
         "rfCacheType": "MEDIA_TYPE_SSD",
         "flashAccDevices": [],
         "nvdimmAccDevices": [{
            "devicePath": "/dev/dax0.1",
            "storagePool": "accp2",
            "deviceName": "/dev/dax0.1",
            "maxCapacityInKb": -1
         }],
         "useRmCache": false,
         "optimized": true,
         "packageNumber": 0,
         "optimizedNumOfIOBufs": 3,
         "port": 7072,
         "id": "0"
      }],
      "sdcList": [{
         "node": {
            "ostype": "linux",
            "nodeName": null,
            "nodeIPs": ["10.76.1.6"],
            "domain": null,
            "userName": "root",
            "password": "Password1",
            "liaLdapUsername": null,
            "liaLdapPassword": null,
            "liaPassword": null
         },
         "nodeInfo": null,
         "tunables": null,
         "rollbackVersion": null,
         "guid": null,
```

/* splitter (RecoverPoint) IP address to replicate the I/O */

```
         "splitterRpaIp": null,
         "sdcName": "SDCb",
         "isOnESX": null,
         "id": "0",
```

/* Boolean indicating whether SDC should be set to high performance profile */

```
         "optimized": true
      },
      {
         "node": {
            "ostype": "linux",
            "nodeName": null,
            "nodeIPs": ["10.76.1.4"],
            "domain": null,
            "userName": "root",
            "password": "Password1",
            "liaLdapUsername": null,
            "liaLdapPassword": null,
            "liaPassword": null
         },
         "nodeInfo": null,
         "tunables": null,
         "rollbackVersion": null,
         "guid": null,
         "splitterRpaIp": null,
```

```
      "sdcName": "SDC3",
      "isOnESX": null,
      "id": "0",
      "optimized": false
   },
   {
      "node": {
         "ostype": "linux",
         "nodeName": null,
         "nodeIPs": ["10.76.1.2"],
         "domain": null,
         "userName": "non_root(sudo)",
         "password": "Password1",
         "liaLdapUsername": null,
         "liaLdapPassword": null,
         "liaPassword": null
      },
      "nodeInfo": null,
      "tunables": null,
      "rollbackVersion": null,
      "guid": null,
      "splitterRpaIp": null,
      "sdcName": null,
      "isOnESX": null,
      "id": "0",
      "optimized": true
   }],
   "volumes": [{
      "id": "0",
      "name": "vol1",
      "volumeSizeInKb": "8000000",
   }],
   "protectionDomains": [{
      "name": "domain1",
      "storagePools": [{
         "name": "pool1",
         "mediaType": "MEDIA_TYPE_HDD",
         "externalAccelerationType": "EXTERNAL_ACCELERATION_TYPE_READ",
         "dataLayout": "PERFORMANCE_OPTIMIZED",
         "compressionMethod": null,
         "spefAccPoolName": null,
         "shouldApplyZeroPadding": true,
         "writeAtomicitySize": null,
         "overProvisioningFactor": null,
         "maxCompressionRatio": null,
         "perfProfile": null
      },
      {
         "name": "pool2",
         "mediaType": "MEDIA_TYPE_SSD",
         "externalAccelerationType": "EXTERNAL_ACCELERATION_TYPE_WRITE",
         "dataLayout": "SPACE_EFFICIENT",
         "compressionMethod": "COMPRESSION_METHOD_NONE",
         "spefAccPoolName": "accp2",
         "shouldApplyZeroPadding": true,
         "writeAtomicitySize": null,
         "overProvisioningFactor": null,
         "maxCompressionRatio": null,
         "perfProfile": null
      }],
      "accelerationPools": [{
         "name": "accp2",
         "mediaType": "MEDIA_TYPE_NVDIMM",
         "rfcache": null
      }]
       }],
}
```

Node object

The Node object is used as an attribute of each PowerFlex component (for example, on which node the MDM is installed), and describes how the node will be accessed:

- via SSH (Linux node with root credentials)
- via WMI (Windows node with admin domain and credentials)
- via LIA (unknown node with null domain, null credentials and LIA password)

  For example:

```
//Linux node accessed via SSH
"node": {
   "ostype": "unknown",
   "nodeName": null,
   "nodeIPs": ["10.76.60.48"],
   "domain": null,
   "userName": null,
   "password": null,
         "liaLdapUsername": null,
         "liaLdapPassword": null,
   "liaPassword": null
       },
//unknown node (Linux or Windows) - accessed via LIA
"node": {
   "ostype": "linux",
   "nodeName": null,
   "nodeIPs": ["10.76.60.41"],
   "domain": null,
   "userName": "root",
   "password": "Password1",
         "liaLdapUsername": null,
         "liaLdapPassword": null,
   "liaPassword": null
      },
```

LIA usage

Any post-installation operations MUST be performed via LIA. Such operations (for example, upgrades) may fail unless all nodes are targeted to be performed via LIA.

How to obtain a JSESSION ID for PowerFlex Installer REST API URIs

A JSESSION ID must be used to in order to run PowerFlex Installer REST API URIs. URIs run without a JSESSION ID will fail. Once you have a JSESSION ID, you can use the PowerFlex Installer REST API URIs.

About this task

The following steps can be used to obtain a JSESSION ID.

Steps

1. Run the following URI:

```
https://localhost/setup
Content: "username=admin&password=Password1&submit=Login"
POST
```

   where the user name in the above example is "admin" and the password is "Password1".

2. Find the JSESSION ID in the URI output. For example:

```
JSESSIONID=969F624A761937AE80E6CC9E91756B10
```

3. Set the following headers, using the JSESSION ID obtained in the previous steps. For example:

```
"Content-Type":"application/json"
"Cookie":"JSESSIONID=969F624A761937AE80E6CC9E91756B10"
```

Results

The JSESSION ID will be added to the URIs. You may now run any PowerFlex Installer REST API URI.

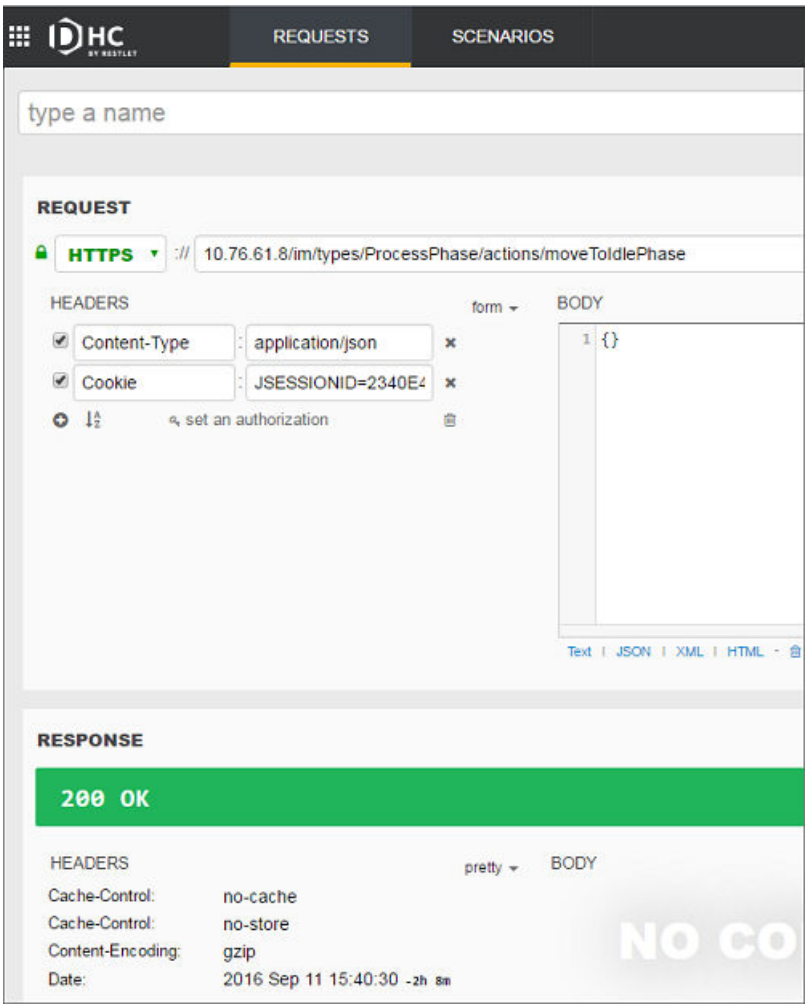


Figure 9. Example of PowerFlex Installer REST API URI when JSESSION ID is configured as a header

Error messages

When there is an error in the command, there are two possible error messages returned.

There are two possible error messages displayed when there is a mistake in the REST API command:

Simple error message

Simple error messages contain the following fields:

- `errorCode`
- `httpStatusCode`

- message

For example:

```
{
    "errorCode": 0,
    "httpStatusCode": 400,
    "message": "Request validation Error. Got illegal parameter: accessMode"
}
```

Detailed error message

Detailed error messages contain the following fields:

- errorCode
- httpStatusCode
- message
- details
- info

For example:

```
{
    "message": "Error with details",
    "httpStatusCode": 500,
    "errorCode": 0,
    "details": [{
        "errorMessage": "Add SDS device IO error",
        "error": "TGT_ADD_DEV_IO_ERROR",
        "rc": 73,
        "info": [{
            "type": "EET_DEV_KEY",
            "objId": null,
            "str": null,
            "num": null,
            "mdmKey": null,
            "protectionDomainKey": null,
            "storagePoolKey": null,
            "accelerationPoolKey": null,
            "sdsKey": null,
            "deviceKey": {
                "id": "fef4903e00010000",
                "sdsKey": {
                    "id": "01baf28400000001",
                    "name": null,
                    "hostname": null,
                    "port": null
                },
                "path": "/dev/sdg",
                "name": "sdg"
            },
            "volumeKey": null
        }]
    },
    {
        "errorMessage": "Add SDS device IO error",
        "error": "TGT_ADD_DEV_IO_ERROR",
        "rc": 73,
        "info": [{
            "type": "EET_DEV_KEY",
            "objId": null,
            "str": null,
            "num": null,
            "mdmKey": null,
            "protectionDomainKey": null,
            "storagePoolKey": null,
            "accelerationPoolKey": null,
            "sdsKey": null,
            "deviceKey": {
```

```
                "id": "fef4903f00010001",
                "sdsKey": {
                    "id": "01baf28400000001",
                    "name": null,
                    "hostname": null,
                    "port": null
                },
                "path": "/dev/sdh",
                "name": "sdh"
            },
            "volumeKey": null
        }]
    }]
}
```

Object names

Define object names according to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex GUI.

It is recommended that you give each object a meaningful name associated with its operational role.

NOTE: Because the system is a single object, its name is automatically unique.

Deploy a system using the REST API

Follow this workflow and these commands for PowerFlex deployment using the REST API. The *PowerFlex REST API Reference Guide* contains details about each of the PowerFlex Installer REST API commands.

Prerequisites

Follow the deployment checklist and preparation guidelines per operating system in this guide, before you deploy the system using the REST API.

About this task

When using the PowerFlex Installer REST API to deploy a system, use JSON objects instead of the CSV file, and the appropriate steps of PowerFlex Installer REST API commands to perform these activities:

1. Upload the software packages.
2. Create the JSON configuration from the CSV file.
3. Deploy the system.

The deployment process consists of the following phases: query, upload, install, and configure. For each phase, monitor its progress and when each one is completed, move to the next phase.

NOTE: Any post installation operations MUST be performed using LIA. Operations such as upgrades may fail unless all nodes have been targeted to be accessed using LIA.

To deploy the system, use the following work flow.

Steps

1. Upload packages to the PowerFlex Installer:

```
POST /im/types/InstallationPackages/instances/actions/uploadPackages
```

2. Create the JSON configuration from the CSV file:

```
/im/types/Configuration/instances/actions/parseFromCSV
```

Headers:

- -u *<GATEWAY_USERNAME>*:*<GATEWAY_PASSWORD>*
- -F "file=@*<FILENAME>*", where <FILENAME> must contain the full path

For example:

```
curl -v -k -X POST -u admin:Password1 -F "file=@c:/deployment/
VxFlex_OS_Complete_Config.csv"

https://254.222.10.123/im/types/Configuration/instances/actions/parseFromCSV
```

This command returns the configuration that is used in the next step.

3. Start deployment using the JSON configuration (the query phase starts automatically):

```
/im/types/Configuration/actions/install
```

Deployment consists of the following phases: query, upload, install, and configure.

4. For each phase, check its progress to monitor that it is completed:

```
/im/types/Command/instances
```

5. When a phase is completed, move to the next phase:

```
/im/types/ProcessPhase/actions/moveToNextPhase
```

6. Repeat the two previous steps until the configuration phase is completed.

IV

PowerFlex VASA

PowerFlex Installer VASA installation procedures

12

PowerFlex VASA provider and vVols

Configure PowerFlex to expose VMware's vSphere API for Storage Awareness (VASA).

Using vVols with PowerFlex

Install the VASA provider in order to support using vVols with PowerFlex.

In order to use vVols with PowerFlex, you must install the VASA provider. In Linux-based systems, you can choose to install the VASA provider during installation of PowerFlex using the PowerFlex Installer. At that point, select whether to use one or three VASA providers. If you want your VASA support to be highly available (HA), which is the mode intended for production use, you should install three VASA providers. Using one VASA provider as a standalone is intended only for lab environments. In the case of an existing PowerFlex system, you can add VASA support by extending the existing cluster by adding one or three VASA providers. This is also done by creating a CSV file and using the PowerFlex Installer.

If you choose to use three VASA providers, this requires additional information to be included in the CSV file for installation in the PowerFlex Installer.

After installing the VASA provider, you must register the VASA provider in the vCenter. For more information, see the *Configure and Customize Dell PowerFlex* guide.

Once the VASA provider is installed, the PowerFlex Gateway will trigger Secure Remote Services/SNMP alerts in case any of the installed VASA providers is down or has any health problems.

PowerFlex VASA provider limitations and prerequisites

Note the following limitations when using PowerFlex VASA provider. For additional details on the VASA provider server requirements, see the Getting to Know Dell PowerFlex Guide.

Limitations

- The VASA provider can be mapped to a single instance of PowerFlex.
- A cluster may contain a single VASA provider or three VASA providers.
- The VASA provider must be installed only on a node which contains an SDC. The VASA provider nodes can be standalone or shared with other PowerFlex components.
- The VASA VM can be hosted on the same node with SDS nodes. The PowerFlex Gateway and MDM hosts can't host a VASA VM.
- The VASA provider should be registered with a single vCenter.
- PowerFlex limits the maximum snapshot hierarchy of a vVol to 31.
- PowerFlex limits the maximum number of volumes to 130K (8,192 mapped volumes per ESXi).

Installing VASA provider — Linux

The VASA provider can be installed as part of the PowerFlex installation by adding information to the CSV file, or as part of an extend operation after deployment. The CSV file is uploaded to the PowerFlex Installer during the deployment/extend procedure.

In a Linux-based system, the VASA provider can be installed during deployment of PowerFlex, or by extending an existing PowerFlex. The following fields must be included in the CSV file:

- Is Vasa Provider

- Vasa Provider IP (optional)

For detailed information about these fields, see CSV topology spreadsheets.

Follow the general deployment instructions outlined in Deployment workflow—Linux.

After the VASA provider is installed, see the *Configure and Customize Dell PowerFlex* guide for post-installation VASA provider and vVols configurations.

V

Post-deployment tasks

Perform post-deployment tasks described in this section.

After performing these tasks, continue to the "Configure and Customize PowerFlex" section for further configuration steps as required.

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Post-deployment tasks

Perform these post-deployment tasks immediately after deployment.

Port usage

If you are using a firewall to protect your network, and need to enable ports and IP addresses, refer to the *PowerFlex Security Configuration Guide*. This guide contains information about port usage for all components, and what to modify in case port numbers need to be changed.

Configure the SDS for optimal memory usage

For certain system configurations, the SDS needs to be configured to be able to use more than one Non-Uniform Memory Access (NUMA) domain.

About this task

In many cases, memory is split into a few NUMA domains. For example, in a two-CPU-socket server configuration, each of the CPUs is in a different NUMA domain. The SDS is configured by default to have affinity to socket 0 in a server or VM, and therefore, by default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0.

In ESXi in HCI configurations, the NUMA definition requires special attention. When configuring the SVM, there are cases where the SVM includes two vNUMAs. For example, if the number of vCPUs required for the SVM is larger than the number of physical cores in the CPU, ESXi will configure the SVM with two vNUMAs. Other cases are ESXi HCI with replication, where the SVM includes both the SDR and the SDS. The SDS will be configured on vNUMA 0 and the SDR on vNUMA 1. The total memory will be split equally between the two vNUMAs, and therefore, the SDS may not have enough memory on NUMA 0.

The memory size required for the SDS depends on the configuration, and on other factors. In order to determine the required size for the SDS, and whether NUMA configuration is required, follow the size recommendations provided in the *Getting to Know Dell PowerFlex Guide*.

If the memory supported by NUMA 0 (usually half the total memory) is less than the amount of memory required by the SDS, you need to allow the SDS access to the memory in the other NUMA. For example, if a node has 96 GiB of physical memory, divided into 48 GiB on each CPU, and the SDS requires 58 GiB of memory, this NUMA modification is required, because 48 GiB is not sufficient for the SDS.

This operation will have a small effect on performance (less than 10%), and for this reason, is not the default setting.

NOTE: This setting can also be modified on operational SDSs that are expected to use more memory in the future, for example, due to adding SDS devices. Note that this procedure requires a graceful SDS restart in order to take effect.

Steps

1. Use one of the following options to edit the NUMA configuration:

Option	Description
Use the echo command	On the node where the SDS is deployed, issue the command: `echo "numa_memory_affinity=0" >> /opt/emc/scaleio/sds/cfg/conf.txt`
Use a text editor	a. On the node where the SDS is deployed, navigate to the file: `/opt/emc/scaleio/sds/cfg/conf.txt` b. Open the file with a text editor, and add the following line to the file: *numa_memory_affinity=0* c. Save the file.

2. Gracefully restart the SDS:
   a. Use one of the PowerFlex management interfaces to place the SDS in instant maintenance mode.
   b. Restart the SDS process:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
/opt/emc/scaleio/sds/bin/create_service.sh
```

   c. Exit instant maintenance mode.
   d. Wait for the rebuild to finish before performing other tasks.

Results

Configuration of memory usage for the SDS is complete.

Next steps

Repeat this procedure on all SDSs that require this modification.

Configure the SDS for NVDIMM health awareness

Optionally, configure the SDS to enable periodic monitoring of bad blocks in NVDIMM devices.

Prerequisites

Root access permissions are required for this procedure.

About this task

To enable this feature in the PowerFlex system, perform this procedure on every SDS that uses NVDIMM acceleration devices. This feature is disabled by default.

Steps

1. In command line, run the command:

```
echo "acc_dev_ndctl_error_detection_enable=1" >> /opt/emc/scaleio/sds/cfg/conf.txt
```

2. Restart the SDS:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
```

```
/opt/emc/scaleio/sds/bin/create_service.sh
```

3. Wait until system rebuild/rebalance is complete, and then repeat steps 1 and 2 on the next SDS, until all SDSs are configured.

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Licensing

PowerFlex Licensing

PowerFlex installations are enabled to be fully functional for non-production environments. However, a license is required to use PowerFlex in a production environment.

PowerFlex licenses are purchased by total physical device capacity (in TB). You can activate your licensed capacity over multiple PowerFlex systems—each system with its unique installation ID.

For example, your purchase order may have been for 1000 TB. Your License Authorization Code (LAC) will entitle you to activate all, or part of that. You can activate 500 TB for one PowerFlex system, and leave the rest for another activation, for the same, or for a different system.

To obtain a license for production use, and to receive technical support, open a service ticket with EMC Support at https://support.emc.com.

When you purchase a license entitlement for PowerFlex, an email containing the LAC is sent to you or your purchasing department with a link to the Dell EMC Software Licensing Central website. The LAC is needed to complete the entitlement activation process on the licensing site, which is part of the online support portal located at Dell support site.

If you cannot find the LAC email, you can use the Software Licensing Central website to find your license entitlements.

Activate an entitlement and download the license file using the License Authorization Code

Use this procedure to activate the entitlements and download the license file when you have access to the License Authorization Code (LAC) email. The entitlements are the usage rights that you have purchased for the specific host machine.

Prerequisites

Ensure that the following are readily available.

- LAC email
- Sales order number

Steps

1. Identify the PowerFlex system's installation ID:
   - To identify the ID using the PowerFlex CLI, log in to the SCLI and run:

     ```
     scli --query_license
     ```

     The installation ID is displayed:

     ```
     Installation ID: 0123456789abcdef
     ```

   - To identify the ID using the GUI, in the GUI main window click the menu next to the user name (upper-right corner) and select **About**.

     The installation ID is displayed in the **About** window. PowerFlex
2. Click the link in the LAC email, and log in.
   The link takes you directly to the **Software Licensing Central** page.
3. Enter the LAC and click **Search**.

An online wizard assists you with the entitlement activation process.

4. On the **Select Products** page, select the product to activate and then click **Start the Activation Process**.
5. On the **Company Details** page, confirm (or update) your company's information and then click **Select a Machine**.
6. On the **Select a Machine** page, select the machine on which to activate the product and then click **Save Machine & Continue**.

   Click **Search** to locate an existing machine (one on which EMC product was previously activated), or add a new machine.

   NOTE: In the context of the activatation process, a machine is a PowerFlex system, which can include multiple servers.

7. On the **Enter Details** page, enter the quantity of the entitlement in TB to activate on the machine and the PowerFlex installation ID identified previously.

   To allocate the available capacity over multiple machines, select less than the full amount available and repeat the activation process on the other machines.

8. Click **Next**.
9. On the **Review** page, review your selections.

   NOTE: The license key will be emailed to the username logged in to the licensing system. To send it to more recipients, click **Email to more people** and enter their email addresses.

10. Click **Activate**.

Next steps

After you receive the license key, continue by installing the license in the system.

Activate an entitlement without an LAC

The License Authorization Code is needed to complete the entitlement activation process for your PowerFlex system. If you cannot find the LAC email, use this procedure to activate the entitlements and download the license file. The entitlements are the usage rights that you have purchased for the specific host machine.

Prerequisites

Ensure that the sales order number for the PowerFlex system is readily available.

Steps

1. Identify the PowerFlex system's installation ID:
   - To identify the ID using the PowerFlex CLI, log in to the SCLI and run:

     ```
     scli --query_license
     ```

     The installation ID is displayed:

     ```
     Installation ID: 0123456789abcdef
     ```

   - To identify the ID using the PowerFlex user interface, in the user interface main window, click the menu next to the username (upper-right corner) and select **About**.

     The installation ID is displayed in the **About** window.

2. Log in to the support site, at Dell Support site.

   If you are a new user, create a user account.
3. From the **Support Tasks** list, select **Manage Licenses and Usage Intelligence**.
4. On the **Manage Licenses** page, select **PowerFlex Software**.

   The browser directs you to **Software Licensing Central**.
5. On the **Powerlink Licensing** page, click **View Entitlements**.
6. On the **Powerlink Licensing - Search Entitlements** page, type the sales order number, and then click **Search Entitlements**.

7. Select the entitlement to activate from the list of entitlements, and then select **Options** > **Activate**.

   An online wizard assists you with the entitlement activation process.

8. On the **Company Details** page, confirm (or update) the information of your company and then click **Select a Machine**.
9. On the **Select a Machine** page, select the machine on which to activate the product and then click **Save Machine & Continue**.

   Click **Search** to locate an existing machine (one on which a Dell Technologies product was previously activated), or add a new machine.

   NOTE: In the context of the activation process, a machine is a PowerFlex system, which can include multiple servers.

10. On the **Enter Details** page, enter the quantity of the entitlement in TB to activate on the machine and the PowerFlex installation ID identified previously.

    To allocate the available capacity over multiple machines, select less than the full amount available and repeat the activation process on the other machines.

11. Click **Next**.
12. On the **Review** page, review your selections.

    NOTE: The license key is emailed to the username that is logged in to the licensing system. To send it to more recipients, click **Email to more people** and enter their email addresses.

13. Click **Activate**.

Next steps

After you receive the license key, continue by installing the license in the system.

Install the license

After activating the entitlements, install the PowerFlex license in the system.

About this task

Steps

1. Locate the email containing the PowerFlex license key.
2. Click the link in the email and download the license file to the primary MDM server.
3. Install the license:

   ```
   scli --set_license --license_file <license_file>
   ```

   Where *<license_file>* is the full path to the license file.

   For example:

   ```
   scli --set_license --license_file /tmp/0239SH4SS89023T6.lic
   ```

   The PowerFlex license is now installed in the MDM cluster.

   NOTE: Actual command syntax is operating-system dependent. For more information, see the *PowerFlex CLI Reference Guide*.

Licensing error messages

The following table lists error messages that may be generated by the system, and their troubleshooting solutions.

Table 36. Licensing error messages

Error Message	Description	Solution
The license key is invalid or does not match this version. Contact Support.	The license key is invalid.	Contact support.
The current system configuration exceeds the license entitlements.	More capacity has been installed than the license allows.	Reduce capacity, or extend the license capacity.
Operation could not be completed. The license capacity has been exceeded.	When you try to add an SDS or device, it will cause the licensed capacity to be exceeded.	Do not add the SDS or device, or extend the license capacity.
The license key is too long	The license file is larger than expected.	Check the accuracy of the license key.
The license has expired	The duration of the license has ended.	Extend the duration of the license.
The license installation ID does not match the ID of this system	When the Installation ID was entered in the ELM, it may have been incorrect.	Contact support.
The license contains a mismatch of the SWID. Contact Support.	The license key is invalid.	Contact support.
The issuer of the license you are attempting to add does not match that of the product	The license key is invalid.	Contact support.
The license contains a mismatch of the capacity values for basic and advanced features. Contact Support.	The capacity licensed for basic features is not equal to the capacity licensed for advanced feature.	Contact support.

View license information

You can view current license information using either the PowerFlex CLI or the GUI.

- To view license information using the CLI, run:

```
scli --query_license
```

NOTE: Actual command syntax is operating-system dependent. For more information, see the *PowerFlex CLI Reference Guide*.

- To view license information from the PowerFlex GUI:
  1. From the **Preferences** menu (upper-right corner), select **Systems Settings**.

     The System Settings window displays license and certificate information.

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How to modify an IP network for a PowerFlex system

Modify IP addresses or add IP addresses to a PowerFlex system.

NOTE: This procedure can be used on PowerFlex v2.x and higher systems.

Add IP addresses to a PowerFlex system

Optionally, add IP addresses to a three-node MDM cluster PowerFlex system after deployment. This topic explains how to do so in an ESXi environment.

Prerequisites

1. Prepare the IP addresses and networking required for your environment.
2. Configure the new IP addresses/networks on all systems.
3. Verify connectivity between all components using the new network.

NOTE: Use the ping command to check connectivity, including the relevant MTU size in the command.

About this task

The following example shows how to configure a system for an IPv6 network.

Table 37. Network and IPv6 addresses

Network	IPv6 Addresses
SVM Management	2a01:6f40:2003:104:10:54:80:121 (Gateway) 2a01:6f40:2003:104:10:54:80:122 2a01:6f40:2003:104:10:54:80:127 2a01:6f40:2003:104:10:54:80:128
ESXi Management	2a01:6f40:2003:104:10:54:80:124 2a01:6f40:2003:104:10:54:80:125 2a01:6f40:2003:104:10:54:80:126
SVM Data **NOTE:** Data IP addresses are used for MDMs, SDSs, and SDRs A virtual IP address is only used for the MDM Virtual IP address.	2a01:6f40:2003:192:192:168:73:6 2a01:6f40:2003:192:192:168:73:5 (Gateway) 2a01:6f40:2003:192:192:168:73:7 2a01:6f40:2003:192:192:168:73:8 2a01:6f40:2003:192:192:168:73:10 (Virtual IP address)

Table 37. Network and IPv6 addresses (continued)

Network	IPv6 Addresses
ESXi Data (Data IP addresses are used for SDCs)	2a01:6f40:2003:192:192:168:73:1 2a01:6f40:2003:192:192:168:73:2 2a01:6f40:2003:192:192:168:73:3
SDR External IP Addresses	2a01:6f40:2003:192:172:168:73:6 2a01:6f40:2003:192:172:168:73:7 2a01:6f40:2003:192:172:168:73:8

Steps

1. Open a CLI session with the primary MDM:

```
scli --login --username <NAME> [--password <PASSWORD>] [--ldap_authentication | --
native_authentication] [(--approve_certificate_once | --approve_certificate]
   --accept_banner_by_scripts_only
```

2. Query the cluster, and keep a note of the output for later use:

```
scli --query_cluster
```

Output example:

```
Cluster:
    Name: SIO-DEVOPS, Mode: 3_node, State: Normal, Active: 3/3, Replicas: 2/2
    Virtual IPs: 192.168.73.10
Master MDM:
    Name: Manager2, ID: 0x1e143dda22790f31
        IPs: 192.168.73.7, Management IPs: 10.54.80.127, Port: 9011, Virtual IP
interfaces: eth1
        Version: 2.6.10000
Slave MDMs:
    Name: Manager1, ID: 0x1788ff920173be80
        IPs: 192.168.73.6, Management IPs: 10.54.80.122, Port: 9011, Virtual IP
interfaces: eth1
        Status: Normal, Version: 2.6.10000
Tiebreakers:
    Name: TB1, ID: 0x4ec52e035f72ea42
        IPs: 192.168.73.8, Port: 9011
        Status: Normal, Version: 2.6.10000
```

3. Switch MDM cluster mode to single mode:

```
scli --switch_cluster_mode --cluster_mode 1_node --remove_slave_mdm_id <SLAVE_ID> --
remove_tb_id <TIE_BREAKER_ID>
```

For example:

```
scli --switch_cluster_mode --cluster_mode 1_node --remove_slave_mdm_id
0x1788ff920173be80 --remove_tb_id 0x4ec52e035f72ea42
```

The secondary and tiebreaker have been reverted to standby mode.

4. Remove the standby MDMs mentioned in the previous step:

```
scli --remove_standby_mdm --remove_mdm_id <STANDBY_ID>
scli --remove_standby_mdm --remove_mdm_id <STANDBY_ID>
```

For example:

```
scli --remove_standby_mdm --remove_mdm_id 0x4ec52e035f72ea42
scli --remove_standby_mdm --remove_mdm_id 0x1788ff920173be80
```

5. Add the removed secondary MDM back to the system as a standby MDM, using the old and the additional IP addresses.

```
scli --add_standby_mdm --mdm_role <ROLE> --new_mdm_ip <DATA_IP_ADDRESSES>  --
new_mdm_management_ip <MANAGEMENT_IP_ADDRESSES> --allow_asymmetric_ips --force_clean
--new_mdm_virtual_ip_interface <INTERFACE_NAMES>
```

For example:

```
scli --add_standby_mdm --mdm_role manager --new_mdm_ip
192.168.73.6,2a01:6f40:2003:192:192:168:73:6  --new_mdm_management_ip
10.54.80.122,2a01:6f40:2003:104:10:54:80:122 --allow_asymmetric_ips --force_clean --
new_mdm_virtual_ip_interface eth1,eth3
```

6. Add the removed tiebreaker back to the system as a standby MDM, using the old and the additional IP addresses.

```
scli --add_standby_mdm --mdm_role tb --new_mdm_ip <DATA_IP_ADDRESSES> --
allow_asymmetric_ips --force_clean
```

For example:

```
scli --add_standby_mdm --mdm_role tb --new_mdm_ip
192.168.73.8,2a01:6f40:2003:192:192:168:73:8 --allow_asymmetric_ips --force_clean
```

7. Switch MDM cluster operation mode back to three-node cluster mode.

```
scli --switch_cluster_mode --cluster_mode 3_node --add_slave_mdm_id <ID> --add_tb_id
<ID>
```

For example:

```
scli --switch_cluster_mode --cluster_mode 3_node --add_slave_mdm_id
0x0041f4ba31fbde70 --add_tb_id 0x6a1d2a0645dfd6c2
```

8. To verify cluster functionality, switch MDM ownership:
   a. Switch ownership:

   ```
   scli --switch_mdm_ownership --new_master_mdm_id <ID>
   ```

   For example:

   ```
   scli --switch_mdm_ownership --new_master_mdm_id 0x0041f4ba31fbde70
   ```

   b. Query the MDM cluster to get the current cluster state/health:

   ```
   scli --query_cluster
   ```

9. Repeat steps 3 to 8 for the secondary MDM (the one that was initially the primary MDM).
   All the IP addresses have now been added, and the initial primary MDM is now the primary MDM again, in a three-node MDM cluster.
10. Add virtual IP addresses to the MDM cluster. In the command, include both the existing IP addresses, and the new ones.

```
scli --modify_cluster_virtual_ips --cluster_virtual_ip <VIRTUAL_IP_ADDRESSES>
```

For example:

```
scli --modify_cluster_virtual_ips --cluster_virtual_ip
192.168.73.10,2a01:6f40:2003:192:192:168:73:10
```

11. Update the SDCs with the old and new virtual IP addresses. On each ESXi host in the PowerFlex system, perform the following steps:

a. Find the IniGuidStr of the host:

```
esxcli system module parameters list -m scini
```

b. Add the new IP addresses to the ESXi host. In the command, include both the old and the new virtual IP addresses:

```
esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<INI_GUID_STRING>
IoctlMdmIPStr=<IP_ADDRESSES>"
```

For example:

```
esxcli system module parameters set -m
scini -p "IoctlIniGuidStr=86dbe022-d0fd-4a4c-927a-cbf7eb4549f2
IoctlMdmIPStr=192.168.73.10,2a01:6f40:2003:192:192:168:73:10"
```

c. After modifying the IP addresses, a reboot cycle is needed for the changes to take effect. For all ESXi SDCs, gracefully reboot one node at a time.

Procedures for gracefully rebooting a node are described in "Gracefully shut down or reboot a node" in the "Reference" section of the *Upgrade PowerFlex Guide*.

12. Add new IP addresses to SDSs and SDRs.

a. Add the new IP addresses to SDSs, using the command:

```
scli --add_sds_ip (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_sds_ip <IP>
  [--sds_ip_role {sdc_only | sds_only | all}] [--force_add_ip]
```

For example:

```
scli --mdm_ip 192.168.73.7 --add_sds_ip --sds_ip 192.168.73.4 --new_sds_ip
192.168.73.24
```

b. Add SDRs, using the command:

```
scli --add_sdr_ip (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_sdr_ip <IP> [--sdr_ip_role  <ROLE>]
```

For example:

```
scli --mdm_ip 192.168.73.7 --add_sdr_ip --sdr_ip 192.168.73.4 -- new_sdr_ip
192.168.73.24
```

c. Verify that the new SDS IP addresses are being used, using the command:

```
scli --query_sds_connectivity_status (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

For example:

```
scli --query_sds_connectivity_status --protection_domain_name pd_18
```

d. Verify that the new SDR IP addresses are being used, using the command:

```
scli --query_all_sdr
```

For example:

```
scli --query_all_sdr
```

Example output:

```
Query-all-SDR returned 3 SDR nodes.

Protection Domain ID: a0e7ba3500000000
SDR ID: 24fc609400000000 Name: sdr49 State: Connected, Joined IP:
192.168.73.4,192.168.73.24 Port: 11088 Version: 3.5.0
```

```
SDR ID: 24fc609500000001 Name: sdr50 State: Connected, Joined IP: 192.168.73.5
Port: 11088 Version: 3.5.0
SDR ID: 24fc609600000002 Name: sdr51 State: Connected, Joined IP: 192.168.73.6
Port: 11088 Version: 3.5.0
```

e. Verify that the SDCs are using the new SDS IP addresses by running the following command on each SDC:

```
drv_cfg --query_tgts
```

f. If required, remove the old IP addresses from the SDSs, using the command:

```
scli --remove_sds_ip (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP_ADDRESS> [--
sds_port <PORT>]) --sds_ip_to_remove <IP_ADDRESS>
```

For example:

```
scli --mdm_ip 192.168.73.7 --remove_sds_ip
 --sds_ip 192.168.73.24 --sds_ip_to_remove 192.168.73.4
```

g. If required, remove the old IP addresses from each SDR, using the command:

```
scli --remove_sdr_ip (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP_ADDRESS> [--
sdr_port <PORT>]) --sdr_ip_to_remove <IP_ADDRESS>
```

For example:

```
scli --mdm_ip 192.168.73.7 --remove_sdr_ip
 --sdr_ip 192.168.73.24 --sdr_ip_to_remove 192.168.73.4
```

Next steps

After you have added a new IP network to the system, you can use this procedure to remove the old IP network, if it is no longer used. A typical scenario is moving systems from IPv4 networks to IPv6 based networks.

- In the relevant MDM steps, the old MDM IP addresses need to be omitted during the removal process.
- In the relevant SDC steps, after the old MDM virtual IP addresses are no longer in use, remove them from the `esxcli system module parameters set -m scini -p` command in the `IoctlMdmIPStr=<IP_ADDRESSES>` section.
- For the SDSs, use the `scli --remove_sds_ip` command to remove the old IP addresses.

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What to do next

This section describes the next steps for using PowerFlex following deployment.

What to do next

Once you have deployed PowerFlex, proceed to the post-installation tasks in the *Configure and Customize PowerFlex Guide*, and perform mandatory and optional configuration tasks.